Electrical Machines

Previous Year Questions • 1991–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Electrical Machines (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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UNIT

1

Magnetic Circuits

Topics

1. Magnetically Coupled Circuits, Dot Convention
2. Energy in Coupled circuits
3. Electric and Magnetic Materials

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Unit Snapshot*

Stream: EE	**Questions:** 17	**MCQ:** 8	**MSQ:** 1	**NAT:** 7
1M: 3	**2M:** 10	**Descriptive:** 4	**Avg Weightage** ~2M	**Range:** 1M(2016)–2M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q1.25.1 MSQ 2025 2M

Consider two coupled circuits, having self-inductances L_1 and L_2, that carry non-zero currents I_1 and I_2, respectively. The mutual inductance between the circuits is M with unity coupling coefficient. The stored magnetic energy of the coupled circuits is minimum at which of the following value(s) of $\frac{I_1}{I_2}$?

A. $-\frac{M}{L_1}$ **B.** $-\frac{M}{L_2}$ **C.** $-\frac{L_1}{M}$ **D.** $-\frac{L_2}{M}$

Q1.21.1 MCQ 2021 2M

The input impedance, $Z_{in}(s)$, for the network shown is

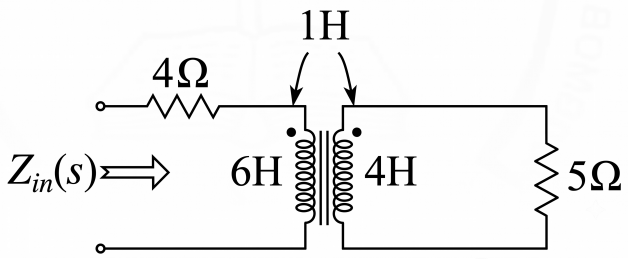


Fig. Q1.21.1

A. $\frac{23s^2 + 46s + 20}{4s + 5}$ **B.** $6s + 4$

C. $7s + 4$ **D.** $\frac{25s^2 + 46s + 20}{4s + 5}$

Q1.19.1 NAT 2019 2M

The magnetic circuit shown below has a uniform cross-sectional area and an air gap of 0.2 cm. The mean path length of the core is 40 cm. Assume that leakage and fringing fluxes are negligible. When the core relative permeability is assumed to be infinite, the magnetic flux density computed in the air gap is 1 tesla. With the same Ampere-turns, if the core relative permeability is assumed to be 1000 (linear), the flux density in tesla (rounded off to three decimal places) calculated in the air gap is _____.

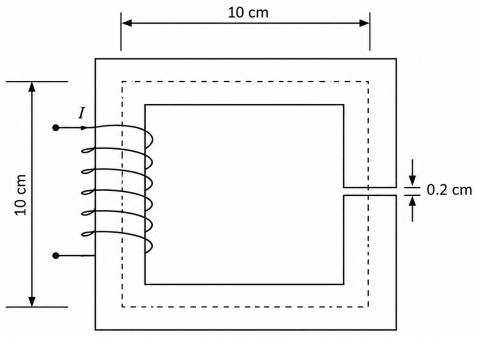


Fig. Q1.19.1

Q1.18.1 MCQ 2018 2M

A transformer with a toroidal core of permeability μ is shown in the figure. Assuming a uniform flux density across the circular core cross-section of radius $r \ll R$, and neglecting any leakage flux, the best estimate for the mean radius R is

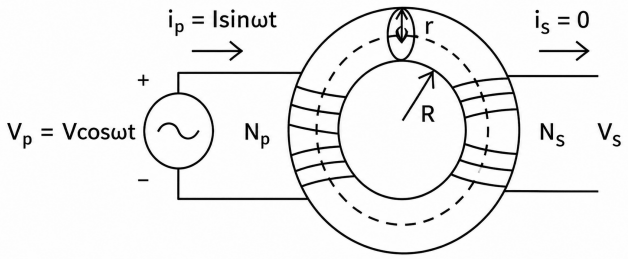


Fig. Q1.18.1

A. $\frac{\mu V r^2 N_p^2 \omega}{I}$ **B.** $\frac{\mu I r^2 N_p N_s \omega}{V}$

C. $\frac{\mu V r^2 N_p^2 \omega}{2I}$ **D.** $\frac{\mu I r^2 N_p^2 \omega}{2V}$

Q1.16.1 NAT 2016 2M

The flux linkage (λ) and current (i) relation for an electromagnetic system is $\lambda = \frac{\sqrt{i}}{g}$. When $i = 2$ A and g (air-gap length) = 10 cm, the magnitude of mechanical force on the moving part, in N, is _____.

Q1.15.1 NAT 2015 2M

Two identical coils each having inductance L are placed together on the same core. If an overall inductance of αL is obtained by interconnecting these two coils, the minimum value of α is _____.

Common Data Questions

Statement for Linked Questions (Q1.07.1 and Q1.07.2): An inductor designed with 400 turns coil wound on an iron core of 16 cm^2 cross sectional area and with a cut of an air gap length of 1 mm. The coil is connected to a 230 V, 50 Hz ac supply. Neglect coil resistance, core loss, iron reluctance and leakage inductance. ($\mu_0 = 4\pi \times 10^{-7}$ H/m)

Q1.07.1 MCQ 2007 2M

The current in the inductor is

A. 18.08 A **B.** 9.04 A **C.** 4.56 A **D.** 2.28 A

Q1.07.2 MCQ 2007 2M

The average force on the core to reduce the air gap will be

A. 832.29 N **B.** 1666.22 N **C.** 3332.47 N **D.** 6664.84 N

Q1.04.1 MCQ 2004 1M

For a linear electromagnetic circuit, the following statement is true.

A. Field energy is equal to the co-energy

B. Field energy is greater than the co-energy

C. Field energy is lesser than the co-energy

D. Co-energy is zero

Q1.01.1 Descriptive 2001 5M

Determine the resonance frequency and the Q-factor of the circuit shown in figure.

Data: $R = 10\,\Omega$, $C = 3\,\mu$F, $L_1 = 40$ mH, $L_2 = 10$ mH and $M = 10$ mH.

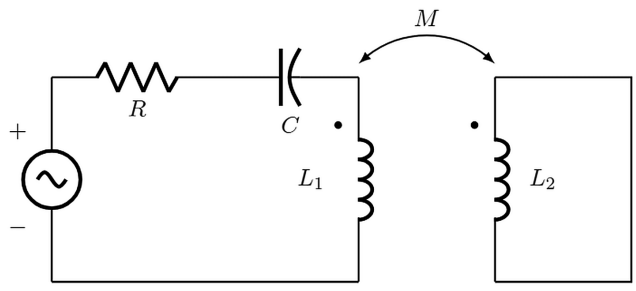


Fig. Q1.01.1

Q1.01.2 MCQ 2001 2M

The hysteresis loop of a magnetic material has an area of 5 cm^2, with the scales given as 1 cm = 2 AT and 1 cm = 50 mWb. At 50 Hz, the total hysteresis loss is

A. 15 W **B.** 20 W **C.** 25 W **D.** 50 W

Q1.97.1 NAT 1997 5M

The given figure shows a magnetic circuit formed by an ideal core material. Determine the magnetic flux density in the air gap.

_____

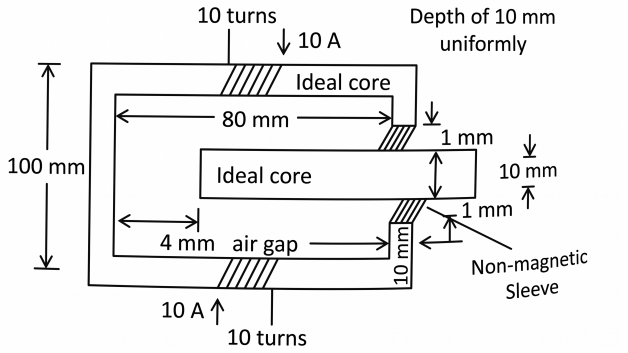


Fig. Q1.97.1

Q1.97.2 NAT 1997 2M

Two identical coils of negligible resistance, when connected in series across a 200 V, 50 Hz source, draw a current of 10 A. When the terminals of one of the coils are reversed, the current drawn is 8 A. The coefficient of coupling between the two coils is _____

Q1.95.1 NAT 1995 5M

The figure below shows the details of an electromagnet with an operating coil rated for 230 V and 50 Hz. The coil has 1500 turns. Find the average force of attraction on the armature of the electromagnet and the current drawn by the coil. Omit winding resistance and core losses, and assume infinite permeability for the core.

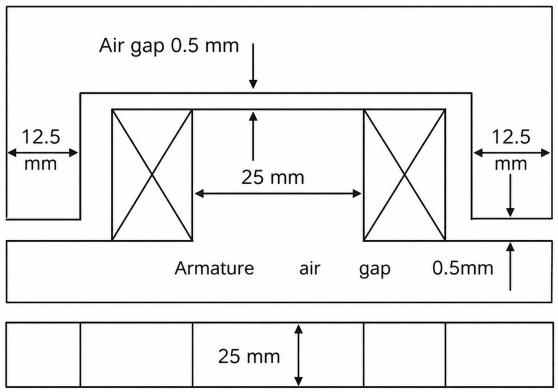


Fig. Q1.95.1

Q1.92.1 MCQ 1992 1M

The developed electromagnetic force and/or torque in electro-mechanical energy conversion systems acts in a direction that tends

A. to increase the stored energy at constant *mmf*

B. to decrease the stored energy at constant flux

C. to decrease the co-energy at constant *mmf*

D. to decrease the stored energy at constant *mmf*

Q1.91.1 NAT 1991 2M

A toroidal iron ring has a uniform cross-sectional area of 50 mm^2 and a mean magnetic path length of 100 mm. The ring has an air-gap of 1 mm, and it is excited with a dc current of 1 A through a coil of 100 turns wound uniformly along its length. The iron may be assumed to be a perfect magnetic material. The effect of fringing at the gap may be assumed to increase the effective area of magnetic flux at the gap by 10%. Evaluate

(i) The exciting mmf of the coil

(ii) The effective reluctance of the magnetic circuit.

(iii) The inductance of the coil

(iv) The energy stored in the magnetic field under the above excitation.

Q1.91.2 MCQ 1991 1M

The energy stored in the magnetic field of solenoid 30 cm long and 3 cm diameter wound with 1000 turns of wire carrying a current of 10 A is

A. 0.015 joule **B.** 0.15 joule

C. 0.5 joule **D.** 1.15 joule

Answer Key

Q1.25.1 A, D	Q1.21.1 A	Q1.19.1 0.83 to 0.84
Q1.18.1 D	Q1.16.1 186 to 190	Q1.15.1 0
Q1.07.1 D	Q1.07.2 A	Q1.04.1 A
Q1.01.1 530.52, 10	Q1.01.2 C	Q1.97.1 0.0628
Q1.97.2 0.111	Q1.95.1 303.3	Q1.92.1 B
Q1.91.1 100, 1.44×10^7, 0.69, 0.345	Q1.91.2 B	

UNIT

II

Transformers

Topics

1. Single Phase Transformer: Construction
2. Equivalent Circuit and Phasor Diagram
3. Open circuit and Short circuit tests
4. Voltage Regulation and Efficiency of Transformer
5. Effect of Frequency in Transformer
6. Parallel Operation of Transformers
7. Auto Transformer
8. Three Phase Transformer - Connections and Vector Groups
9. Scott - T Transformer
10. Harmonics in Transformer
11. Electromechanical Energy Conversion Principles

Unit Snapshot*

Stream: EE	**Questions:** 108	**MCQ:** 80	**MSQ:** 2	**NAT:** 26
1M: 50	**2M:** 53	**Descriptive:** 5	**Avg Weightage** ~4M	**Range:** 1M(2024)–7M(2015)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MSQ 2026 1M

A 220V/12V single-phase transformer is designed for use in India and rated 100 VA at 50 Hz. Later, this unit is shipped to the USA where it is used as a 110V/6V transformer at 60 Hz. Which of the following statements is/are correct?

- **A.** No-load current drawn would be smaller for operation in the USA compared to that in India
- **B.** For the same load current, the losses would be higher in the USA compared to that in India
- **C.** The peak magnetic flux density in the core would be higher while operating in the USA compared to that in India
- **D.** The eddy current losses in the core would be approximately 44% higher while operating in the USA compared to that in India

Q2.26.2 MCQ 2026 2M

Three single-phase 11kV/3.3kV transformers are connected to form a three-phase transformer bank with connections as shown.

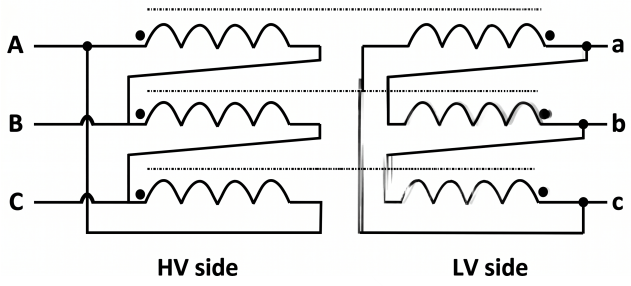


Fig. Q2.26.2

Considering the ABC phase sequence, what is the vector group of the transformer?

A. Dd0 **B.** Dd4 **C.** Dd6 **D.** Dd10

Q2.26.3 MCQ 2026 1M

A 15 kVA, 1100 V/220 V single-phase two-winding transformer is configured as a 1.32 kV/1.1 kV autotransformer. What will be the rating of the autotransformer?

A. 60 kVA **B.** 75 kVA **C.** 90 kVA **D.** 100 kVA

Q2.25.1 MCQ 2025 2M

The transformer connection given in the figure is part of a balanced 3-phase circuit where the phase sequence is "abc". The primary to secondary turns ratio is 2 : 1. If $(I_a + I_b + I_c = 0)$, then the relationship between I_A and I_{ad} will be

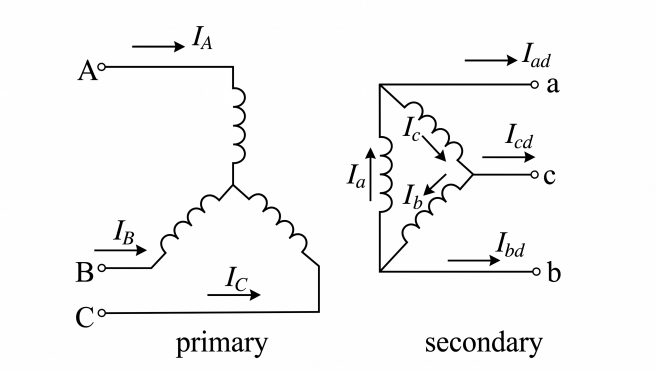


Fig. Q2.25.1

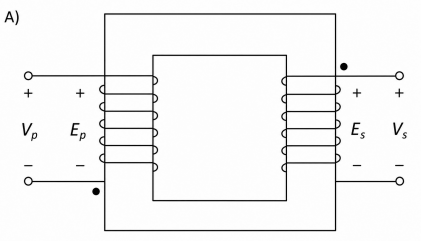
- **A.** $\frac{|I_A|}{|I_{ad}|} = \frac{1}{2\sqrt{3}}$ and I_{ad} lags I_A by 30°
- **B.** $\frac{|I_A|}{|I_{ad}|} = \frac{1}{2\sqrt{3}}$ and I_{ad} leads I_A by 30°
- **C.** $\frac{|I_A|}{|I_{ad}|} = 2\sqrt{3}$ and I_{ad} lags I_A by 30°
- **D.** $\frac{|I_A|}{|I_{ad}|} = 2\sqrt{3}$ and I_{ad} leads I_A by 30°

Q2.24.1 MCQ 2024 1M

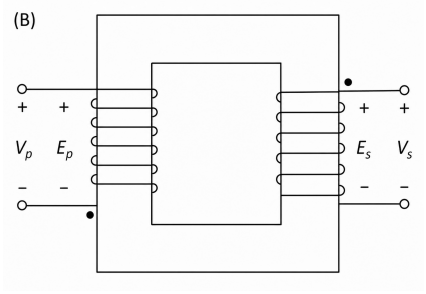
Which of the following options represents possible voltage polarities in a single phase two winding transformer?

Here, V_p is the applied primary voltage, E_p is the induced primary voltage, V_s is the open circuit secondary voltage, and E_s is the induced secondary voltage.

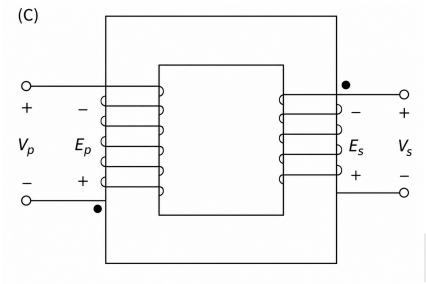
A.



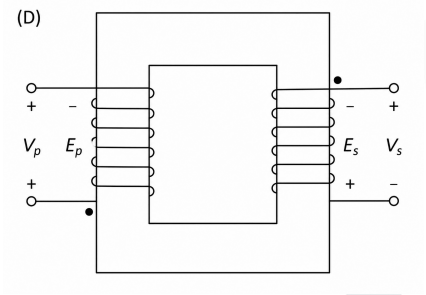
B.



C.



D.



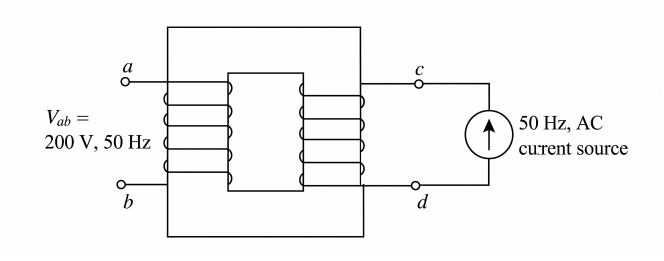


Fig. Q2.23.1

Q2.21.1 NAT 2021 2M

An air-core radio-frequency transformer, as shown, has a primary winding and a secondary winding. The mutual inductance M between the windings of the transformer is _____ μH.
(Round off to 2 decimal places.)

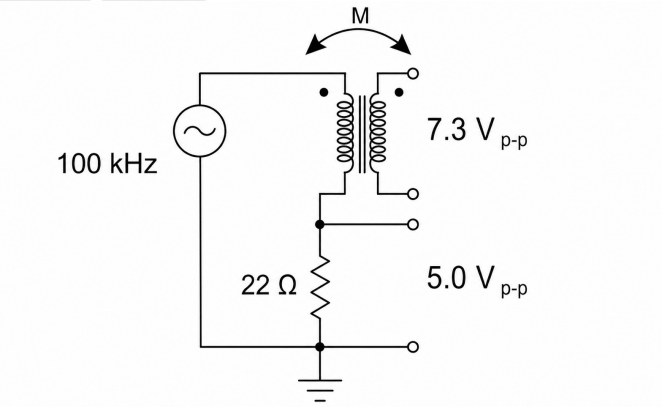


Fig. Q2.21.1

Q2.23.1 NAT 2023 2M

When the winding c-d of the single-phase 50 Hz two winding transformer is supplied from an AC current source of frequency 50 Hz, the rated voltage of 200 V (rms) 50 Hz is obtained at the open-circuit terminals a-b. The cross sectional area of the core is 5000 mm^2 and the average length of the core traversed by the mutual flux is 500 mm.The maximum allowable flux density in the core is B_{max} = 1 Wb/m^2 and the relative permeability of the core material is 5000. The leakage impedance of the winding a-b and the winding c-d in 50 Hz are $(5 + j100\pi \times 0.16)$ Ω and $(11.25 + j100\pi \times 0.36)$ Ω, respectively. Considering the magnetizing characteristics to be linear and neglecting core loss, the self-inductance of the winding a-b in millihenry is _____ (round to the 1 decimal place).

Q2.21.2 MCQ 2021 2M

In a single-phase transformer, the total iron loss is 2500 W at nominal voltage of 440 V and frequency 50 Hz. The total iron loss is 850 W at 220 V and 25 Hz. Then, at nominal voltage and frequency, the hysteresis loss and eddy current loss respectively are

A. 1600 W and 900 W
B. 900 W and 1600 W
C. 250 W and 600 W
D. 600 W and 250 W

Q2.20.1 NAT 2020 2M

Windings 'A', 'B' and 'C' have 20 turns each and are wound on the same iron core as shown, along with winding 'X' which has 2 turns. The figure shows the sense (clockwise/anti-clockwise) of each of the windings only and does not reflect the exact number of turns. If windings

'A', 'B' and 'C' are supplied with balanced 3-phase voltages at 50 Hz and there is no core saturation, the no-load RMS voltage (in V, rounded off to 2 decimal places) across winding 'X' is _____.

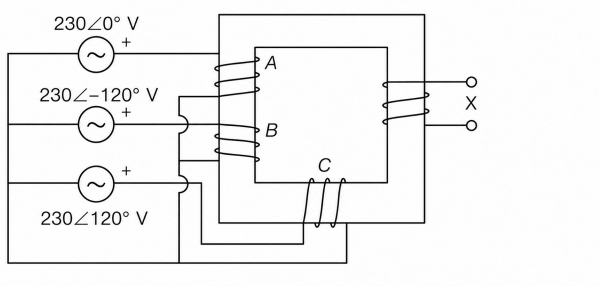


Fig. Q2.20.1

Q2.20.2 NAT | 2020 | 1M

A single-phase, 4 kVA, 200 V/100 V, 50 Hz transformer with a laminated CRGO steel core has a rated no-load loss of 450 W. When the high-voltage winding is excited with 160 V, 40 Hz sinusoidal ac supply, the no-load losses are found to be 320 W. When the high-voltage winding of the same transformer is supplied from a 100 V, 25 Hz sinusoidal ac source, the no-load losses will be _____ W (rounded off to 2 decimal places).

Q2.19.1 NAT | 2019 | 2M

The magnetic circuit shown below has a uniform cross-sectional area and an air gap of 0.2 cm. The mean path length of the core is 40 cm. Assume that leakage and fringing fluxes are negligible. When the core relative permeability is assumed to be infinite, the magnetic flux density computed in the air gap is 1 tesla. With same Ampere-turns, if the core relative permeability is assumed to be 1000 (linear), the flux density in tesla (Round off to three decimal places) calculated in the air gap is _____.

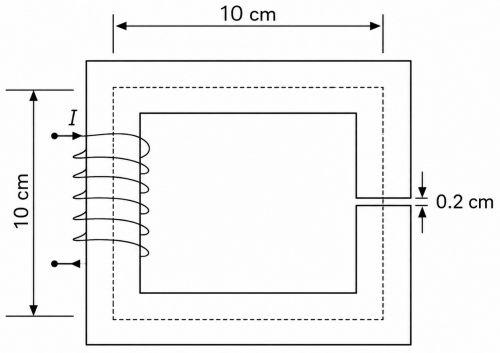


Fig. Q2.19.1

Q2.19.2 NAT | 2019 | 2M

A single-phase transformer of rating 25 kVA, supplies a 12 kW load at power factor of 0.6 lagging. The additional load at unity power factor in kW (round off to two decimal places) that may be added before this transformer exceeds its rated kVA is _____.

Q2.19.3 MCQ | 2019 | 1M

A 5 kVA, 50 V/100 V, single-phase transformer has a secondary terminal voltage of 95 V when loaded. The regulation of the transformer is

A. 4.5% **B.** 9% **C.** 5% **D.** 1%

Q2.18.1 NAT | 2018 | 2M

A 3-phase 900 kVA, 3 kV/$\sqrt{3}$ kV (Δ/Y), 50 Hz transformer has primary (high voltage side) resistance per phase of 0.3 Ω and secondary (low voltage side) resistance per phase of 0.02 Ω. Iron loss of the transformer is 10 kW. The full load % efficiency of the transformer operated at unity power factor is _____ (up to 2 decimal places).

Q2.18.2 MCQ | 2018 | 1M

A single-phase 100 kVA, 1000 V/100 V, 50 Hz transformer has a voltage drop of 5% across its series impedance at full load. Of this, 3% is due to resistance. The percentage regulation of the transformer at full load with 0.8 lagging power factor is

A. 4.8 **B.** 6.8 **C.** 8.8 **D.** 10.8

Q2.17.1 NAT | 2017 | 2M

A three-phase, three winding Δ/Δ/Y (1.1 kV/6.6 kV/400 V) transformer is energized from AC mains at the 1.1 kV side. It supplies 900 kVA load at 0.8 power factor lag from the 6.6 kV winding and 300 kVA load at 0.6 power factor lag from the 400 V winding. The RMS line current in ampere drawn by the 1.1 kV winding from the mains is _____. (Give the answer up to one decimal place.)

Q2.17.2 NAT | 2017 | 1M

If the primary line voltage rating is 3.3 kV (Y side) of a 25 kVA, Y-Δ transformer (the per phase turns ratio is 5 : 1), then the line current rating of the secondary side (in Ampere) is _____.

Q2.16.1 MCQ 2016 1M

If an ideal transformer has an inductive load element at port 2 as shown in the figure below, the equivalent inductance at port 1 is

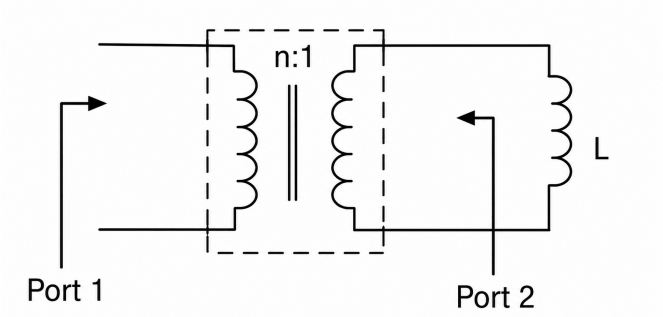


Fig. Q2.16.1

A. nL **B.** n^2L **C.** $\frac{n}{L}$ **D.** $\frac{n^2}{L}$

Q2.16.2 MCQ 2016 2M

If the star side of the star-delta transformer shown in the figure is excited by a negative sequence voltage, then

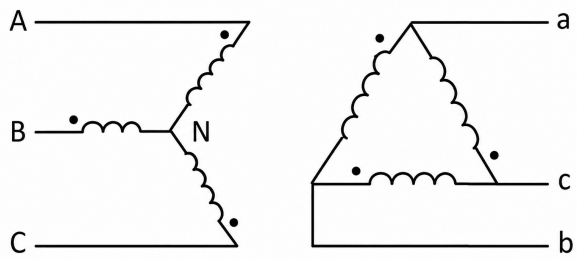


Fig. Q2.16.2

A. V_{AB} leads V_{ab} by 60° **B.** V_{AB} lags V_{ab} by 60°
C. V_{AB} leads V_{ab} by 30° **D.** V_{AB} lags V_{ab} by 30°

Q2.16.3 NAT 2016 1M

The following figure shows the connection of an ideal transformer with primary to secondary turns ratio of 1 : 100. The applied primary voltage is 100 V (rms), 50 Hz, AC. The rms value of the current I, in ampere, is _____.

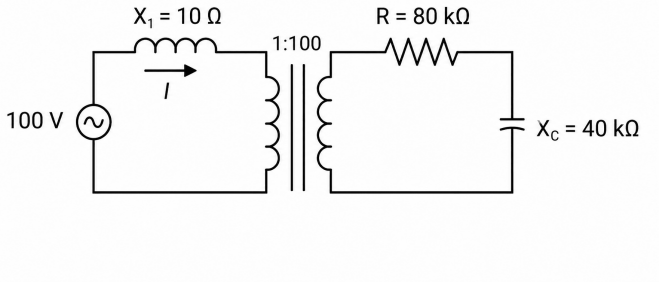


Fig. Q2.16.3

Q2.16.4 MCQ 2016 2M

Three single-phase transformers are connected to form a delta-star three-phase transformer of 110 kV/11 kV. The transformer supplies at 11 kV a load of 8 MW at 0.8 p.f. lagging to a nearby plant. Neglect the transformer losses. The ratio of phase currents in delta side to star side is

A. $1 : 10\sqrt{3}$ **B.** $10\sqrt{3} : 1$ **C.** $1 : 10$ **D.** $\sqrt{3} : 10$

Q2.16.5 NAT 2016 2M

A single-phase, 2 kVA, 100/200 V transformer is reconnected as an auto-transformer such that its kVA rating is maximum. The new rating, in kVA, is _____.

Q2.16.6 NAT 2016 2M

A single-phase 400 V, 50 Hz transformer has an iron loss of 5000 W at the rated condition. When operated at 200 V, 25 Hz, the iron loss is 2000 W. When operated at 416 V, 52 Hz, the value of the hysteresis loss divided by the eddy current loss is _____.

Q2.16.7 MCQ 2016 2M

A single-phase, 22 kVA, 2200 V/220 V, 50 Hz, distribution transformer is to be connected as an auto-transformer to get an output voltage of 2420 V. Its maximum kVA rating as an auto-transformer is

A. 22 **B.** 24.2 **C.** 242 **D.** 2420

Q2.15.1 NAT 2015 2M

A 200/400 V, 50 Hz, two-winding transformer is rated at 20 kVA. Its windings are connected as an auto-transformer of rating 200/600 V. A resistive load of 12 Ω is connected to the high voltage (600 V) side of the auto-transformer. The value of equivalent load resistance (in Ohm) as seen from low voltage side is _____.

Q2.15.2 NAT 2015 2M

Two single-phase transformers T_1 and T_2 each rated at 500 kVA are operated in parallel. Percentage impedances of T_1 and T_2 are $(1 + j6)$ and $(0.8 + j4.8)$, respectively. To share a load of 1000 kVA at 0.8 lagging power factor, the contribution of T_2 (in kVA) is _____.

Q2.15.3 MCQ 2015 1M

Find the transformer ratios a and b such that the impedance (Z_{in}) is resistive and equals 2.5 Ω when the network is excited with a sine wave voltage of angular frequency of 5000 rad/s.

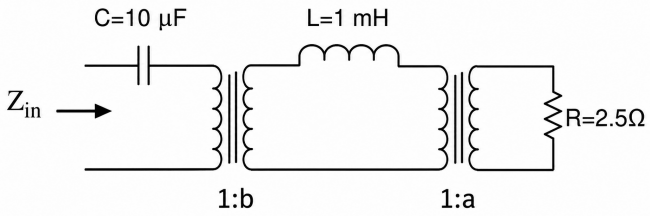


Fig. Q2.15.3

A. $a = 0.5,\ b = 2.0$ **B.** $a = 2.0,\ b = 0.5$
C. $a = 1.0,\ b = 1.0$ **D.** $a = 4.0,\ b = 0.5$

Q2.15.4 MCQ 2015 2M

A three-winding transformer is connected to an AC voltage source as shown in the figure. The number of turns are as follows: $N_1 = 100$, $N_2 = 50$, $N_3 = 50$. If the magnetizing current is neglected, and the currents in two windings are $\overline{I_2} = 2\angle 30°$ A and $\overline{I_3} = 2\angle 150°$ A, then what is the value of the current $\overline{I_1}$ in Ampere?

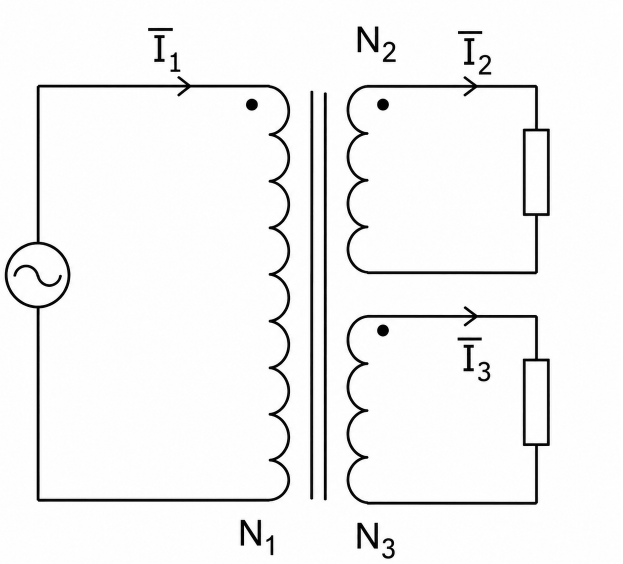


Fig. Q2.15.4

A. $1\angle 90°$ **B.** $1\angle 270°$ **C.** $4\angle 90°$ **D.** $4\angle 270°$

Q2.15.5 NAT 2015 2M

Two three-phase transformers are realized using single-phase transformers, as shown in the figure.

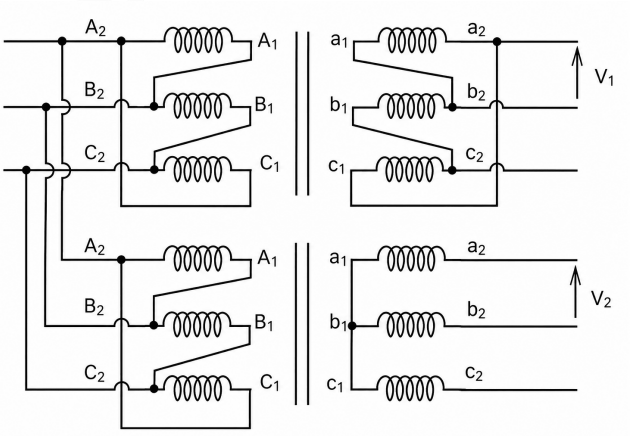


Fig. Q2.15.5

The phase difference (in degree) between voltages V_1 and V_2 is _____.

Q2.15.6 MCQ 2015 2M

A balanced (positive sequence) three-phase AC voltage source is connected to a balanced, star connected load through a star-delta transformer as shown in the figure. The line-to-line voltage rating is 230 V on the star side,

and 115 V on the delta side. If the magnetizing current is neglected and $\overline{I_s} = 100\angle 0°$ A, then what is the value of $\overline{I_p}$ in Ampere?

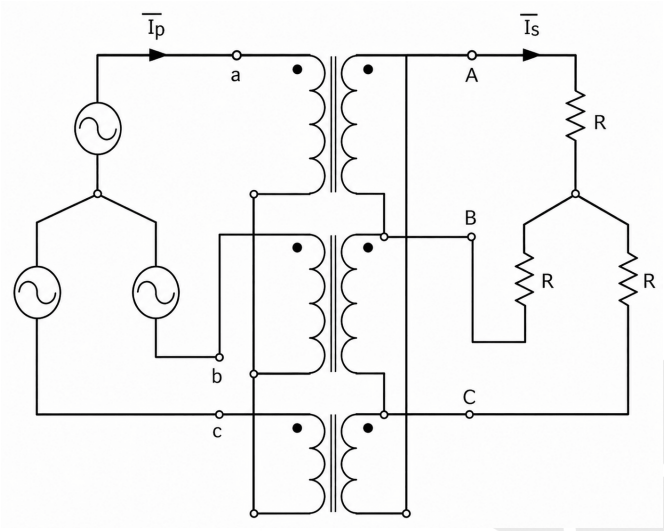


Fig. Q2.15.6

A. $50\angle 30°$ **B.** $50\angle -30°$ **C.** $50\sqrt{3}\angle 30°$ **D.** $200\angle 30°$

Q2.15.7 MCQ 2015 1M

The primary mmf is least affected by the secondary terminal conditions in a

A. power transformer **B.** potential transformer
C. current transformer **D.** distribution transformer

Q2.15.8 MCQ 2015 1M

The self inductance of the primary winding of a single phase, 50 Hz, transformer is 800 mH, and that of the secondary winding is 600 mH. The mutual inductance between these two windings is 480 mH. The secondary winding of this transformer is short circuited and the primary winding is connected to a 50 Hz, single phase, sinusoidal voltage source. The current flowing in both the windings is less than their respective rated currents. The resistance of both windings can be neglected. In this condition, what is the effective inductance (in mH) seen by the source?

A. 416 **B.** 440 **C.** 200 **D.** 920

Q2.14.1 MCQ 2014 2M

The core loss of a single phase, 230/115 V, 50 Hz power transformer is measured from 230 V side by feeding the primary (230 V side) from a variable voltage variable frequency source while keeping the secondary open circuited. The core loss is measured to be 1050 W for 230 V, 50 Hz input. The core loss is again measured to be 500 W for 138 V, 30 Hz input. The hysteresis and eddy current losses of the transformer for 230 V, 50 Hz input are respectively,

A. 508 W and 542 W **B.** 468 W and 582 W
C. 498 W and 552 W **D.** 488 W and 562 W

Q2.14.2 NAT 2014 1M

A single phase, 50 kVA, 1000 V/100 V two winding transformer is connected as an autotransformer as shown in the figure.

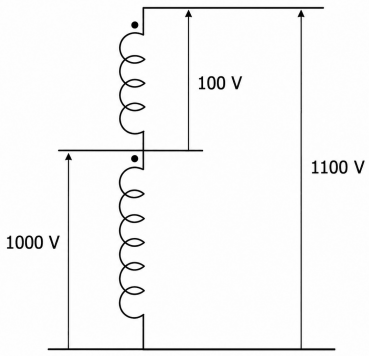


Fig. Q2.14.2

The kVA rating of the autotransformer is _____.

Q2.14.3 MCQ 2014 2M

An open circuit test is performed on 50 Hz transformer, using variable frequency source and keeping V/f ratio constant, to separate its eddy current and hysteresis losses. The variation of core loss/frequency as function of frequency is shown in the figure

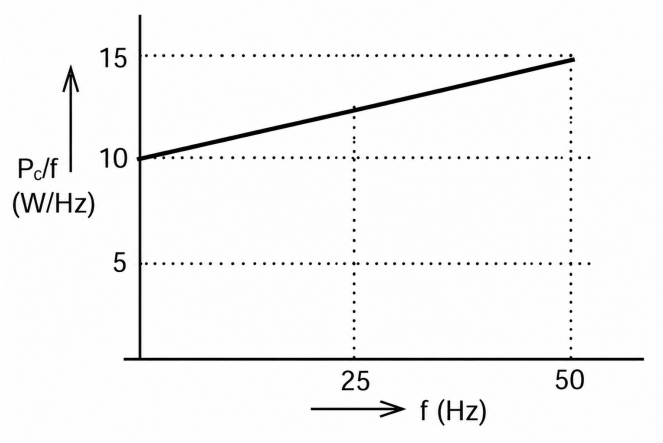


Fig. Q2.14.3

The hysteresis and eddy current losses of the transformer at 25 Hz respectively are

A. 250 W and 2.5 W
B. 250 W and 62.5 W
C. 312.5 W and 62.5 W
D. 312.5 W and 250 W

Q2.14.4 MCQ 2014 2M

The parameters measured for a 220V/110V, 50 Hz, single-phase transformer are:

Self inductance of primary winding = 45 mH

Self inductance of secondary winding = 30 mH

Mutual inductance between primary and secondary winding = 20 mH

Using the above parameters, the leakage (L_{l1}, L_{l2}) and magnetizing (L_m) inductances as referred to primary side in the equivalent circuit respectively, are

A. 5 mH, 20 mH and 40 mH

B. 5 mH, 80 mH and 40 mH

C. 25 mH, 10 mH and 20 mH

D. 45 mH, 30 mH and 20 mH

Q2.14.5 NAT 2014 2M

The load shown in the figure absorbs 4 kW at a power factor of 0.89 lagging.

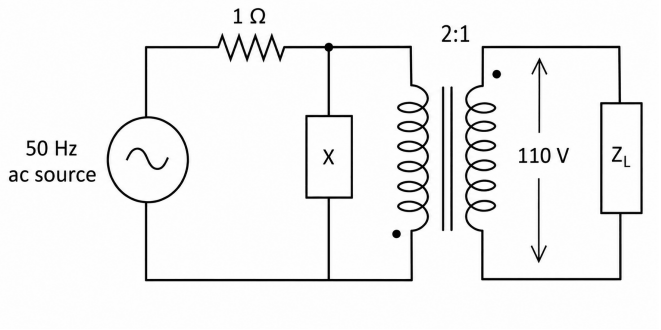


Fig. Q2.14.5

Assuming the transformer to be ideal, the value of the reactance X to improve the input power factor to unity is _____.

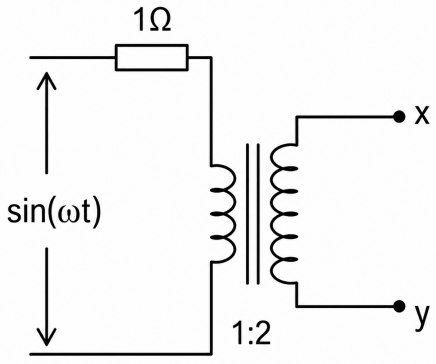
Q2.14.6 MCQ 2014 1M

For a single phase, two winding transformer, the supply frequency and voltage are both increased by 10%. The percentage changes in the hysteresis loss and eddy current loss, respectively, are

A. 10 and 21
B. −10 and 21
C. 21 and 10
D. −21 and 10

Q2.14.7 MCQ 2014 1M

Assuming an ideal transformer, the Thevenin's equivalent voltage and impedance as seen from the terminals x and y for the circuit in figure are



A. $2\sin(\omega t)$, 4Ω
B. $1\sin(\omega t)$, 1Ω
C. $1\sin(\omega t)$, 2Ω
D. $2\sin(\omega t)$, 0.5Ω

Q2.14.8 MCQ 2014 1M

For a specified input voltage and frequency, if the equivalent radius of the core of a transformer is reduced by half, the factor by which the number of turns in the primary should change to maintain the same no load current is

A. 1/4 **B.** 1/2 **C.** 2 **D.** 4

Q2.13.1 MCQ 2013 1M

A single-phase transformer has no-load loss of 64 W, as obtained from an open-circuit test. When a short-circuit test is performed on it with 90% of the rated currents flowing in its both LV and HV windings, the measured loss is 81 W. The transformer has maximum efficiency when operated at

A. 50.0% of the rated current.

B. 64.0% of the rated current.

C. 80.0% of the rated current.

D. 88.8% of the rated current.

Q2.13.2 MCQ 2013 2M

The following arrangement consists of an ideal transformer and an attenuator which attenuates by a factor of 0.8. An ac voltage V_{WX1} = 100 V is applied across WX to get an open circuit voltage V_{YZ1} across YZ. Next, an ac voltage V_{YZ2} = 100 V is applied across YZ to get an open circuit voltage V_{WX2} across WX. Then, V_{YZ1}/V_{WX1}, V_{WX2}/V_{YZ2} are respectively,

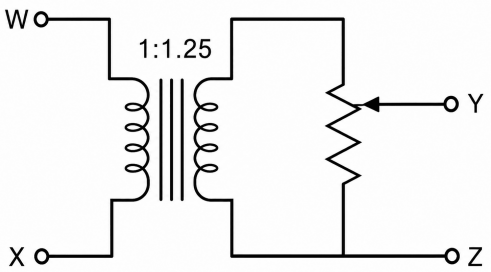


Fig. Q2.13.2

A. 125/100 and 80/100 **B.** 100/100 and 80/100
C. 100/100 and 100/100 **D.** 80/100 and 80/100

Q2.12.1 MCQ 2012 2M

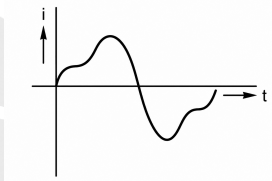
A single phase 10 kVA, 50 Hz transformer with 1 kV primary winding draws 0.5 A and 55 W, at rated voltage and frequency, on no load. A second transformer has a core with all its linear dimensions $\sqrt{2}$ times the corresponding dimensions of the first transformer. The core material and lamination thickness are the same in both transformers. The primary windings of both transformers have the same number of turns. If a rated voltage of 2 kV at 50 Hz is applied to the primary of the second transformer, then the no load current and power, respectively, are

A. 0.7 A, 77.8 W **B.** 0.7 A, 155.6 W
C. 1 A, 110 W **D.** 1 A, 220 W

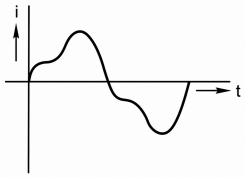
Q2.11.1 MCQ 2011 1M

A single-phase air core transformer, fed from a rated sinusoidal supply, is operating at no load. The steady state magnetizing current drawn by the transformer from the supply will have the waveform

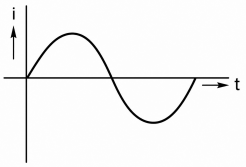
A.



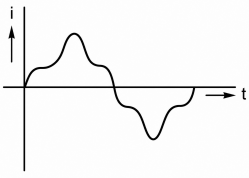
B.



C.



D.



Q2.10.1 MCQ 2010 2M

A balanced star-connected and purely resistive load is connected at the secondary of a star-delta transformer as shown in the figure. The line-to-line voltage rating of the transformer is 110 V/220 V. Neglecting the non-idealities of the transformer, the impedance Z of the equivalent star-connected load, referred to the primary side of the transformer, is:

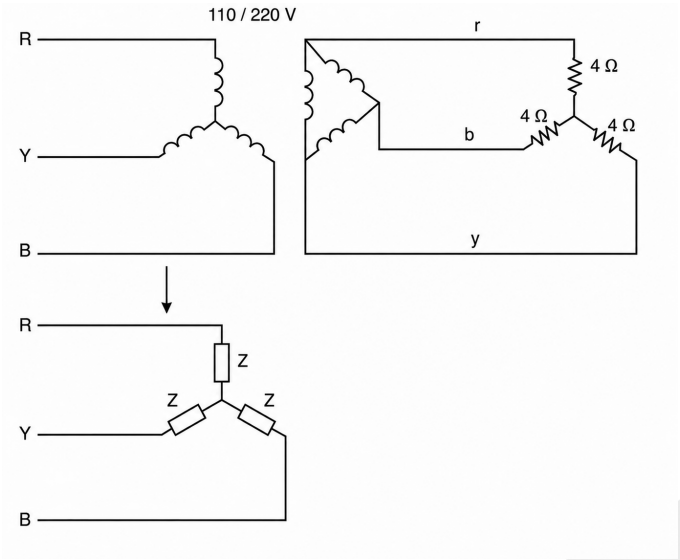
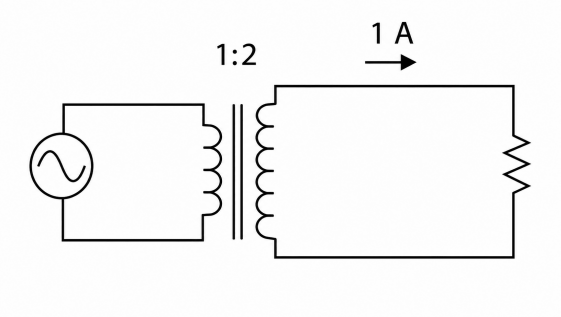


Fig. Q2.10.1

A. $(3 + j0)\ \Omega$ **B.** $(0.866 - j0.5)\ \Omega$
C. $(0.866 + j0.5)\ \Omega$ **D.** $(1 + j0)\ \Omega$

Q2.10.2 MCQ 2010 1M

A single-phase transformer has a turns ratio of 1 : 2, and is connected to a purely resistive load as shown in the figure. The magnetizing current drawn is 1 A and the secondary current is 1 A. If core losses and leakage reactances are neglected, the primary current is



A. 1.41 A **B.** 2 A **C.** 2.24 A **D.** 3 A

Q2.09.1 MCQ 2009 1M

The single phase, 50Hz, iron core transformer in the circuit has both vertical arms with a cross sectional area of $20cm^2$ and both horizontal arms with a cross sectional area of $10cm^2$. If the two windings shown were wound instead on opposite horizontal arms, the mutual inductance will

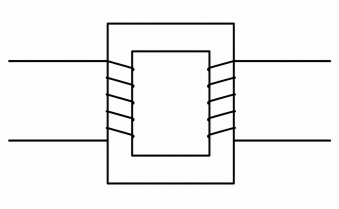


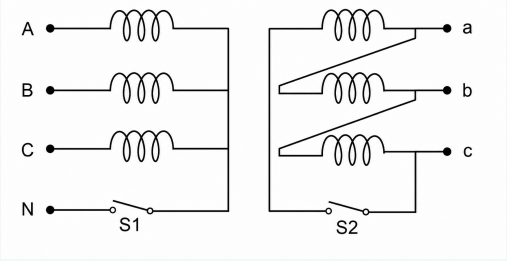
Fig. Q2.09.1

A. double **B.** remain same
C. be halved **D.** become one quarter

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Questions (Q2.09.2 and Q2.09.3): The star-delta transformer shown above is excited on the star side with a balanced, 4-wire, 3-phase, sinusoidal voltage supply of rated magnitude. The transformer is under no load condition.



Q2.09.2 MCQ 2009 2M

With both S1 and S2 open, the core flux waveform will be

A. a sinusoid at fundamental frequency

B. flat-topped with third harmonic

C. peaky with third-harmonic

D. none of these

Q2.09.3 MCQ 2009 2M

With S2 closed and S1 open, the current waveform in the delta winding will be

A. a sinusoid at fundamental frequency

B. flat-topped with third harmonic

C. only third-harmonic

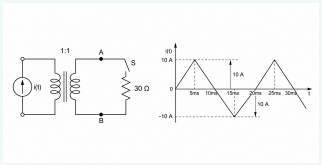
D. none of these

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Questions (Q2.09.4 and Q2.09.5):

The circuit diagram shows a two-winding, lossless transformer with no leakage flux, excited from a current source, $i(t)$, whose waveform is also shown. The transformer has a magnetizing inductance of $400/\pi$ mH.



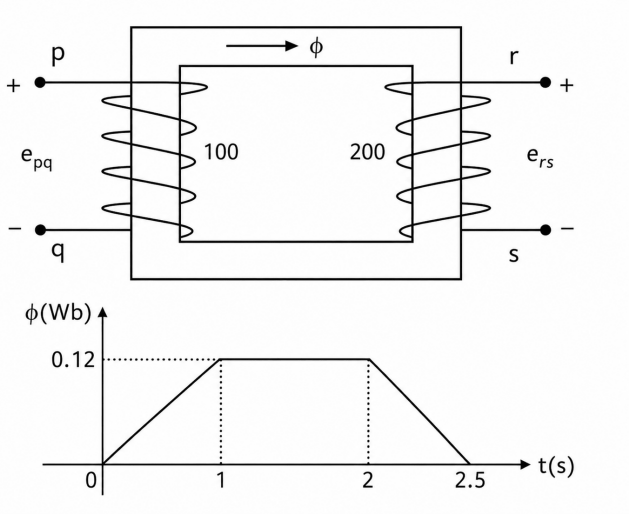


Fig. Q2.08.2

Q2.09.4 MCQ 2009 2M

The peak voltage across A and B, with S open is

A. $400/\pi$ V **B.** 800 V **C.** $4000/\pi$ V **D.** $800/\pi$ V

Q2.09.5 MCQ 2009 2M

If the waveform of $i(t)$ is changed to $i(t) = 10\sin(100\pi t)$ A, the peak voltage across A and B with S closed is

A. 400 V **B.** 240 V **C.** 320 V **D.** 160 V

Q2.08.1 MCQ 2008 1M

It is desired to measure parameters of 230 V/115 V, 2 kVA, single-phase transformer. The following watt-meters are available in a laboratory:

W_1 : 250 V, 10 A, Low Power Factor
W_2 : 250 V, 5 A, Low Power Factor
W_3 : 150 V, 10 A, High Power Factor
W_4 : 150 V, 5 A, High Power Factor

The watt-meters used in open circuit test and short circuit test of the transformer will respectively be

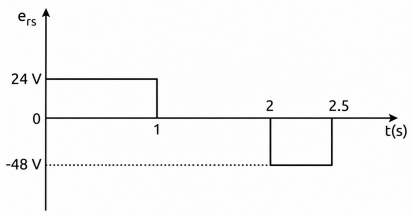
A. W_1 and W_2 **B.** W_2 and W_4
C. W_1 and W_4 **D.** W_2 and W_3

Q2.08.2 MCQ 2008 2M

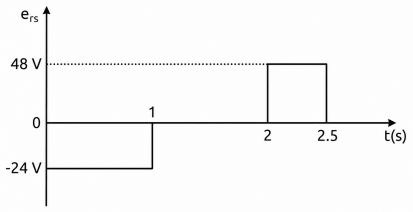
The core of a two-winding transformer is subjected to a magnetic flux variation as indicated in the figure.

The induced emf (e_{rs}) in the secondary winding as a function of time will be of the form

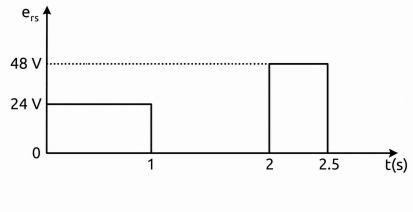
A.



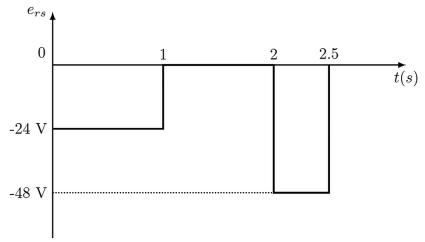
B.



C.



D.



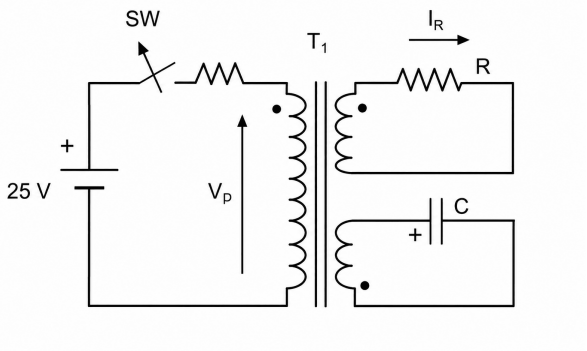


Fig. Q2.07.1

A. −25 V, 0.0 A **B.** very large voltage, very large current
C. 5.0 V, 0.5 A **D.** −5.0 V, −0.5 A

Q2.07.2 MCQ 2007 2M

A single-phase 50 kVA, 250V/500V two winding transformer has an efficiency of 95% at full load, unity power factor. If it is reconfigured as a 500V/750V autotransformer, its efficiency at its new rated load at unity power factor will be

A. 95.752% **B.** 97.851% **C.** 98.276% **D.** 99.241%

Q2.07.3 MCQ 2007 1M

In a transformer, zero voltage regulation at full load is

A. not possible

B. possible at unity power factor load

C. possible at leading power factor load

D. possible at lagging power factor load

Q2.06.1 MCQ 2006 1M

In transformers, which of the following statements is valid?

A. In an open circuit test, copper losses are obtained while in short circuit test, core losses are obtained

B. In an open circuit test, current is drawn at high power factor

C. In a short circuit test, current is drawn at zero power factor

D. In an open circuit test, current is drawn at low power factor

Q2.08.3 MCQ 2008 1M

Three single-phase transformers are connected to form a 3-phase transformer bank. The transformers are connected in the following manner:

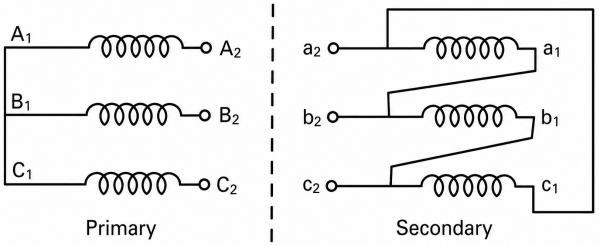


Fig. Q2.08.3

The transformer connection will be represented by

A. Yd0 **B.** Yd1 **C.** Yd6 **D.** Yd11

Q2.07.1 MCQ 2007 2M

In the figure, transformer T_1 has two secondaries, all three windings having the same number of turns and with polarities as indicated. One secondary is shorted by a 10 Ω resistor R, and the other by a 15 μF capacitor. The switch SW is opened ($t = 0$) when the capacitor is charged to 5 V with the left plate as positive. At $t = 0^+$ the voltage V_p and current I_R are

Q2.06.2 MCQ 2006 2M

Two transformers are to be operated in parallel such that they share load in proportion to their kVA ratings. The rating of the first transformer is 500 kVA and its pu leakage impedance is 0.05 pu. If the rating of second transformer is 250 kVA, then its pu leakage impedance is

A. 0.20 **B.** 0.10 **C.** 0.05 **D.** 0.025

Common Data Questions

Statement for Linked Questions (Q2.06.3 and Q2.06.4):
A 300 kVA transformer has 95% efficiency at full load 0.8 pf lagging and 96% efficiency at half load, unity pf.
The iron loss (P_i) and copper loss (P_c) in kW, under full load operation are

Q2.06.3 MCQ 2006 2M

The iron loss (P_i) and copper loss (P_c) in kW, under full load operation are

A. $P_c = 4.12,\ P_i = 8.51$ **B.** $P_c = 6.59,\ P_i = 9.21$
C. $P_c = 8.51,\ P_i = 4.12$ **D.** $P_c = 12.72,\ P_i = 3.07$

Q2.06.4 MCQ 2006 2M

What is the maximum efficiency (in %) at unity pf load ?

A. 95.1 **B.** 96.2 **C.** 96.4 **D.** 98.1

Q2.05.1 MCQ 2005 1M

Which three-phase connection can be used in a transformer to introduce a phase difference of 30° between its output and corresponding input line voltages

A. Star – Star **B.** Star – Delta
C. Delta – Delta **D.** Delta – Zigzag

Q2.05.2 MCQ 2005 1M

The equivalent circuit of a transformer has leakage reactance X_1, X_2' and magnetizing reactance X_M. Their magnitudes satisfy

A. $X_1 > X_2' > X_M$ **B.** $X_1 < X_2' < X_M$
C. $X_1 = X_2' > X_M$ **D.** $X_1 = X_2' < X_M$

Q2.04.1 MCQ 2004 1M

A 500 kVA, 3-phase transformer has iron losses of 300 W and full load copper losses of 600 W. The percentage load at which the transformer is expected to have maximum efficiency is

A. 50.0% **B.** 70.7% **C.** 141.4% **D.** 200.0%

Q2.04.2 MCQ 2004 2M

A 50 kVA, 3300/230 V single-phase transformers is connected as an autotransformer shown in figure. The nominal rating of the autotransformer will be

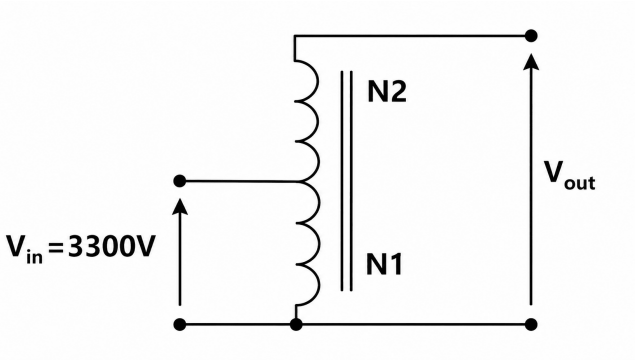


Fig. Q2.04.2

A. 50.0 kVA **B.** 53.5 kVA **C.** 717.4 kVA **D.** 767.4 kVA

Q2.04.3 MCQ 2004 2M

The resistance and reactance of a 100 KVA 11000|400 V, Δ-Y distribution transformer are 0.02 and 0.07 pu respectively. The phase impedance of the transformer referred to the primary is

A. $(0.02 + j0.07)\Omega$ **B.** $(0.55 + j1.925)\Omega$
C. $(15.125 + j52.94)\Omega$ **D.** $(72.6 + j254.1)\Omega$

Q2.03.1 MCQ 2003 1M

A single phase transformer has a maximum efficiency of 90% at full load and unity power factor. Efficiency at half load at the same power factor is

A. 86.7% **B.** 88.26% **C.** 88.9% **D.** 87.8%

Q2.03.2 MCQ 2003 2M

Fig.Q2.03.2 shows an ideal single-phase transformer. The primary and secondary coils are wound on the core as shown. Turns ratio $\left(\frac{N_1}{N_2}\right) = 2$. The correct phasors of voltages E_1, E_2, currents I_1, I_2 and core flux Φ are as shown in

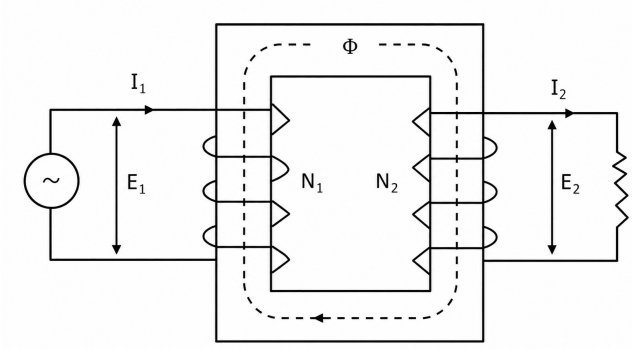
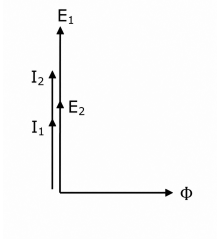


Fig. Q2.03.2

A.



B.

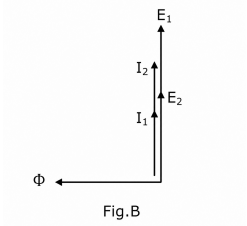


Fig.B

C.

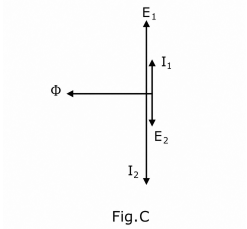


Fig.C

D.

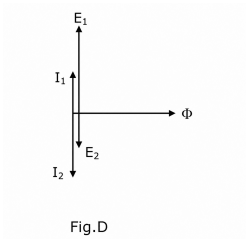


Fig.D

Q2.03.3 MCQ 2003 2M

Fig.Q2.03.3 shows a Δ-Y connected 3-phase distribution transformer used to step down the voltage from 11000 V to 415 V line-to-line. It has two switches S_1 and S_2. Under normal conditions, S_1 is closed and S_2 is open. Under certain special conditions, S_1 is open and S_2 is closed. In such a case, the magnitude of the voltage across the LV terminals a and c is

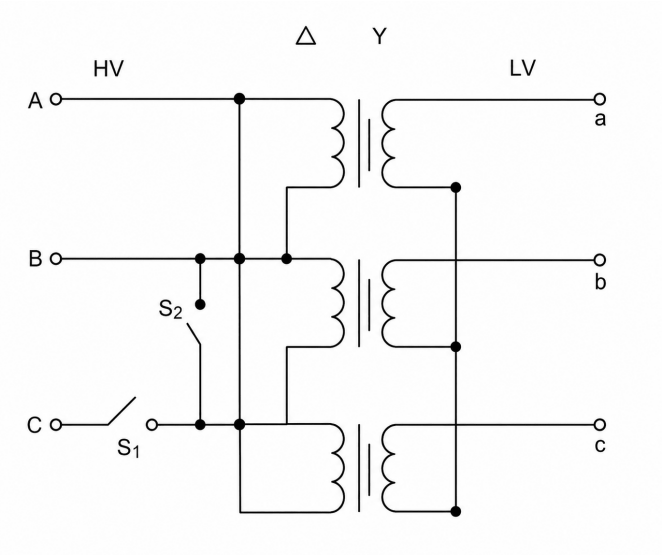


Fig. Q2.03.3

A. 240 V **B.** 480 V **C.** 415 V **D.** 0 V

Q2.03.4 MCQ 2003 2M

Fig Q2.03.4 shows an ideal three-winding transformer. The three windings 1, 2, 3 of the transformer are wound on the same core as shown. The turn's ratio $N_1 : N_2 : N_3$ is $4:2:1$. A resistor of 10Ω is connected across winding-2. A capacitor of reactance 2.5Ω is connected across winding-3. Winding-1 is connected across a 400 V, ac supply. If the supply voltage phasor $V_1 = 400\angle 0°$, the supply current phasor I_1 is given by

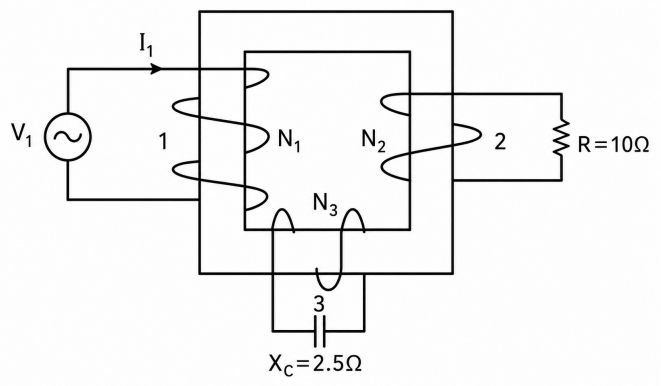


Fig. Q2.03.4

A. $(-10 + j10)$ A **B.** $(-10 - j10)$ A
C. $(10 + j10)$ A **D.** $(10 - j10)$ A

Q2.02.1 MCQ 2002 1M

A 1 kVA, 230V/100V, single phase, 50 Hz transformer having negligible winding resistance and leakage inductance is operating under saturation, while 250 V, 50 Hz sinusoidal supply is connected to the high voltage winding. A resistive load is connected to the low voltage winding which draws rated current. Which one of the following quantities will not be sinusoidal?

A. Voltage induced across the low voltage winding

B. Core flux

C. Load current

D. Current drawn from the source

Q2.02.2 MCQ 2002 1M

A 400V/200V/200V, 50 Hz three winding transformer is connected as shown in Fig.1.9. The reading of the voltmeter, V, will be

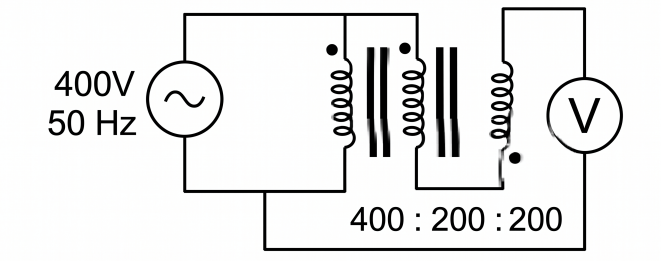


Fig. Q2.02.2

A. 0 V **B.** 400 V **C.** 600 V **D.** 800 V

Q2.02.3 NAT 2002 2M

A single phase 6300 kVA, 50 Hz, 3300V/400V distribution transformer is connected between two 50 Hz supply systems, A and B as shown in Figure below, the transformer has 12 and 99 turns in the low and high voltage windings respectively. The magnetizing reactance of the transformer referred to the high voltage side is 500 Ω. The leakage reactance of the high and low voltage windings are 1.0 Ω and 0.012Ω respectively. Neglect the winding resistance and core losses of the transformer. The Thevenin voltage of system A is 3300V while that of system B is 400V. the short circuit reactance of systems A and B are 0.5Ω and 0.010 Ω respectively. If no power is transferred between A and B, so that the two system voltages are in phase, find the magnetizing ampere turns of the transformer.

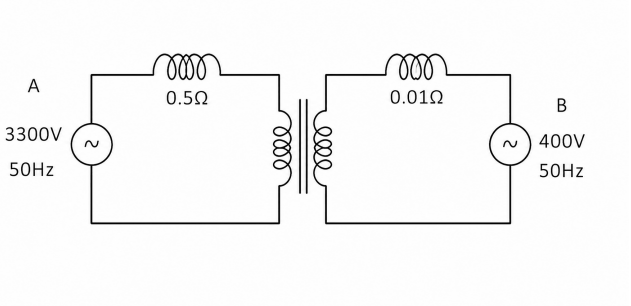


Fig. Q2.02.3

Q2.01.1 MCQ 2001 1M

A single-phase transformer is to be switched to the supply to have minimum inrush current. The switch should be closed at

A. maximum supply voltage

B. zero supply voltage

C. $\frac{1}{\sqrt{2}}$ maximum supply voltage

D. $\frac{1}{2}$ maximum supply voltage

Q2.01.2 MCQ 2001 1M

The core flux of a practical transformer with a resistive load

A. is strictly constant with load changes

B. increases linearly with load

C. increases as the square root of the load

D. decreases with increased load

Q2.01.3 MCQ 2001 2M

A 3-phase transformer has rating of 20 MVA, 220 kV (star) – 33 kV (delta) with leakage reactance of 12%. The transformer reactance (in ohms) referred to each phase of the L.V. delta-connected side is

A. 23.5 **B.** 19.6 **C.** 18.5 **D.** 8.7

Q2.01.4 NAT 2001 5M

An ideal transformer has a linear B/H characteristic with a finite slope and a turns ratio of 1 : 1. The primary of the transformer is energized with an ideal current source, producing the signal i as shown in Fig.Q2.01.4. Sketch the shape (neglecting the scale factor) of the following signals, labeling the time axis clearly:

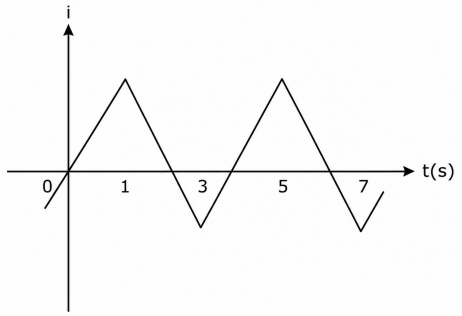


Fig. Q2.01.4

A. the core flux ϕ_{oc} with the secondary of the transformer open

B. the open-circuited secondary terminal voltage $v_2(t)$

C. the short-circuited secondary current $i_2(t)$, and

D. the core flux ϕ_{sc} with the secondary of the transformer short-circuited.

Q2.01.5 MCQ 2001 2M

The hysteresis loop of a magnetic material has an area of 5 cm^2 with the scales given as 1 cm = 2 AT and 1 cm = 50 mWb. At 50 Hz, the total hysteresis loss is

A. 15 W **B.** 20 W **C.** 25 W **D.** 50 W

Q2.00.1 NAT 2000 5M

In a single-phase, three-winding transformer, the turns ratio for primary : secondary : tertiary windings is 20 : 4 : 1. With the lagging currents of 50 A at a power factor of 0.6 in the tertiary winding, find the primary current and power-factor.

Q2.00.2 MCQ 2000 2M

A 3-phase delta/star transformer is supplied at 6000 V on the delta-connected side. The terminal voltage on the secondary side when supplying full load at 0.8 lagging power-factor is 415 V. The equivalent resistance and reactance drops for the transformer are 1% and 5% respectively. The turns ratio of the transformer is:

A. 14 **B.** 24 **C.** 42 **D.** 20

Q2.00.3 MCQ 2000 1M

In a constant voltage transformer (CVT), the output voltage remains constant due to

A. Capacitor **B.** Input Voltage
C. Saturation **D.** Tapped windings

Q2.99.1 MCQ 1999 1M

The windings of a Q kVA, $\frac{V_1}{V_2}$ volt, three-phase, Delta connected, core type transformer are reconnected to work as a single phase transformer. The maximum voltage and the power ratings of the new configuration are,

A. $\frac{3V_1}{3V_2}$, $3Q$ **B.** $\frac{V_1}{V_2}$, $\frac{Q}{3}$
C. $\frac{\sqrt{3}V_1}{\sqrt{3}V_2}$, $2Q$ **D.** $\frac{2V_1}{2V_2}$, $\frac{2Q}{3}$

Q2.99.2 MCQ 1999 2M

A 10 kVA, 400 V/200 V single-phase transformers with 10% impedance draws a steady short circuit line current of

A. 50 A **B.** 150 A **C.** 250 A **D.** 350 A

Q2.99.3 MCQ 1999 2M

The percentage resistance and percentage reactance of a 10 kVA, 400 V/200 V, 3-phase transformer are 2% and 10% respectively. If the constant losses in the machine are 1%, the maximum possible percentage efficiency of the transformer is:

A. 98.32 **B.** 97.25 **C.** 96.85 **D.** 96.12

Q2.99.4 MCQ 1999 2M

A 400 V/100 V, 10 kVA two-winding transformer is re-connected as an auto-transformer across a suitable voltage source. The maximum rating of such an arrangement could be

A. 50 kVA **B.** 15 kVA **C.** 12.5 kVA **D.** 8.75 kVA

Q2.99.5 MCQ 1999 2M

A 10 kVA, 400 V/200 V, single phase transformer with a percentage resistance of 3% and percentage reactance of 6% is supplying a current of 50 A to a resistive load. The value of the load voltage is:

A. 194 V **B.** 390 V **C.** 192 V **D.** 196 V

Q2.99.6 NAT 1999 5M

Two single-phase transformers A and B have the following parameters:
Transformer A: 400 V/200 V, percentage resistance and percentage reactance 3% and 4% respectively.
Transformer B: 5 kVA, 400 V/200 V, percentage resistance and percentage reactance are 4% and 3% respectively.
These two transformers are connected in parallel and they share a common load of 12 kW at a power factor of 0.8 lagging. Determine the active and reactive power delivered by transformer A.

Q2.98.1 MCQ 1998 1M

The efficiency of a 100 kVA transformer is 0.98 at full load as well as at half load. For this transformer at full load the copper loss.

A. is less than core loss **B.** is equal to core loss
C. is more than core loss **D.** None of the above

Q2.98.2 MCQ 1998 1M

The magnetizing current in a transformer is rich in

A. 3^{rd} harmonic **B.** 5^{th} harmonic
C. 7^{th} harmonic **D.** 13^{th} harmonic

Q2.98.3 MCQ 1998 2M

A 50 Hz transformer having equal hysteresis and eddy current losses at rated excitation is operated at 45 Hz at 90% of its rated voltage. Compared to rated operating point, the core losses under this condition:

A. reduce by 10% **B.** reduce by 19%
C. reduce by 14.5% **D.** remain unchanged

Q2.98.4 NAT 1998 5M

In a 50 kVA, 11 kV/400 V transformers, the iron and copper losses are 500 W and 600 W respectively under rated conditions. Calculate the efficiency on unity power factor at full load. Find the load for maximum efficiency and the iron and copper losses corresponding to this load.

Q2.97.1 MCQ 1997 1M

The low voltage winding of a 400/230 V, 1–phase, 50 Hz transformer is to be connected to a 25 Hz supply at 25 Hz, the supply voltage should be

A. 230 V **B.** 460 V **C.** 115 V **D.** 65 V

Q2.97.2 MCQ 1997 1M

A voltage $v = 400\sin(314.16\,t)$ is applied to a 1-phase transformer on no-load. If the no load current of the transformer is $2\sin(314.16\,t - 85°)$, then the magnetization branch impedance will be approximately equal to

A. $141\angle 90°$ **B.** $200\angle -85°$
C. $200\angle 85°$ **D.** $282\angle -80°$

Q2.97.3 NAT 1997 2M

A 3-phase transformer bank consists of three identical 2300/230 V, 15 kVA single-phase transformers connect in delta/delta. The bank supplies a 20 kVA, unity p.f. 3-phase load. If one of the single-phase transformers develops a fault and is removed, the load carried by each of the two transformers now operating in open delta will be _____ kVA.

Q2.96.1 MCQ 1996 1M

An Auto-transformer is used in transmission and distribution

A. when operator is not available

B. when iron losses are to be reduced

C. when efficiency considerations can be ignored

D. when the transformation ratio is small

Q2.96.2 MCQ 1996 1M

Keeping in view the requirement of parallel operation, which of the 3-phase connections given below are possible?

A. delta-delta to delta-star

B. delta-delta to star-delta

C. star-star to delta-delta

D. delta-star to star-delta

Q2.96.3 MCQ 1996 1M

The function of oil in a transformer is

A. to provide insulation and cooling

B. to provide protection against lightning

C. to provide protection against short circuit

D. to provide lubrication

Q2.95.1 MCQ 1995 1M

Supply to one terminal of a delta-wye connected three-phase core type transformer which is on no-load, fails. Assuming magnetic circuit symmetry, voltages on the secondary side will be:

A. 230, 230, 115 **B.** 230, 115, 115
C. 345, 115, 115 **D.** 345, 0, 345

Q2.95.2 NAT 1995 1M

When started by means of an auto transformer with 50% tapping, supply current at start of an induction motor is reduced to _____ of that when started by means of a star-delta starter.

Q2.95.3 NAT 1995 1M

The percentage impedance of a 100 kVA, 11 kV/400 V, delta/wye, 50 Hz transformer is 4.5%. For the circulation of half the full load current during short circuit test, with low voltage terminals shorted, the applied voltage on the high voltage side will be _____

Q2.95.4 NAT 1995 5M

A 4 kVA, 50 Hz, single-phase transformer has a ratio 200/400 V. The data taken on the l.v. side at the rated voltage show that the open circuit input wattage is 80 W. The mutual inductance between the primary and the secondary windings is 1.91 H. What value will be the current taken by the transformer, if the no-load test is conducted on the h.v. side at rated voltage? Neglect the effect of winding resistance and leakage reactances.

Q2.94.1 MCQ 1994 1M

Two transformers of identical voltages but of different capacities are operating in parallel. For satisfactory load sharing

A. Impedances must be equal

B. Per-unit impedances must be equal

C. Per-unit impedances and $\frac{X}{R}$ ratios must be equal

D. Impedances and $\frac{X}{R}$ ratios must be equal

Q2.93.1 MCQ 1993 2M

A 220/440 V, 50 Hz, 5 kVA single phase transformer operates on 220 V, 40 Hz supply with secondary winding. Then

A. the eddy current loss and hysteresis loss of the transformer decrease

B. the eddy current loss and hysteresis loss of the transformer increase

C. the hysteresis loss of the transformer increases while eddy current loss remains the same

D. the hysteresis loss remains the same whereas eddy current loss decreases

Q2.92.1 MCQ 1992 1M

Two transformers of the same type, using the same grade of iron and conductor materials, are designed to work at the same flux and current densities; but the linear dimensions of one are two times those of the other in all respects. The ratio of kVA of the two transformers closely equals

A. 16 **B.** 8 **C.** 4 **D.** 2

Q2.92.2 MCQ 1992 1M

Two transformers of different kVA ratings working in parallel share the load in proportion to their ratings when their.

A. per unit leakage impedances on the same kVA base are the same

B. per unit leakage impedances on their respective ratings are equal

C. ohmic values of the leakage impedances are inversely proportional to their ratings

D. ohmic values of the magnetising reactances are the same

Q2.92.3 MCQ 1992 1M

A Buchholz relay is used for

A. protection of a transformer against all internal faults

B. protection of a transformer against external faults

C. protection of a transformer against both internal and external faults

D. protection of induction motors

Q2.91.1 MCQ 1991 1M

The hysteresis and eddy current losses of a single phase transformer working on 200 V, 50 Hz supply are P_h and P_e respectively. The percentage decrease in these, when operated on a 160 V, 40 Hz supply are :

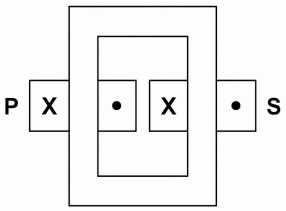
A. 32, 36 **B.** 20, 36 **C.** 25, 20 **D.** 40, 80

Q2.91.2 MSQ 1991 1M

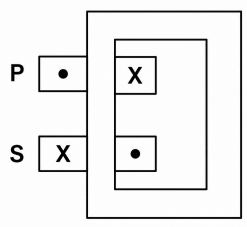
The relative current directions through the Primary (P) and Secondary (S) of a single phase transformer connected to a resistive load on the secondary side are indicated in the various cross-sectional views given in the figure.

Which of these are correct representations

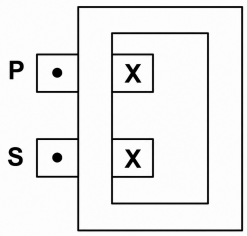
A.



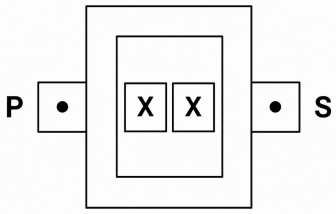
B.



C.



D.



Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q2.26.1 A	Q2.26.2 D	Q2.26.3 C
Q2.25.1 A	Q2.24.1 MTA	Q2.23.1 2150 to 2250
Q2.21.1 51.12	Q2.21.2 B	Q2.20.1 MTA
Q2.20.2 162.5	Q2.19.1 0.83 to 0.84	Q2.19.2 7.10 to 7.30
Q2.19.3 C	Q2.18.1 97.36	Q2.18.2 A
Q2.17.1 625.2	Q2.17.2 37.878	Q2.16.1 B
Q2.16.2 D	Q2.16.3 10	Q2.16.4 A
Q2.16.5 6	Q2.16.6 1.4423	Q2.16.7 C
Q2.15.1 1.33	Q2.15.2 555 to 556	Q2.15.3 B
Q2.15.4 A	Q2.15.5 30	Q2.15.6 A
Q2.15.7 C	Q2.15.8 A	Q2.14.1 A
Q2.14.2 550	Q2.14.3 B	Q2.14.4 B

Q2.14.5 23.5 to 23.7	Q2.14.6 A	Q2.14.7 A
Q2.14.8 C	Q2.13.1 C	Q2.13.2 B
Q2.12.1 B	Q2.11.1 C	Q2.10.1 D
Q2.10.2 C	Q2.09.1 B	Q2.09.2 B
Q2.09.3 C	Q2.09.4 D	Q2.09.5 B
Q2.08.1 D	Q2.08.2 A	Q2.08.3 B
Q2.07.1 D	Q2.07.2 C	Q2.07.3 C
Q2.06.1 D	Q2.06.2 C	Q2.06.3 C
Q2.06.4 B	Q2.05.1 B	Q2.05.2 D
Q2.04.1 B	Q2.04.2 D	Q2.04.3 D
Q2.03.1 D	Q2.03.2 A	Q2.03.3 B
Q2.03.4 C	Q2.02.1 D	Q2.02.2 A
Q2.02.3 652.83	Q2.01.1 A	Q2.01.2 A
Q2.01.3 B	Q2.01.4 *	Q2.01.5 C
Q2.00.1 2.5,0.6	Q2.00.2 B	Q2.00.3 A
Q2.99.1 D	Q2.99.2 C	Q2.99.3 B
Q2.99.4 A	Q2.99.5 A	Q2.99.6 7.466,6.787
Q2.98.1 C	Q2.98.2 A	Q2.98.3 C
Q2.98.4 97.84,500,45.65	Q2.97.1 C	Q2.97.2 C
Q2.97.3 10	Q2.96.1 D	Q2.96.2 D
Q2.96.3 A	Q2.95.1 B	Q2.95.2 0.75
Q2.95.3 247.5	Q2.95.4 0.389	Q2.94.1 C
Q2.93.1 C	Q2.92.1 A	Q2.92.2 B
Q2.92.3 A	Q2.91.1 B	Q2.91.2 A,B

UNIT

III

DC Machines

Topics

A. Types of DC Machines - Separately Excited, Series and Shunt
B. Construction of DC Machines
C. Armature Winding, Equivalent Resistance and Voltage across Brushes
D. Armature Reaction: Concept and Method to Improve
E. Commutation
F. Losses and Testing in DC Machines
G. Starter of DC Motor
H. DC Generator Characteristics - Series and Compound
I. DC Motor Characteristics/Mode of Operation
J. Speed Control and Braking of DC Motors

Unit Snapshot*

Stream: EE	**Questions:** 78	**MCQ:** 45	**MSQ:** 0	**NAT:** 32
1M: 24	**2M:** 42	**Descriptive:** 12	**Avg Weightage** ~3M	**Range:** 2M(2025)–5M(2015)

**See preface for how Avg Weightage and Range are calculated.*

Q3.25.1 MCQ 2025 2M

A DC series motor with negligible series resistance is running at a certain speed driving a load, where the load torque varies as cube of the speed. The motor is fed from a 400 V DC source and draws 40 A armature current. Assume linear magnetic circuit. The external resistance, in Ω, that must be connected in series with the armature to reduce the speed of the motor by half, is closest to

A. 23.28 **B.** 4.82 **C.** 46.7 **D.** 0

Q3.24.1 NAT 2024 2M

A 5 kW, 220 V DC shunt motor has 0.5 Ω armature resistance including brushes. The motor draws a no-load current of 3 A. The field current is constant at 1 A. Assuming that the core and rotational losses are constant and independent of the load, the current (in amperes) drawn by the motor while delivering the rated load, for the best possible efficiency, is _____ (rounded off to 2 decimal places).

Q3.23.1 MCQ 2023 1M

The four stator conductors (A, A′, B and B′) of a rotating machine are carrying DC currents of the same value, the directions of which are shown in the figure (i). The rotor coils a-a' and b-b' are formed by connecting the back ends of conductors 'a' and 'a'' and 'b' and 'b'', respectively, as shown in figure (ii). The e.m.f. induced in coil a-a' and coil b-b' are denoted by $E_{a\text{-}a'}$ and $E_{b\text{-}b'}$, respectively. If the rotor is rotated at uniform angular speed ω rad/s in the clockwise direction then which of the following correctly describes the $E_{a\text{-}a'}$ and $E_{b\text{-}b'}$?

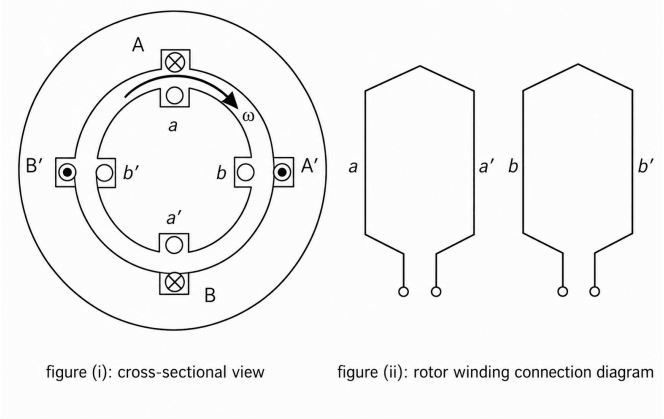


figure (i): cross-sectional view figure (ii): rotor winding connection diagram

Fig. Q3.23.1

A. $E_{a\text{-}a'}$ and $E_{b\text{-}b'}$ have finite magnitudes and are in the same phase

B. $E_{a\text{-}a'}$ and $E_{b\text{-}b'}$ have finite magnitudes with $E_{b\text{-}b'}$ leading $E_{a\text{-}a'}$

C. $E_{a\text{-}a'}$ and $E_{b\text{-}b'}$ have finite magnitudes with $E_{a\text{-}a'}$ leading $E_{b\text{-}b'}$

D. $E_{a\text{-}a'} = E_{b\text{-}b'} = 0$

Q3.23.2 NAT 2023 1M

A separately excited DC motor rated 400 V, 15 A, 1500 RPM drives a constant torque load at rated speed operating from 400 V DC supply drawing rated current. The armature resistance is 1.2 Ω. If the supply voltage drops by 10% with field current unaltered then the resultant speed of the motor in RPM is _____ (Round off to the nearest integer).

Q3.22.1 NAT 2022 2M

A 280 V, separately excited DC motor with armature resistance of 1 Ω and constant field excitation drives a load. The load torque is proportional to the speed. The motor draws a current of 30 A when running at a speed of 1000 rpm. Neglect frictional losses in the motor. The speed, in rpm, at which the motor will run, if an additional resistance of value 10 Ω is connected in series with the armature, is _____. (round off to nearest integer)

Q3.21.1 NAT 2021 2M

A belt-driven DC shunt generator running at 300 RPM delivers 100 kW to a 200 V DC grid. It continues to run as a motor when the belt breaks, taking 10 kW from the DC grid. The armature resistance is 0.025 Ω, field resistance is 50 Ω, and brush drop is 2 V. Ignoring armature reaction, the speed of the motor is _____ RPM. (Round off to 2 decimal places.)

Q3.20.1 MCQ 2020 2M

A 250 V dc shunt motor has an armature resistance of 0.2 Ω and a field resistance of 100 Ω. When the motor is operated on no-load at rated voltage, it draws an armature current of 5 A and runs at 1200 rpm. When a load is coupled to the motor, it draws total line current of 50 A at rated voltage, with a 5% reduction in the air-gap flux due to armature reaction. Voltage drop across the brushes can be taken as 1 V per brush under all operating conditions. The speed of the motor, in rpm, under this loaded condition, is closest to:

A. 1200 **B.** 1000 **C.** 1220 **D.** 900

Q3.19.1 NAT 2019 2M

A 220 V DC shunt motor takes 3 A at no-load. It draws 25 A when running at full-load at 1500 rpm. The armature and shunt resistances are 0.5 Ω and 220 Ω, respectively. The no-load speed in rpm (round off to two decimal places) is _____.

Q3.18.1 NAT 2018 2M

A 200 V DC series motor, when operating from rated voltage while driving a certain load, draws 10 A current and runs at 1000 r.p.m. The total series resistance is 1 Ω. The magnetic circuit is assumed to be linear. At the same supply voltage, the load torque is increased by 44%. The speed of the motor in r.p.m. (rounded to the nearest integer) is _____.

Q3.18.2 NAT 2018 1M

A separately excited dc motor has an armature resistance $R_a = 0.05\,\Omega$. The field excitation is kept constant. At an armature voltage of 100 V, the motor produces a torque of 500 Nm at zero speed. Neglecting all mechanical losses, the no-load speed of the motor (in radian/s) for an armature voltage of 150 V is _____ (up to 2 decimal places).

Q3.17.1 NAT 2017 2M

A 220 V DC series motor runs drawing a current of 30 A from the supply. Armature and field circuit resistances are 0.4 Ω and 0.1 Ω, respectively. The load torque varies as the square of the speed. The flux in the motor may be taken as being proportional to the armature current. To reduce the speed of the motor by 50%, the resistance in ohms that should be added in series with the armature is _____. (Give the answer up to two decimal places.)

Q3.17.2 NAT 2017 2M

A separately excited DC generator supplies 150 A to a 145 V DC grid. The generator is running at 800 RPM. The armature resistance of the generator is 0.1 Ω. If the speed of the generator is increased to 1000 RPM, the current in amperes supplied by the generator to the DC grid is _____. (Give the answer up to one decimal place.)

Q3.17.3 MCQ 2017 2M

A 220 V, 10 kW, 900 rpm separately excited DC motor has an armature resistance $R_a = 0.02\,\Omega$. When the motor operates at rated speed and with rated terminal voltage, the electromagnetic torque developed by the motor is 70 Nm. Neglecting the rotational losses of the machine, the current drawn by the motor from the 220 V supply is

A. 34.2 A **B.** 30 A **C.** 22 A **D.** 4.84 A

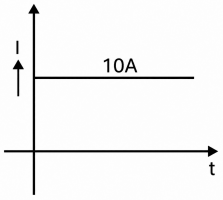
Q3.17.4 NAT 2017 2M

A 120 V DC shunt motor takes 2 A at no load. It takes 7 A on full load while running at 1200 rpm. The armature resistance is 0.8 Ω, and the shunt field resistance is 240 Ω. The no load speed, in rpm, is _____.

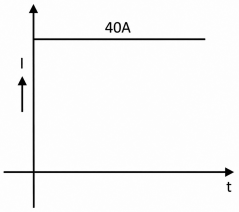
Q3.16.1 MCQ 2016 1M

A 4-pole, lap-connected, separately excited dc motor is drawing a steady current of 40 A while running at 600 rpm. A good approximation for the waveshape of the current in an armature conductor of the motor is given by

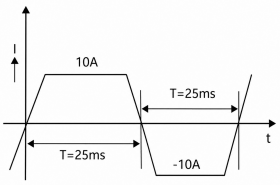
A.



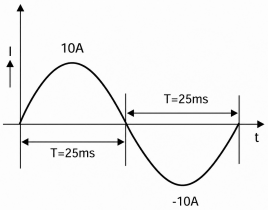
B.



C.



D.



Q3.16.2 NAT 2016 2M

A DC shunt generator delivers 45 A at a terminal voltage of 220 V. The armature and the shunt field resistances are 0.01 Ω and 44 Ω respectively. The stray losses are 375 W. The percentage efficiency of the DC generator is _____.

Q3.15.1 MCQ 2015 1M

A separately excited DC generator has an armature resistance of 0.1 Ω and negligible armature inductance. At rated field current and rated rotor speed, its open-circuit voltage is 200 V. When this generator is operated at half the rated speed, with half the rated field current, an uncharged 1000 μF capacitor is suddenly connected across the armature terminals. Assume that the speed remains unchanged during the transient. At what time (in microsecond) after the capacitor is connected will the voltage across it reach 25 V?

A. 62.25 **B.** 69.3 **C.** 73.25 **D.** 77.3

Q3.15.2 NAT 2015 2M

A separately excited DC motor runs at 1000 rpm on no load when its armature terminals are connected to a 200 V DC source and the rated voltage is applied to the field winding. The armature resistance of this motor is 1 Ω. The no-load armature current is negligible. With the motor developing its full load torque, the armature voltage is set so that the rotor speed is 500 rpm. When the load torque is reduced to 50% of the full load value under the same armature voltage conditions, the speed rises to 520 rpm. Neglecting the rotational losses, the full load armature current (in Ampere) is _____.

Q3.15.3 NAT 2015 2M

A DC motor has the following specifications: 10 hp, 37.5 A, 230 V; flux/pole = 0.01 Wb, number of poles = 4, number of conductors = 666, number of parallel paths = 2. Armature resistance = 0.267 Ω. The armature reaction is negligible and rotational losses are 600 W. The motor operates from a 230 V DC supply. If the motor runs at 1000 rpm, the output torque produced (in Nm) is _____.

Q3.15.4 NAT 2015 1M

A shunt-connected DC motor operates at its rated terminal voltage. Its no-load speed is 200 radian/second. At its rated torque of 500 Nm, its speed is 180 radian/second. The motor is used to directly drive a load whose load torque T_L depends on its rotational speed ω_r (in radian/second), such that $T_L = 2.78 \times \omega_r$. Neglecting rotational losses, the steady-state speed (in radian/second) of the motor, when it drives this load, is _____.

Q3.15.5 MCQ 2015 1M

A 4-pole, separately excited, wave wound DC machine with negligible armature resistance is rated for 230 V and 5 kW at a speed of 1200 rpm. If the same armature coils are reconnected to form a lap winding, what is the rated voltage (in volts) and power (in kW) respectively at 1200 rpm of the reconnected machine if the field circuit is left unchanged?

A. 230 and 5 **B.** 115 and 5
C. 115 and 2.5 **D.** 230 and 2.5

Q3.15.6 NAT 2015 2M

With an armature voltage of 100 V and rated field winding voltage, the speed of a separately excited DC motor driving a fan is 1000 rpm, and its armature current is 10 A. The armature resistance is 1 Ω. The load torque of the fan load is proportional to the square of the rotor speed. Neglecting rotational losses, the value of the armature voltage (in Volt) which will reduce the rotor speed to 500 rpm is _____.

Q3.14.1 NAT 2014 2M

A 15 kW, 230 V dc shunt motor has armature circuit resistance of 0.4 Ω and field circuit resistance of 230 Ω. At no load and rated voltage, the motor runs at 1400 rpm and the line current drawn by the motor is 5 A. At full load, the motor draws a line current of 70 A. Neglect armature reaction. The full load speed of the motor in rpm is _____.

Q3.14.2 NAT 2014 2M

A 250 V dc shunt machine has armature circuit resistance of 0.6 Ω and field circuit resistance of 125 Ω. The machine is connected to 250 V supply mains. The motor is operated as a generator and then as a motor separately. The line current of the machine in both the cases is 50 A. The ratio of the speed as a generator to the speed as a motor is _____.

Q3.14.3 NAT 2014 1M

The no-load speed of a 230 V separately excited dc motor is 1400 rpm. The armature resistance drop and the brush drop are neglected. The field current is kept constant at rated value. The torque of the motor in Nm for an armature current of 8 A is _____.

Q3.14.4 NAT 2014 2M

A separately excited 300 V DC shunt motor under no load runs at 900 rpm drawing an armature current of 2 A. The armature resistance is 0.5 Ω and leakage inductance is 0.01 H. When loaded, the armature current is 15 A. Then the speed in rpm is _____.

Q3.12.1 MCQ 2012 2M

A 220 V, 15 kW, 1000 rpm shunt motor with armature resistance of 0.25 Ω, has a rated line current of 68 A and a rated field current of 2.2 A. The change in field flux required to obtain a speed of 1600 rpm while drawing a line current of 52.8 A and a field current of 1.8 A is

A. 18.18 % increase **B.** 18.18 % decrease
C. 36.36 % increase **D.** 36.36 % decrease

Q3.11.1 MCQ 2011 1M

A 4-point starter is used to start and control the speed of a

A. dc shunt motor with armature resistance control

B. dc shunt motor with field weakening control

C. dc series motor

D. dc compound motor

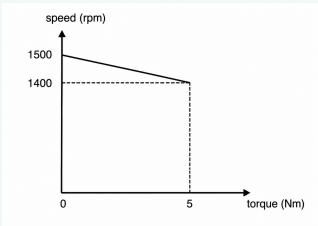
Q3.11.2 MCQ 2011 2M

A 220 V, DC shunt motor is operating at a speed of 1440 rpm. The armature resistance is 1.0 Ω and armature current is 10 A. If the excitation of the machine is reduced by 10 %, the extra resistance to be put in the armature circuit to maintain the same speed and torque will be

A. 1.79 Ω **B.** 2.1 Ω **C.** 3.1 Ω **D.** 18.9 Ω

Common Data Questions

Statement for Linked Questions (Q3.10.1 and Q3.10.2):
A separately excited DC motor runs at 1500 rpm under no-load with 200 V applied to the armature. The field voltage is maintained at its rated value. The speed of the motor, when it delivers a torque of 5 Nm is 1400 rpm as shown in the figure. The rotational losses and armature reaction are neglected.



Q3.10.1 MCQ 2010 2M

The armature resistance of the motor is

A. 2 Ω **B.** 3.4 Ω **C.** 4.4 Ω **D.** 7.7 Ω

Q3.10.2 MCQ 2010 2M

For the motor to deliver a torque of 2.5 Nm at 1400 rpm, the armature voltage to be applied is

A. 125.5 V **B.** 193.3 V **C.** 200 V **D.** 241.7 V

Q3.09.1 MCQ 2009 2M

Figure shows the extended view of a 2 pole dc machine with 10 armature conductors. Normal brush positions are shown by A and B, placed at the interpolar axis. If the brushes are now shifted, in the direction of rotation, to A′ and B′ as shown, the voltage waveform $V_{A'B'}$ will resemble

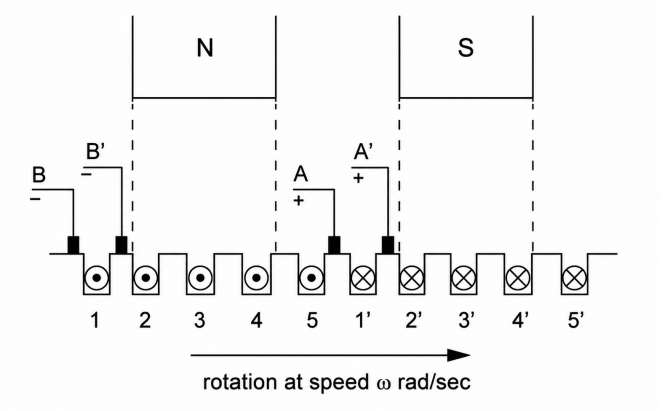
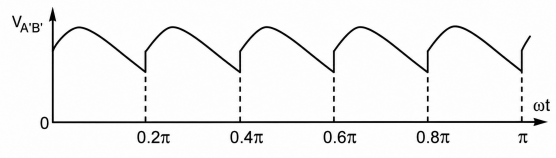
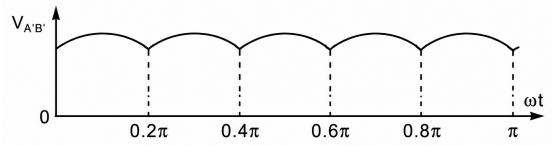


Fig. Q3.09.1

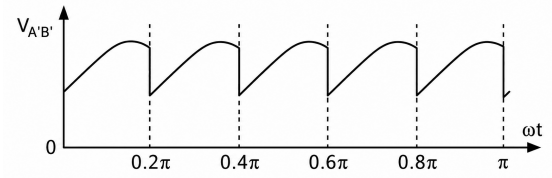
A.



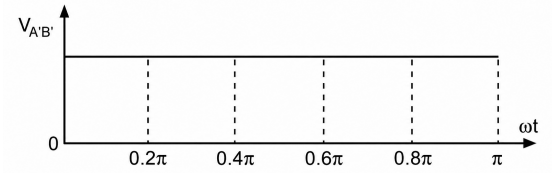
B.



C.



D.



Common Data Questions

Statement for Linked Questions Q3.08.1 and Q3.08.2:
A 240 V, dc shunt motor draws 15 A while supplying the rated load at a speed of 80 rad/s. The armature resistance is 0.5 Ω and the field winding resistance is 80 Ω.

Q3.08.1 MCQ 2008 2M

The net voltage across the armature resistance at the time of plugging will be

A. 6 V **B.** 234 V **C.** 240 V **D.** 474 V

Q3.08.2 MCQ 2008 2M

The external resistance to be added in the armature circuit to limit the armature current to 125 % of its rated value is

A. 31.1 Ω **B.** 31.9 Ω **C.** 15.1 Ω **D.** 15.9 Ω

Q3.07.1 MCQ 2007 1M

The dc motor, which can provide zero speed regulation at full load without any controller, is

A. series **B.** shunt
C. cumulative compound **D.** differential compound

Q3.06.1 MCQ 2006 1M

In a DC machine, which of the following statements is true?

A. Compensating winding is used for neutralizing armature reaction while interpole winding is used for producing residual flux

B. Compensating winding is used for neutralizing armature reaction while interpole winding is used for improving commutation

C. Compensating winding is used for improving commutation while interpole winding is used for neutralizing armature reaction

D. Compensating winding is used for improving commutation while interpole winding is used for producing residual flux

Q3.06.2 MCQ 2006 2M

A 220 V DC machine supplies 20 A at 200 V as a generator. The armature resistance is 0.2 Ω. If the machine is now operated as a motor at same terminal voltage and current but with the flux increased by 10 %, then ratio of motor speed to generator speed is

A. 0.87 **B.** 0.95 **C.** 0.96 **D.** 1.06

Q3.05.1 MCQ 2005 2M

Two magnetic poles revolve around a stationary armature carrying two coils ($c_1 - c_1'$, $c_2 - c_2'$) as shown in Fig. 3.05.1. Consider the instant when the poles are in a position as shown. Identify the correct statement regarding the polarity of the induced emf at this instant in coil sides c_1 and c_2.

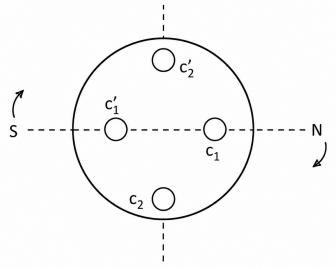


Fig. Q3.05.1

A. • in c_1, no emf in c_2 **B.** ⊗ in c_1, no emf in c_2
C. • in c_2, no emf in c_1 **D.** ⊗ in c_2, no emf in c_1

Q3.05.2 MCQ 2005 2M

A 50 kW dc shunt motor is loaded to draw the rated armature current at any given speed. When driven (i) at half the rated speed by armature voltage control and (ii) at 1.5 times the rated speed by field control, the respective output powers delivered by the motor are approximately

A. 25 kW in (i) and 75 kW in (ii)

B. 25 kW in (i) and 50 kW in (ii)

C. 50 kW in (i) and 75 kW in (ii)

D. 50 kW in (i) and 50 kW in (ii)

Q3.05.3 MCQ 2005 2M

In relation to DC machines, match the following and choose the correct combination.

Group –1 **Performance Variables**	**Group –2** **Proportional to**
(P) Armature emf (E)	(1) Flux (ϕ), speed (ω) and armature current (I_a)
(Q) Developed torque (T)	(2) ϕ and ω only
(R) Developed power (P)	(3) ϕ and I_a only
	(4) I_a and ω only
	(5) I_a only

A. $P-3$ $Q-3$ $R-1$ **B.** $P-2$ $Q-5$ $R-4$
C. $P-3$ $Q-5$ $R-4$ **D.** $P-2$ $Q-3$ $R-1$

Q3.04.1 MCQ 2004 1M

The following motor definitely has a permanent magnet rotor

A. DC commutator motor **B.** Brushless dc motor
C. Stepper motor **D.** Reluctance motor

Q3.04.2 MCQ 2004 2M

The armature resistance of a permanent magnet dc motor is 0.8 Ω. At no load, the motor draws 1.5 A from a supply voltage of 25 V and runs at 1500 rpm. The efficiency of the motor while it is operating on load at 1500 rpm drawing a current of 3.5 A from the same source will be

A. 48.0 % **B.** 57.1 % **C.** 59.2 % **D.** 88.8 %

Q3.04.3 MCQ 2004 2M

A 8 pole, DC generator has a simplex wave-wound armature containing 32 coils of 6 turns each. Its flux per pole is 0.06 Wb. The machine is running at 250 rpm. The induced armature voltage is

A. 96 V **B.** 192 V **C.** 384 V **D.** 768 V

Q3.03.1 MCQ 2003 2M

A dc series motor driving an electric train faces a constant power load. It is running at rated speed and rated voltage. If the speed has to be brought down to 0.25 p.u. the supply voltage has to be approximately brought down to

A. 0.75 p.u. **B.** 0.5 p.u. **C.** 0.25 p.u. **D.** 0.125 p.u.

Q3.03.2 MCQ 2003 2M

When Stator and Rotor windings of a 2-pole rotating electrical machine are excited, each would produce a sinusoidal mmf distribution in the air gap with peak values F_s and F_r respectively. The rotor mmf lags stator mmf by a space angle δ at any instant as shown in Fig. 3.03.2. Thus, half of stator and rotor surfaces will form one pole with the other half forming the second pole. Further, the direction of torque acting on the rotor can be clockwise or counter-clockwise.

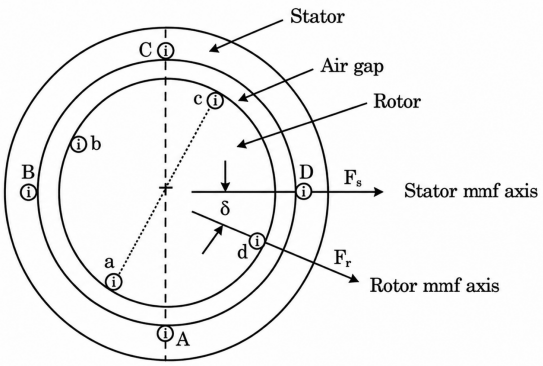


Fig. Q3.03.2

The following Table gives four sets of statements as regards poles and torque. Select the correct set corresponding to the mmf axes as shown in Fig.3.03.2

A.

Stator Surface ABC forms	Stator Surface CDA forms	Rotor Surface cda forms	Rotor Surface abc forms	Torque is
North Pole	South Pole	North Pole	South Pole	Clockwise

B.

Stator Surface ABC forms	Stator Surface CDA forms	Rotor Surface cda forms	Rotor Surface abc forms	Torque is
South Pole	North Pole	North Pole	South Pole	Counter clockwise

C.

Stator Surface ABC forms	Stator Surface CDA forms	Rotor Surface cda forms	Rotor Surface abc forms	Torque is
North Pole	South Pole	South Pole	North Pole	Counter clockwise

D.

Stator Surface ABC forms	Stator Surface CDA forms	Rotor Surface cda forms	Rotor Surface abc forms	Torque is
South Pole	North Pole	South Pole	North Pole	Clockwise

Q3.03.3 MCQ 2003 2M

To conduct load test on a dc shunt motor, it is coupled to a generator which is identical to the motor. The field of the generator is also connected to the same supply source as the motor. The armature of the generator is connected to a load resistance. The armature resistance is 0.02 p.u. armature reaction and mechanical losses can be neglected. With rated voltage across the motor, the load resistance across the generator is adjusted to obtain rated armature current in both motor and generator. The p.u. value of this load resistance is

A. 1.0 **B.** 0.98 **C.** 0.96 **D.** 0.94

Q3.03.4 MCQ 2003 1M

The speed/torque regimes in a dc motor and the control methods suitable for the same are given respectively in Group II and Group I

Group I		Group II	
P	Field Control	1	Below base speed
Q	Armature Control	2	Above base speed
		3	Above base torque
		4	Below base torque

A. $P-1;\ Q-3$ **B.** $P-2;\ Q-1$
C. $P-2;\ Q-3$ **D.** $P-1;\ Q-4$

Q3.02.1 MCQ 2002 1M

A dc series motor fed from rated supply voltage is overloaded and its magnetic circuit is saturated. The torque-speed characteristic of this motor will approximately be represented by which curve of Fig.3.02.1?

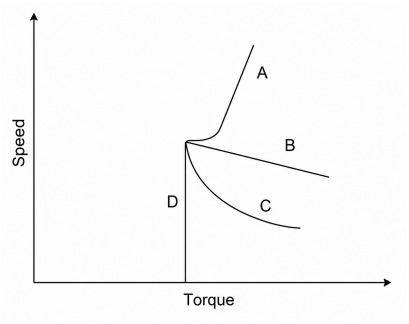


Fig. Q3.02.1

A. Curve A **B.** Curve B **C.** Curve C **D.** Curve D

Q3.02.2 MCQ 2002 2M

A 200 V, 2000 rpm, 10A, separately excited dc motor has an armature resistance of 2Ω. Rated dc voltage is applied to both the armature and field winding of the motor. If the armature draws 5A from the source, the torque developed by the motor is

A. 4.30 Nm **B.** 4.77 Nm **C.** 0.45 Nm **D.** 0.50 Nm

Q3.02.3 NAT 2002 2M

A 230V, 250 rpm, 100A separately excited dc motor has an armature resistance of 0.5Ω. the motor is connected to a 230V dc supply, and the rated dc voltage is applied to the field winding. It is driving a load whose torque speed characteristic is given by $T_L = 500 - 10\omega$, where ω is the rotational speed expressed in rad/sec and T_L is the load torque in Nm. Find the steady state speed at which the motor will drive the load and the armature current drawn by it from the source. Neglect the rotational losses of the machine.

Q3.01.1 NAT 2001 5M

In a dc motor running at 2000 rpm, the hysteresis and eddy current losses are 500W and 200W respectively. If the flux remains constant, calculate the speed at which the total iron losses are halved.

Q3.01.2 NAT 2001 5M

A dc series motor is rated 230V, 1000 rpm, 80 A (refer to Fig.3.01.2). the series field resistance is 0.11Ω, and the armature resistance is 0.14Ω. If the flux at an armature current of 20A is 0.4 times of that under rated condition, calculate the speed at this reduced armature current of 20A.

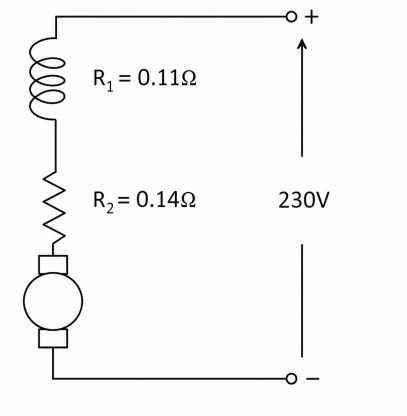


Fig. Q3.01.2

Q3.01.3 MCQ 2001 2M

An electric motor with 'constant output power' will have a torque speed characteristic in the form of a

A. straight line through the origin

B. straight line parallel to the speed axis

C. circle about the origin

D. rectangular hyperbola

Q3.00.1 MCQ 2000 1M

The compensating winding in a DC machine

A. is located in armature slots for compensation of the armature reaction

B. is located on commutating poles for improving the commutation

C. is located on pole shoes for avoiding the flashover at the commutator surface

D. is located on poles shoes to avoid the sparking at the brushes.

Q3.00.2 MCQ 2000 2M

A 240 V dc series motor takes 40A when giving its rated output at 1500 rpm. Its resistance is 0.3 ohms. The value of resistance that must be added to obtain the rated torque at 1000 rpm is:

A. 6 ohms **B.** 5.7 ohms **C.** 2.2 ohms **D.** 1.9 ohms

Q3.00.3 MCQ 2000 2M

A permanent magnet dc commutator motor has a no load speed of 6000 rpm when connected to a 120V dc supply. The armature resistance is 2.5 ohms and other losses may be neglected. The speed of the motor with supply voltage of 60V developing a torque 0.5 Nm, is:

A. 3000 rpm **B.** 2673 rpm **C.** 2836 rpm **D.** 5346 rpm

Q3.00.4 NAT 2000 5M

A belt driven dc shunt generator runs at 1500 rpm delivering 10 kW, at 220V bus bars. The belt breaks, following which the machine operates as a motor drawing 2kW power. What will be its speed as a motor? The armature and field resistances are 0.25 ohms and 55 ohms respectively. Ignore armature reaction and assume the contact drop at each brush to be 1V.

Q3.99.1 MCQ 1999 1M

A 4-pole lap-wound DC generator has a developed power of P watts and voltage of E volts. Two adjacent brushes of the machine are removed as they are worn out. If the machine operates with the remaining brushes, the developed voltage and power that can be obtained from the machine are

A. $E,\ P$ **B.** $\frac{E}{2},\ \frac{P}{2}$ **C.** $E,\ \frac{P}{4}$ **D.** $E,\ \frac{P}{2}$

Q3.99.2 MCQ 1999 2M

A DC shunt motor is running at 1200 rpm, when excited with 220 V DC. Neglecting the losses and saturation, the speed of the motor when connected to a 175V DC supply is:

A. 750 rpm **B.** 900 rpm **C.** 1050 rpm **D.** 1200 rpm

Q3.99.3 NAT 1999 5M

A separately excited DC shunt motor is driving a fan load whose torque is proportional to the square of the speed. When 100 V are applied to the motor, the current taken by the motor is 8A, with the speed being 500 rpm. At what applied voltage does the speed reach 750 rpm and then

what is the current drawn by the armature? Assume the armature circuit resistance to be 1Ω. Neglect brush drop and mechanical losses.

Q3.98.1 MCQ 1998 2M

A 240 V d.c. shunt motor with an armature resistance of 0.5Ω has a full load current of 40 A. Find the ratio of the stalling torque to the full load torque when a resistance of 1Ω is connected in series with the armature?

A. 4 **B.** 12 **C.** 6 **D.** None of the above

Q3.98.2 NAT 1998 5M

A DC shunt generator delivers 60 kW at 240 V and 360 rpm. The armature and field resistances are 0.015Ω and 60Ω respectively. Calculate the speed of the machine running as a shunt motor and taking 60 kW input at 240 V. Allow 1 volt per brush for contact drop.

Q3.97.1 NAT 1997 2M

A 5 kW, 200 V d.c. shunt motor has an armature resistance of 1 ohm and shunt field resistance of 100 ohms. At no-load, the motor draws 6A from a 200 V supply and runs at 1000 rpm. The rotational loss of the machine is . . . W and the no load torque is . . . N-m

Q3.97.2 NAT 1997 2M

A 200 V, 10 kW lap-wound d.c. generator has 10 poles and 500 conductors on its armature. If the pole face covers 80% of the pole pitch, the pole face conductors required to fully compensate for armature reaction will be . . . conductors/pole.

Q3.97.3 NAT 1997 2M

The field coil of a two-pole d.c. series motor is made up of two identical sections. In one case (i) two sections of the field coil are connected in series, and in another case (ii) the two sections are connected in parallel. If the motor takes the rated current in both the cases, then torque (i) : torque (ii) is . . . and speed (i) : speed (ii) is . . . (Neglect the effect of magnetic saturation)

Q3.97.4 NAT 1997 5M

At 50% of full load, the armature current drawn by a d.c. shunt motor is 40 A when connected to a 200 V d.c. mains. By decreasing the field flux, its speed is raised by 20% this also causes a 10% increase in load torque. Calculate the percentage change in field current. The armature resistance including the brushes is 1 Ohm. Neglect saturation and armature reaction.

Q3.96.1 MCQ 1996 1M

A 4 pole generator with 16 coil has a two layer lap winding. The pole pitch is

A. 32 **B.** 16 **C.** 8 **D.** 4

Q3.96.2 MCQ 1996 2M

Out of the following factors for a dc machine,

(i) Interpole

(ii) Armature resistance

(iii) Armature leakage reactance

(iv) Armature reaction

(v) Reduction in field current

The factors that are responsible for decrease in the terminal voltage of a shunt generator are

A. I, II and IV **B.** II, III and V
C. II, IV and V **D.** III, IV and V

Q3.96.3 MCQ 1996 1M

A 4 pole dynamo with wave wound armature has 51 slots containing 20 conductors in each slot. The induced emf is 357 volts and the speed is 8500 rpm. The flux per pole will be

A. 3.5 mWb **B.** 1.2 mWb **C.** 14 mWb **D.** 21 mWb

Q3.95.1 MCQ 1995 1M

A differentially compounded d.c. motor with interpoles and with brushes on the neutral axis is to be driven as a generator in the same direction with the same polarity of the terminal voltage. It will then

A. be a cumulatively compound generator but the interpole coil connections are to be reversed

B. be a cumulatively compounded generator without reversing the interpole coil connections.

C. be a differentially compounded generator without reversing the interpole coil connections

D. be a differentially compounded generator but the interpole coil connections are to be reversed.

Q3.95.2 NAT 1995 5M

A 220V, 1.5 kW, 859 rpm, separately excited dc motor has armature resistance of 2.5 ohms and it draws a current of 8A at rated-load condition. If the field current and the armature voltage are fixed at the value of rated speed at rated load, what will be the no-load speed of the motor? Assume losses remain constant between no-load and full load operation.

Q3.93.1 MCQ 1993 1M

A cumulative compounded long shunt motor is driving a load at rated torque and rated speed. If the series field is shunted by a resistance equal to the resistance of the series field, keeping the torque constant.,

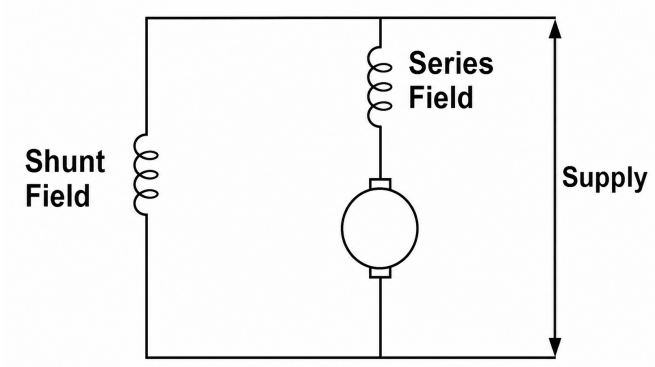


Fig. Q3.93.1

A. the armature current increases

B. the motor speed increases

C. the armature current decreases

D. the motor speed decreases

Q3.92.1 NAT 1992 5M

A separately excited d.c. motor has an armature resistance of 0.5 ohm. It runs off a 250 V d.c. supply drawing an armature current of 20 A at 1500 rpm. The torque developed for an armature current of 10 A will be _____ for the same field current.

Q3.92.2 MCQ 1992 1M

Neglecting all losses, the developed torque (T) of a d.c. separately excited motor, operating under constant terminal voltage, is related to its output power (P) as under

A. $T \propto \sqrt{P}$ **B.** $T \propto P$

C. $T^2 \propto P^3$ **D.** T independent of P

Q3.91.1 NAT 1991 1M

A 10 kW, 6 pole, D.C generator develops an EMF of 200 V at 1500 rpm. The armature has a lap connected winding. The average flux density over a pole pitch is 0.9 Tesla. The length and diameter of the armature are 0.25 m and 0.2 m respectively. Calculate (a) The flux per pole (b) The total number of active conductors in the armature (c) The torque developed by the machine when the armature supplies a current of 50 A.

Q3.91.2 MCQ 1991 1M

Match the windings of a large D.C series motor and their functions:

Windings		Function	
(A)	Series field winding	(P)	To avoid field distortion under the pole
(B)	Shunt field winding	(Q)	To avoid sparking
(C)	Commutating pole winding	(R)	To generate working flux
(D)	Compensating winding	(S)	To avoid run-away speeds on no load

A. $A \rightarrow R,\ B \rightarrow S,\ C \rightarrow Q,\ D \rightarrow P$

B. $A \rightarrow S,\ B \rightarrow R,\ C \rightarrow P,\ D \rightarrow Q$

C. $A \rightarrow P,\ B \rightarrow Q,\ C \rightarrow R,\ D \rightarrow S$

D. $A \rightarrow Q,\ B \rightarrow P,\ C \rightarrow S,\ D \rightarrow R$

Q3.91.3 Descriptive 1991 1M

A separately excited d.c. motor has to be run at 125% of rated speed when delivering 50% torque. The excitation will be _____.

Q3.91.4 MCQ 1991 1M

Level compounded D.C. generator is used for which of the following applications?

A. Electric Arc Welding **B.** Electric Traction

C. Battery Charging **D.** Lighting

Answer Key

Q3.25.1 A	Q3.24.1 27.19 to 27.39	Q3.23.1 D
Q3.23.2 1340 to 1345	Q3.22.1 483	Q3.21.1 275.10 to 275.30
Q3.20.1 C	Q3.19.1 1577 to 1583	Q3.18.1 825
Q3.18.2 600	Q3.17.1 10.75	Q3.17.2 550
Q3.17.3 B	Q3.17.4 1241.5 to 1242.0	Q3.16.1 C
Q3.16.2 86.84	Q3.15.1 B	Q3.15.2 8

Q3.15.3 57.789	Q3.15.4 179.5 to 180.5	Q3.15.5 B
Q3.15.6 47.5	Q3.14.1 1240.63	Q3.14.2 1.27
Q3.14.3 12.55	Q3.14.4 880.43	Q3.12.1 D
Q3.11.1 B	Q3.11.2 A	Q3.10.1 B
Q3.10.2 B	Q3.09.1 A	Q3.08.1 D
Q3.08.2 A	Q3.07.1 D	Q3.06.1 B

Q3.06.2 A	Q3.05.1 A	Q3.05.2 B
Q3.05.3 D	Q3.04.1 C	Q3.04.2 A
Q3.04.3 C	Q3.03.1 B	Q3.03.2 C
Q3.03.3 C	Q3.03.4 B	Q3.02.1 B
Q3.02.2 A	Q3.02.3 300.8,27.15	Q3.01.1 1140
Q3.01.2 2678.57	Q3.01.3 D	Q3.00.1 C

Q3.00.2 D	Q3.00.3 B	Q3.00.4 1387
Q3.99.1 C	Q3.99.2 D	Q3.99.3 156
Q3.98.1 A	Q3.98.2 343.15	Q3.97.1 784,7.486
Q3.97.2 4	Q3.97.3 2 and <1	Q3.97.4 28
Q3.96.1 D	Q3.96.2 C	Q3.96.3 B
Q3.95.1 B	Q3.95.2 940	Q3.93.1 A
Q3.92.1 15.28	Q3.92.2 B	Q3.91.1 0.023,340,63.69
Q3.91.2 A	Q3.91.3 DECREASES BY 20%	Q3.91.4 D

UNIT

IV

Three Phase Induction Machines

Topics

A. Principle of Operation of Rotating Machines
B. Construction of 3-Phase Induction Motor
C. Equivalent Circuit of Induction Motor
D. Power Flow
E. Torque-Slip, Torque-Speed Characteristics and Performance of IM
F. Stability and Testing – No-Load (OC) and Blocked-Rotor (SC) Tests
G. Starting of Induction Motors
H. Speed Control and Double Cage Rotor
I. Dynamic Braking
J. Plugging, Cogging and Crawling
K. Single-Phase Induction Motors – Operating Principle

Unit Snapshot*

Stream: EE	**Questions:** 96	**MCQ:** 67	**MSQ:** 1	**NAT:** 26
1M: 40	**2M:** 42	**Descriptive:** 14	**Avg Weightage** ~3M	**Range:** 1M(2020)–5M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 NAT 2026 2M

A balanced three-phase supply is given to a 30 kW, 4-pole, 400 V, 50 Hz, wound rotor induction motor with Y-connected stator and rotor windings. The motor is driving a constant torque load. With shorted slip rings, the machine runs at 1476 rpm.

When an external non-inductive resistance of 0.27 Ω per phase is connected in series in the rotor circuit, the steady-state speed drops to 1404 rpm.

Neglecting rotational losses, the actual per phase rotor winding resistance is _____ Ω.

(Round off to two decimal places)

Q4.26.2 MCQ 2026 1M

Which one of the following options is correct regarding the typical double-squirrel-cage structure used in induction motors?

A. At starting, a larger portion of the rotor current flows in the outer cage

B. At starting, a larger portion of the rotor current flows in the inner cage

C. At rated speed, most of the rotor current flows in the outer cage

D. The purpose of the double-squirrel-cage structure is to lower the effective rotor resistance at starting

Q4.25.1 MCQ 2025 2M

A 3-phase, 400 V, 4 pole, 50 Hz star connected induction motor has the following parameters referred to the stator:

$$R_r' = 1\,\Omega, \qquad X_s = X_r' = 2\,\Omega$$

Stator resistance, magnetizing reactance and core loss of the motor are neglected. The motor is run with constant V/f control from a drive. For maximum starting torque, the voltage and frequency output, respectively, from the drive, is closest to,

A. 400 V and 50 Hz **B.** 200 V and 25 Hz
C. 100 V and 12.5 Hz **D.** 300 V and 37.5 Hz

Q4.25.2 MCQ 2025 1M

The operating region of the developed torque (T_{em}) and speed (ω) of an induction motor drive is given by the shaded region OQRE in the figure. The load torque (T_L) characteristic is also shown. The motor drive moves from the initial operating point O to the final operating point S. Which one of the following trajectories will take the shortest time?

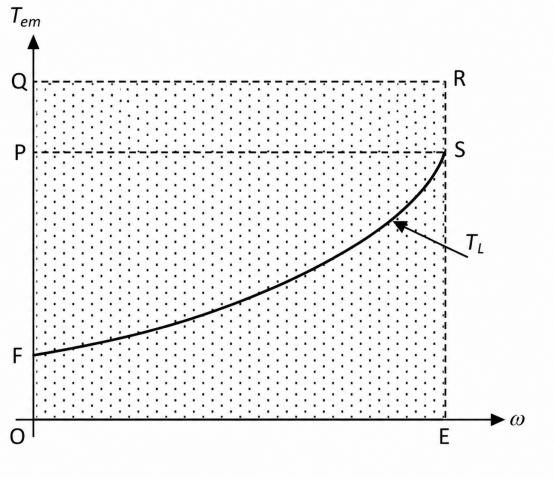


Fig. Q4.25.2

A. $O - Q - R - S$ **B.** $O - P - S$
C. $O - E - S$ **D.** $O - F - S$

Q4.24.1 NAT 2024 2M

A 3-phase star connected slip ring induction motor has the following parameters referred to the stator:

$$R_s = 3\,\Omega, \quad X_s = 2\,\Omega, \quad X_r' = 2\,\Omega, \quad R_r' = 2.5\,\Omega$$

The per phase stator to rotor effective turns ratio is 3 : 1. The rotor winding is also star connected. The magnetizing reactance and core loss of the motor can be neglected. To have maximum torque at starting, the value of the extra resistance in ohms (referred to the rotor side) to be connected in series with each phase of the rotor winding is _____ (rounded off to 2 decimal places).

Q4.24.2 MCQ 2024 1M

A three phase, 50 Hz, 6 pole induction motor runs at 960 rpm. The stator copper loss, core loss, and the rotational loss of the motor can be neglected. The percentage efficiency of the motor is

A. 92 **B.** 94 **C.** 96 **D.** 98

Q4.23.1 NAT 2023 2M

A three phase 415 V, 50 Hz, 6-pole, 960 RPM, 4 HP squirrel cage induction motor drives a constant torque load at rated speed operating from rated supply and delivering rated output. If the supply voltage and frequency are reduced by 20%, the resultant speed of the motor in RPM

(neglecting the stator leakage impedance and rotational losses) is _____ (Round off to the nearest integer).

Q4.23.2 MCQ 2023 1M

The following details present various modes of induction machine operation and the ranges of slip:

List-I (Mode of operation)

a. Running in generator mode

b. Running in motor mode

c. Plugging in motor mode

List-II (Range of Slip)

p. From 0.0 to 1.0

q. From 1.0 to 2.0

r. From −1.0 to 0.0

The correct matching between the elements in List-I with those of List-II is

A. a-r, b-p, and c-q **B.** a-r, b-q, and c-p
C. a-p, b-r, and c-q **D.** a-q, b-p, and c-r

Q4.22.1 NAT 2022 1M

The frequencies of the stator and rotor currents flowing in a three-phase 8-pole induction motor are 40 Hz and 1 Hz, respectively. The motor speed, in rpm, is _____. *(round off to nearest integer)*

Q4.22.2 NAT 2022 2M

A 4-pole induction motor with inertia of 0.1 kg-m^2 drives a constant load torque of 2 Nm. The speed of the motor is increased linearly from 1000 rpm to 1500 rpm in 4 seconds as shown in the figure below. Neglect losses in the motor. The energy, in joules, consumed by the motor during the speed change is _____. *(round off to nearest integer)*

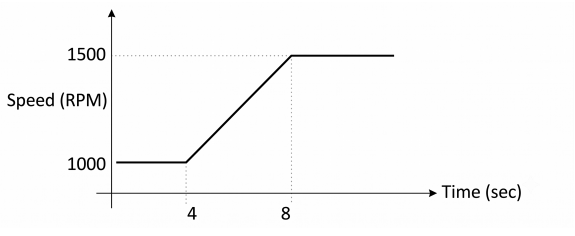


Fig. Q4.22.2

Q4.22.3 NAT 2022 2M

A 3-phase, 415 V, 4-pole, 50 Hz induction motor draws 5 times the rated current at rated voltage at starting. It is required to bring down the starting current from the supply to 2 times of the rated current using a 3-phase autotransformer. If the magnetizing impedance of the induction motor and no load current of the autotransformer is neglected, then the transformation ratio of the autotransformer is given by _____. *(round off to two decimal places).*

Q4.21.1 MCQ 2021 1M

Consider the table given:

Constructional feature	Machine type	Mitigation
(P) Damper bars	**(S)** Induction motor	**(X)** Hunting
(Q) Skewed rotor slots	**(T)** Transformer	**(Y)** Magnetic locking
(R) Compensating winding	**(U)** Synchronous machine	**(Z)** Armature reaction
	(V) DC machine	

The correct combination that relates the constructional feature, machine type and mitigation is

A. P-V-X, Q-U-Z, R-T-Y **B.** P-U-X, Q-S-Y, R-V-Z
C. P-T-Y, Q-V-Z, R-S-X **D.** P-U-X, Q-V-Y, R-T-Z

Q4.21.2 NAT 2021 2M

An 8-pole, 50 Hz, three-phase, slip-ring induction motor has an effective rotor resistance of 0.08 Ω per phase. Its speed at maximum torque is 650 RPM. The additional resistance per phase that must be inserted in the rotor to achieve maximum torque at start is _____ Ω. (Round off to 2 decimal places.) Neglect the magnetizing current and stator leakage impedance. Consider equivalent circuit parameters referred to the stator.

Q4.21.3 NAT 2021 1M

The power input to a 500 V, 50 Hz, 6-pole, 3-phase induction motor running at 975 RPM is 40 kW. The total stator losses are 1 kW. If the total friction and windage losses are 2.025 kW, then the efficiency is _____ %.

Q4.20.1 NAT 2020 1M

A three-phase, 50 Hz, 4-pole induction motor runs at no-load with a slip of 1%. With full load, the slip increases to 5%. The % speed regulation of the motor (rounded off to 2 decimal places) is _____.

Q4.19.1 NAT 2019 2M

A delta-connected, 3.7 kW, 400 V (line), three-phase, 4-pole, 50-Hz squirrel-cage induction motor has the following equivalent circuit parameters per phase referred to the stator: $R_1 = 5.39\ \Omega$, $R_2 = 5.72\ \Omega$, $X_1 = X_2 = 8.22\ \Omega$. Neglect shunt branch in the equivalent circuit. The starting line current in amperes (round off to two decimal places) when it is connected to a 100 V (line), 10 Hz, three-phase AC source is _____.

Q4.19.2 MCQ 2019 1M

The parameter of an equivalent circuit of a three-phase induction motor affected by reducing the rms value of the supply voltage at the rated frequency is

A. rotor resistance **B.** rotor leakage reactance
C. magnetizing reactance **D.** stator resistance

Q4.17.1 MCQ 2017 1M

A 4 pole induction machine is working as an induction generator. The generator supply frequency is 60 Hz. The rotor current frequency is 5 Hz. The mechanical speed of the rotor in RPM is

A. 1350 **B.** 1650 **C.** 1950 **D.** 2250

Q4.17.2 NAT 2017 2M

A star-connected, 12.5 kW, 208 V (line), 3-phase, 60 Hz squirrel cage induction motor has following equivalent circuit parameters per phase referred to the stator: $R_1 = 0.3\ \Omega$, $R_2 = 0.3\ \Omega$, $X_1 = 0.41\ \Omega$, $X_2 = 0.41\ \Omega$. Neglect the shunt branch in the equivalent circuit. The starting current (in Amperes) for this motor when connected to an 80 V (line), 20 Hz, 3-phase AC source is _____.

Q4.17.3 MCQ 2017 1M

A 3-phase, 4-pole, 400 V, 50 Hz squirrel-cage induction motor is operating at a slip of 0.02. The speed of the rotor flux in mechanical rad/sec, sensed by a stationary observer, is closest to

A. 1500 **B.** 1470 **C.** 157 **D.** 154

Q4.16.1 MCQ 2016 1M

In a constant V/f induction motor drive, the slip at maximum torque.

A. is directly proportional to the synchronous speed.

B. remains constant with respect to the synchronous speed.

C. has an inverse relation with the synchronous speed.

D. has no relation with the synchronous speed.

Q4.16.2 NAT 2016 2M

The starting line current of a 415 V, 3-phase, delta connected induction motor is 120 A, when the rated voltage is applied to its stator winding. The starting line current at a reduced voltage of 110 V, in ampere, is _____.

Q4.15.1 NAT 2015 2M

A 3-phase 50 Hz square wave (6-step) VSI feeds a 3-phase, 4 pole induction motor. The VSI line voltage has a dominant 5^{th} harmonic component. If the operating slip of the motor with respect to fundamental component voltage is 0.04, the slip of the motor with respect to 5^{th} harmonic component of voltage is _____.

Q4.15.2 NAT 2015 1M

Figure shows the per-phase equivalent circuit of a two-pole three-phase induction motor operating at 50 Hz. The "air-gap" voltage, V_g across the magnetizing inductance, is 210 V rms, and the slip, s, is 0.05. The torque (in Nm) produced by the motor is _____.

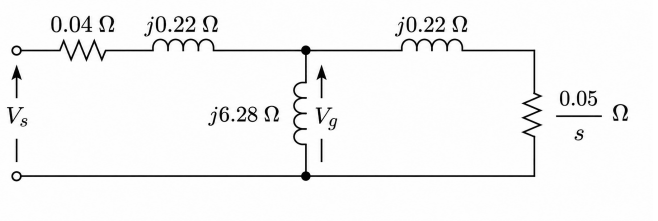


Fig. Q4.15.2

Q4.15.3 NAT 2015 2M

A 220 V, 3-phase, 4-pole, 50 Hz induction motor of wound rotor type is supplied at rated voltage and frequency. The stator resistance, magnetizing reactance, and core loss are negligible. The maximum torque produced by the rotor is 225% of full load torque and it occurs at 15% slip. The actual rotor resistance is 0.03 Ω/phase. The value of external resistance (in Ohm) which must be inserted in a rotor phase if the maximum torque is to occur at start is _____.

Q4.14.1 NAT 2014 1M

An 8-pole, 3-phase, 50 Hz induction motor is operating at a speed of 700 rpm. The frequency of the rotor current of the motor in Hz is _____.

Q4.14.2 MCQ 2014 2M

A 3 phase, 50 Hz, six pole induction motor has a rotor resistance of 0.1 Ω and reactance of 0.92 Ω. Neglect the voltage drop in stator and assume that the rotor resistance is constant. Given that the full load slip is 3%, the ratio of maximum torque to full load torque is

A. 1.567 **B.** 1.712 **C.** 1.948 **D.** 2.134

Q4.14.3 MCQ 2014 1M

A three-phase, 4-pole, self excited induction generator is feeding power to a load at a frequency f_1. If the load is partially removed, the frequency becomes f_2. If the speed of the generator is maintained at 1500 rpm in both the cases, then

A. $f_1,\ f_2 > 50$ Hz and $f_1 > f_2$

B. $f_1 < 50$ Hz and $f_2 > 50$ Hz

C. $f_1,\ f_2 < 50$ Hz and $f_2 > f_1$

D. $f_1 > 50$ Hz and $f_2 < 50$ Hz

Q4.14.4 MCQ 2014 1M

In a constant V/f control of induction motor, the ratio V/f is maintained constant from 0 to base frequency, where V is the voltage applied to the motor at fundamental frequency f. Which of the following statements relating to low frequency operation of the motor is TRUE?

A. At low frequency, the stator flux increases from its rated value.

B. At low frequency, the stator flux decreases from its rated value.

C. At low frequency, the motor saturates.

D. At low frequency, the stator flux remains unchanged at its rated value.

Q4.14.5 MCQ 2014 2M

A three-phase slip-ring induction motor, provided with a commutator winding, is shown in the figure. The motor rotates in clockwise direction when the rotor windings are closed.

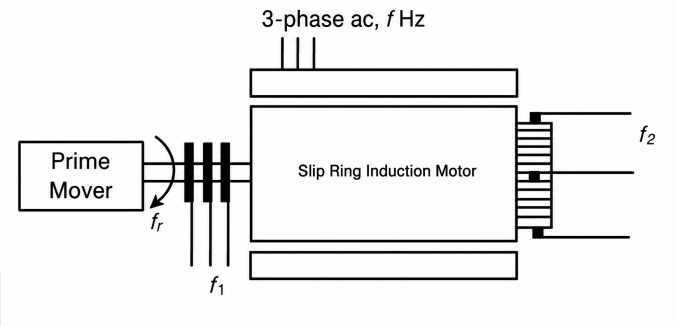


Fig. Q4.14.5

If the rotor winding is open circuited and the system is made to run at rotational speed f_r with the help of prime-mover in anti-clockwise direction, then the frequency of voltage across slip rings is f_1 and frequency of voltage across commutator brushes is f_2. The values of f_1 and f_2 respectively are

A. $f + f_r$ and f **B.** $f - f_r$ and f
C. $f - f_r$ and $f + f_r$ **D.** $f + f_r$ and $f - f_r$

Q4.14.6 MCQ 2014 1M

The torque-speed characteristics of motor (T_M) and load (T_L) for two cases are shown in the figures (a) and (b). The load torque is equal to motor torque at points P, Q, R and S.

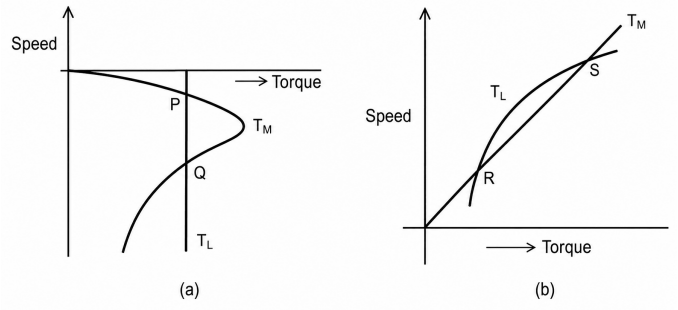


Fig. Q4.14.6

The stable operating points are

A. P and R **B.** P and S
C. Q and R **D.** Q and S

Q4.13.1 MCQ 2013 1M

Leakage flux in an induction motor is

A. flux that leaks through the machine

B. flux that links both stator and rotor windings

C. flux that links none of the windings

D. flux that links the stator winding or the rotor winding but not both

Q4.13.2 MCQ 2013 2M

A 4-pole induction motor, supplied by a slightly unbalanced three-phase 50 Hz source, is rotating at 1440 rpm. The electrical frequency in Hz of the induced negative sequence current in the rotor is

A. 100 **B.** 98 **C.** 52 **D.** 48

Q4.12.1 MCQ 2012 1M

The slip of an induction motor normally does not depend on

A. rotor speed **B.** synchronous speed

C. shaft torque **D.** core-loss component

Q4.12.2 MCQ 2012 2M

The locked rotor current in a 3-phase, star connected 15 kW, 4-pole, 230 V, 50 Hz induction motor at rated conditions is 50 A. Neglecting losses and magnetizing current, the approximate locked rotor line current drawn when the motor is connected to a 236 V, 57 Hz supply is

A. 58.5 A **B.** 45.0 A **C.** 42.7 A **D.** 55.6 A

Q4.11.1 MCQ 2011 2M

A three-phase 440 V, 6 pole, 50 Hz, squirrel cage induction motor is running at a slip of 5%. The speed of stator magnetic field with respect to rotor magnetic field and speed of rotor with respect to stator magnetic field are

A. zero, −5 rpm **B.** zero, 955 rpm

C. 1000 rpm, −5 rpm **D.** 1000 rpm, 955 rpm

Q4.10.1 MCQ 2010 1M

A balanced three-phase voltage is applied to a star-connected induction motor, the phase to neutral voltage being V. The stator resistance, rotor resistance referred to the stator, stator leakage reactance, rotor leakage reactance referred to the stator and the magnetizing reactance are denoted by r_s, r_r', x_s, x_r' and X_m respectively. The magnitude of the starting current of the motor is given by:

A. $\dfrac{V}{\sqrt{(r_s + r_r')^2 + (x_s + x_r')^2}}$ **B.** $\dfrac{V}{\sqrt{r_r'^2 + (x_r' + X_m)^2}}$

C. $\dfrac{V}{\sqrt{(r_s + r_r')^2 + (X_m + x_r')^2}}$ **D.** $\dfrac{V}{\sqrt{r_s^2 + (X_m + x_r')^2}}$

Q4.10.2 MCQ 2010 2M

A separately excited dc machine is coupled to a 50 Hz, three-phase, 4-pole induction machine as shown in the figure. The dc machine is energized first and the machines rotate at 1600 rpm. Subsequently the induction machine is also connected to a 50 Hz three-phase source, the phase sequence being consistent with the direction of rotation. In steady state,

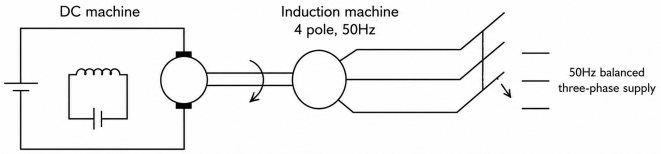


Fig. Q4.10.2

A. both machines act as generators

B. the dc machine acts as a generator, and the induction machine acts as a motor

C. the dc machine acts as a motor, and the induction machine acts as a generator

D. both machines act as motors

Q4.09.1 MCQ 2009 1M

A 3-phase squirrel cage induction motor supplied from a balanced 3-phase source drives a mechanical load. The torque-speed characteristics of the motor (solid curve) and of the load (dotted curve) are shown. Of the two equilibrium points A and B, which of the following options correctly describe the stability of A and B?

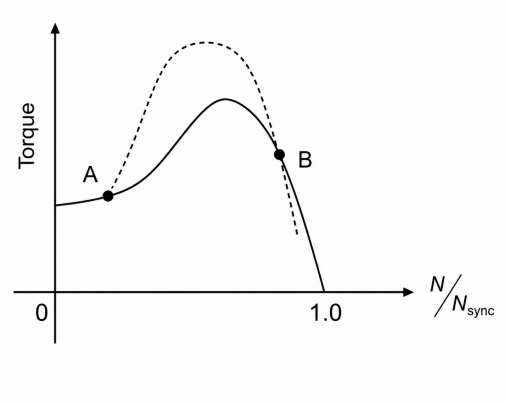


Fig. Q4.09.1

A. A is stable B is unstable

B. A is unstable B is stable

C. Both are stable

D. Both are unstable

Q4.08.1 MCQ 2008 2M

A 400 V, 50 Hz, 30 hp, three-phase induction motor is drawing 50 A current at 0.8 power factor lagging. The stator and rotor copper losses are 1.5 kW and 900 W respectively. The friction and windage losses are 1050 W and the core losses are 1200 W. The air-gap power of the motor will be

A. 23.06 kW **B.** 24.11 kW **C.** 25.01 kW **D.** 26.21 kW

Q4.08.2 MCQ 2008 2M

A 400 V, 50 Hz, 4 pole, 1400 rpm, star connected squirrel cage induction motor has the following parameters referred to the stator:

$$R_r' = 1.0\,\Omega, \quad X_s = X_r' = 1.5\,\Omega$$

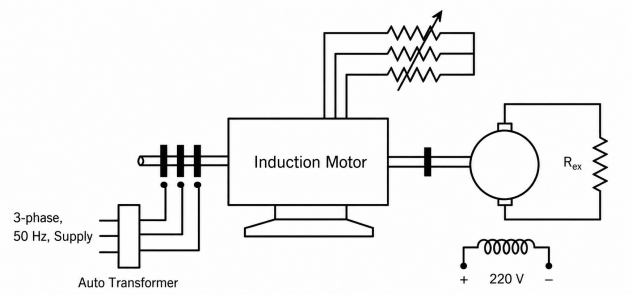
Neglect the stator resistance, as well as the core and rotational losses of the motor.

The motor is controlled from a 3-phase voltage source inverter with constant V/f control. The stator line-to-line voltage (rms) and frequency to obtain the maximum torque at starting will be:

A. 20.6 V, 2.7 Hz **B.** 133.3 V, 16.7 Hz
C. 266.6 V, 33.3 Hz **D.** 323.3 V, 40.3 Hz

Common Data Questions

Statement for Linked Questions (Q4.08.3 and Q4.08.4): A 3-phase, 440 V, 50 Hz, 4-pole, slip ring induction motor is fed from the rotor side through an auto-transformer and the stator is connected to a variable resistance as shown in the figure.



The motor is coupled to a 220 V, separately excited, dc generator feeding power to fixed resistance of 10 Ω. Two-wattmeter method is used to measure the input power to induction motor. The variable resistance is adjusted such that the motor runs at 1410 rpm and the following readings were recorded:

$$W_1 = 1800\,\text{W}, \qquad W_2 = -200\,\text{W}$$

Q4.08.3 MCQ 2008 2M

The speed of rotation of stator magnetic field with respect to rotor structure will be

A. 90 rpm in the direction of rotation

B. 90 rpm in the opposite direction of rotation

C. 1500 rpm in the direction of rotation

D. 1500 rpm in the opposite direction of rotation

Q4.08.4 MCQ 2008 2M

Neglecting all losses of both the machines, the dc generator power output and the current through resistance (R_{ex}) will respectively be

A. 96 W, 3.10 A **B.** 120 W, 3.46 A
C. 1504 W, 12.26 A **D.** 1880 W, 13.71 A

Q4.07.1 MCQ 2007 1M

The electromagnetic torque T_e of a drive and its connected load torque T_L are as shown below. Out of the operating points A, B, C and D, the stable ones are

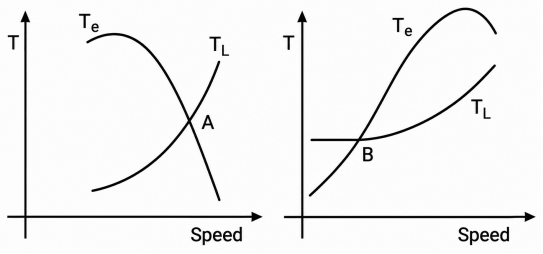


Fig. Q4.07.1

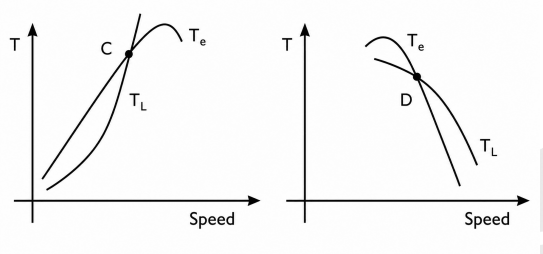


Fig. Q4.07.1

A. A, C, D **B.** B, C **C.** A, D **D.** B, C, D

Q4.07.2 MCQ 2007 2M

A three-phase squirrel cage induction motor has a starting torque of 150% and a maximum torque of 300% with respect to rated torque at rated voltage and rated frequency. Neglect the stator resistance and rotational losses. The value of slip for maximum torque is

A. 13.48% **B.** 16.24% **C.** 18.92% **D.** 26.79%

Common Data Questions

Statement for Linked Questions (Q4.07.3,Q4.07.4 and Q4.07.5): A three phase squirrel cage induction motor has a starting current of seven times the full load current and full load slip of 5%.

Q4.07.3 MCQ 2007 2M

If an autotransformer is used for reduced voltage starting to provide 1.5 per unit starting torque, the autotransformer ratio (%) should be

A. 57.77% **B.** 72.56% **C.** 78.25% **D.** 81.33%

Q4.07.4 MCQ 2007 2M

If a star-delta starter is used to start this induction motor, the per unit starting torque will be

A. 0.607 **B.** 0.816 **C.** 1.225 **D.** 1.616

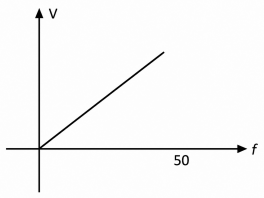
Q4.07.5 MCQ 2007 2M

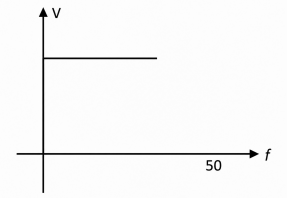
If a starting torque of 0.5 per unit is required then the per unit starting current should be

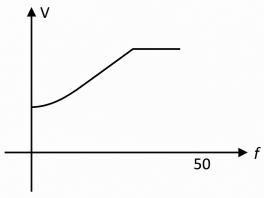
A. 4.65 **B.** 3.75 **C.** 3.16 **D.** 2.13

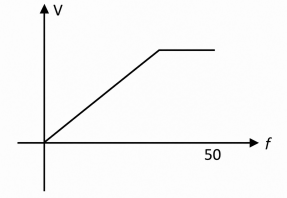
Q4.06.1 MCQ 2006 1M

The speed of a 3-phase, 440 V, 50 Hz induction motor is to be controlled over a wide range from zero speed to 1.5 times the rated speed using a 3-phase voltage source inverter. It is desired to keep the flux in the machine constant in the constant torque region by controlling the terminal voltage as the frequency changes. The inverter output voltage vs frequency characteristic should be

A.


B.


C.


D.


Q4.06.2 MCQ 2006 2M

The speed of a 4-pole induction motor is controlled by varying the supply frequency while maintaining the ratio of supply voltage to supply frequency (V/f) constant. At rated frequency of 50 Hz and rated voltage of 400 V its speed is 1440 rpm. Find the speed at 30 Hz, if the load torque is constant.

A. 882 rpm **B.** 864 rpm **C.** 840 rpm **D.** 828 rpm

Q4.06.3 MCQ 2006 2M

A 3 phase, 4 pole, 400 V, 50 Hz, star connected induction motor has following circuit parameters $r_1 = 1.0\ \Omega$, $r_2' = 0.5\ \Omega$, $x_1 = x_2' = 1.2\ \Omega$, $x_m = 35\ \Omega$

The starting torque when the motor is started direct-on-line is (use approximate equivalent circuit model)

A. 63.6 Nm **B.** 74.3 Nm
C. 190.8 Nm **D.** 222.9 Nm

Q4.06.4 MCQ 2006 2M

A 3 phase, 10 kW, 400 V, 4 pole, 50 Hz, star connected induction motor draws 20 A on full load. Its no load and blocked rotor test data are given below

No Load test:	400 V	6 A	1002 W
Blocked Rotor test:	90 V	15 A	762 W

Neglecting copper loss in the No Load test and core loss in the Blocked Rotor test, estimate the motor's full load efficiency.

A. 76% **B.** 81% **C.** 82.4% **D.** 85%

Q4.05.1 MCQ 2005 1M

On the torque/speed curve of induction motor shown in Fig.4.05.1, four points of operation are marked as W, X, Y, and Z. Which one of them represents the operation at a slip greater than 1?

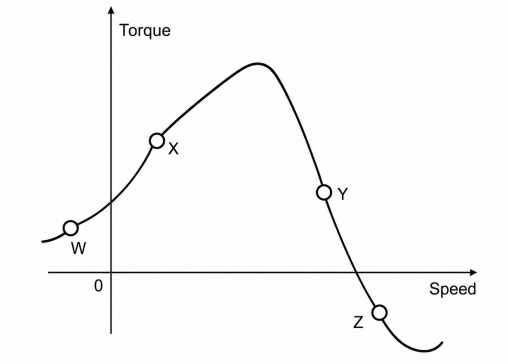


Fig. Q4.05.1

A. W **B.** X **C.** Y **D.** Z

Q4.05.2 MCQ 2005 1M

For an induction motor, operating at a slip s, the ratio of gross power output to air gap power is equal to:

A. $(1-s)^2$ **B.** $(1-s)$ **C.** $\sqrt{(1-s)}$ **D.** $(1-\sqrt{s})$

Q4.05.3 MCQ 2005 2M

Under no load condition, if the applied voltage to an induction motor is reduced from the rated voltage to half the rated value,

A. the speed decreases and the stator current increases

B. both the speed and the stator current decrease

C. the speed and the stator current remain practically constant

D. there is negligible change in the speed but the stator current decreases

Q4.05.4 MCQ 2005 2M

A three-phase cage induction motor is started by direct-on-line (DOL) switching at the rated voltage. If the starting current drawn is 6 times the full load current, and the full load slip is 4%, the ratio of the starting developed torque to the full load torque is approximately equal to

A. 0.24 **B.** 1.44 **C.** 2.40 **D.** 6.00

Q4.05.5 MCQ 2005 2M

Determine the correctness or otherwise of the following Assertion [a] and the Reason [r].

Assertion: Under V/f control of induction motor, the maximum value of the developed torque remains constant over a wide range of speed in the sub-synchronous region.

Reason: The magnetic flux is maintained almost constant at the rated value by keeping the ratio V/f constant over the considered speed range.

A. Both [a] and [r] are true and [r] is the correct reason for [a]

B. Both [a] and [r] are true but [r] is not the correct reason for [a]

C. Both [a] and [r] are false

D. [a] is true but [r] is false

Q4.04.1 MCQ 2004 1M

The direction of rotation of a 3-phase induction motor is clockwise when it is supplied with 3-phase sinusoidal voltage having a phase sequence of A-B-C. For counter-clockwise rotation of the motor, the phase sequence of the power supply should be

A. B-C-A **B.** C-A-B
C. A-C-B **D.** B-C-A or C-A-B

Q4.04.2 MCQ 2004 2M

A 400 V, 15 kW, 4 pole, 50 Hz, a Y-connected induction motor has a full load slip of 4%. The output torque of the machine at full load is

A. 1.66 Nm **B.** 95.50 Nm
C. 99.47 Nm **D.** 624.73 Nm

Q4.04.3 MCQ 2004 2M

The synchronous speed for the seventh space harmonic mmf wave of a 3-phase, 8 pole, 50 Hz induction machine is

A. 107.14 rpm in forward direction

B. 107.14 rpm in reverse direction

C. 5250 rpm in forward direction

D. 5250 rpm in reverse direction

Q4.03.1 MCQ 2003 1M

No-load test on a 3-phase induction motor was conducted at different supply voltages and a plot of input power versus voltage was drawn. This curve was extrapolated to intersect the y-axis. This intersection point yields

A. Core loss **B.** Stator copper loss
C. Stray load loss **D.** Friction and windage loss

Q4.03.2 MCQ 2003 2M

Following are some of the properties of rotating electrical machines

P: Stator winding current is dc, rotor-winding current is ac
Q: Stator winding current is ac, rotor-winding current is dc
R: Stator winding current is ac, rotor-winding current is ac
S: Stator has salient poles and rotor has commutator
T: Rotor has salient poles and sliprings and stator is cylindrical
U: Both stator and rotor have poly-phase windings

DC machines, synchronous machines, and Induction machines exhibit some of the above properties, as given in the following table. Indicate the correct combination from this table

	DC machines	Synchronous machines	Induction machines
(a)	P,S	Q,T	R,U
(b)	Q,U	P,T	R,S
(c)	P,S	R,U	Q,T
(d)	R,S	Q,U	P,T

Q4.03.3 MCQ 2003 2M

A 3-phase Induction Motor is driving a constant torque load at rated voltage and frequency. If both voltage and frequency are halved, the following statements relate to the new condition if stator resistance, leakage reactance, and core loss are ignored.

P: The difference between synchronous speed and actual speed remains same
Q: The air-gap flux remains same
R: The stator current remains same
S: The p.u. slip remains same

Among the statements above, the correct ones are

A. All **B.** P, Q and R
C. Q, R and S **D.** P and S

Q4.03.4 MCQ 2003 2M

An ac induction motor is used for a speed control application. It is driven from an inverter with a constant V/f control. The motor nameplate details are as follows

V : 415 V Ph : 3 f : 50 Hz N : 2850 rpm

The motor is run with the inverter output frequency set at 40 Hz, and with half the rated slip. The running speed of the motor is

A. 2400 rpm **B.** 2280 rpm
C. 2340 rpm **D.** 2790 rpm

Q4.02.1 MCQ 2002 1M

If a 400 V, 50 Hz, star connected, 3 phase squirrel cage induction motor is operated from a 400 V, 75 Hz supply, the torque that the motor can now provide while drawing rated current from the supply?

A. reduces

B. increases

C. remains the same

D. increase or reduces depending upon the rotor resistance

Q4.02.2 MCQ 2002 2M

The rotor of a three phase, 5 kW, 400 V, 50 Hz, slip ring induction motor is wound for 6 poles while its stator is wound for 4 poles. The approximate average no load steady state speed when this motor is connected to 400 V, 50 Hz supply is

A. 1500 rpm **B.** 500 rpm **C.** 0 rpm **D.** 1000 rpm

Q4.02.3 NAT 2002 2M

A 440 V, 50 Hz, 6 pole and 960 rpm star connected induction machine has the following per phase parameters referred to the stator:

$$R_s = 0.6\,\Omega \qquad R_r' = 0.3\,\Omega \qquad X_s = 1\,\Omega$$

The magnetizing reactance is very high and is neglected. The machine is connected to the 440 V, 50 Hz supply and a certain mechanical load is coupled to it. It is found that the magnitude of the stator current is equal to the rated current of the machine but the machine is running at a speed higher than its rated speed. Find the speed at which the machine is running. Also find the torque developed by the machine.

Q4.00.1 MCQ 2000 1M

A 3-phase, 4-pole squirrel cage induction motor has 36 stator and 28 rotor slots. The number of phases in the rotor is:

A. 3 **B.** 9 **C.** 7 **D.** 8

Q4.00.2 MCQ 2000 2M

The power input to a 415 V, 50 Hz, 6 pole, 3-phase induction motor running at 975 rpm is 40 kW. The stator losses are 1 kW and friction and windage losses total 2 kW. The efficiency of the motor is

A. 92.5% **B.** 90% **C.** 91% **D.** 88%

Q4.00.3 MCQ 2000 2M

A three phase, wound rotor induction motor is to be operated with slip energy recovery in the constant torque mode, when it delivers an output power P_o at slip s, then theoretically, the maximum power that is available for recovery at the rotor terminals, is equal to

A. P_o **B.** $P_o s$ **C.** $\frac{P_o}{(1-s)}$ **D.** $\frac{P_o s}{(1-s)}$

Q4.00.4 NAT 2000 5M

A 230 V, 20 hp, 60 Hz, 6-pole, 3-phase induction motor driving a constant torque load at rated frequency, rated voltage and rated horse-power, has a speed of 1175 rpm and an efficiency of 92.1%. Determine the new operating speed if a system disturbance causes 10% drop in voltage and 6% drop in frequency. Assume that friction, windage and stray power losses remain constant.

Q4.99.1 MCQ 1999 1M

The following starting method for an induction motor is inferior in view of the poor starting torque per ampere of the line current drawn:

A. Direct line starting

B. Auto transformer method of starting

C. Series inductor method of starting

D. Star-Delta method of starting

Q4.99.2 NAT 1999 5M

A 3 kW, 400V/200V, Delta/Star, 50 Hz, three phase, 6-pole induction motor is found to draw a line current of 25 A at a power factor of 0.4, when a blocked rotor test is conducted at the rated voltage. Determine the stator and rotor winding resistances in ohms per phase, if the torque developed by the motor under the above condition is 25 N-m.

Q4.98.1 MCQ 1998 1M

A 3 phase squirrel cage induction motor has a full load efficiency of 0.8 and a maximum efficiency of 0.9. It is operated at a slip of 0.6 by applying a reduced voltage. The efficiency of the motor at this operating point is:

A. less than 0.4 **B.** greater than 0.6
C. in the range of 0.8 ± 0.1 **D.** None of the above

Q4.98.2 MCQ 1998 2M

Test	Machine
(A) No load and blocked rotor test	(P) Transformer
(B) Sumpner's test	(Q) Induction machine
(C) Swinburne's test	(R) Synchronous machine
	(S) DC machine

Q4.98.3 NAT 1998 5M

A 50 hp, 6 pole, 50 Hz slip-ring induction motor runs at 960 rpm on full load with a rotor current of 40 A. allowing 300 W for the copper loss in the short circuiting gear and 1200 W for mechanical losses, find the resistance R_2 per phase of the 3 phase rotor winding.

Q4.97.1 NAT 1997 2M

A 3-phase, 20 kW, 400 V, 1470 rpm, 50 Hz squirrel cage induction motor develops a torque of 100 N-m at a speed of 1400 rpm. If the motor is connected to a 30 Hz supply, for keeping the same air-gap flux, the supply voltage should be V and for the same load torque, the new speed will be . . . rpm.

Q4.97.2 NAT 1997 5M

A 50 kV A, 400 V, 3-phase, 50 Hz squirrel cage induction motor has full load slip of 5%. Its standstill impedance is 0.866 ohms/phase. It is started using a tapped autotransformer. If the maximum allowable supply current at the time of starting is 1000 A, then calculate the tap position and the ratio of starting torque to full load torque.

Q4.97.3 MCQ 1997 1M

If an induction machine is run at above synchronous speed, it acts as

A. a synchronous motor **B.** an induction generator
C. an inductor motor **D.** none of these

Q4.96.1 MCQ 1996 1M

Unbalanced supply voltage given to a 3-phase, delta-connected induction motor will cause

A. zero sequence currents

B. less heating of the rotor

C. negative sequence component current

D. all of these

Q4.96.2 MCQ 1996 1M

When the supply voltage to an induction motor is reduced by 10%, the maximum torque will decrease by approximately

A. 5% **B.** 10% **C.** 20% **D.** 40%

Q4.96.3 MCQ 1996 1M

An induction motor having full load torque of 60 Nm when delta-connected develops a starting torque of 120 Nm. For the same supply voltage, if the motor is changed to star-connection, the starting torque developed will be

A. 40 Nm **B.** 60 Nm **C.** 90 Nm **D.** 120 Nm

Q4.96.4 MCQ 1996 3M

Out of the several characteristics shown in Figure, identify the appropriate ones to match the following for a 3-phase induction motor.

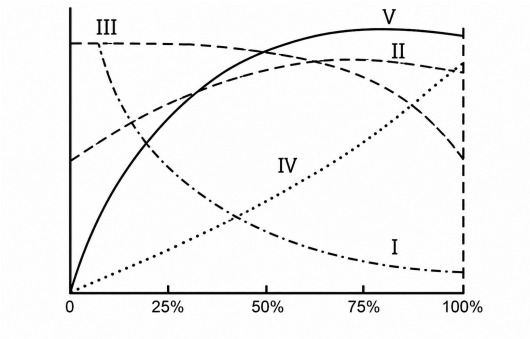


Fig. Q4.96.4

X-axis	Y-axis		
(a) Load	efficiency	(P)	curve-I
(b) Speed	current	(Q)	curve-II
(c) Speed	power factor	(R)	curve-III
		(S)	curve-IV
		(T)	curve-V

Q4.96.5 MCQ 1996 1M

In an induction motor, if the air gap is increased

A. speed will reduce

B. efficiency will improve

C. power factor will be lowered

D. breakdown torque will reduce

Q4.95.1 MCQ 1995 1M

An induction motor is fed from a balanced three phase supply at rated voltage and frequency through a bank of three single phase transformers connected in delta-delta. One unit of the bank develops fault and is removed. Then,

A. single phasing will occur and the machine fails to start

B. single phasing will not occur but the motor terminal voltages will become unbalanced and the machine can be loaded to the extent of 57.7% of its rating.

C. the machine can be loaded to the extent of 57.7% of its rating with balanced supply at its terminals.

D. the machine can be loaded to the extent of $66\frac{2}{3}\%$ with balanced supply at its terminals.

Q4.95.2 NAT 1995 1M

An induction motor runs stably under constant torque load at 1250 rpm off a 50 Hz supply. Its number of poles is _____

Q4.95.3 MCQ 1995 5M

Out of the several characteristics shown in Figure, identify the appropriate ones to match the following.

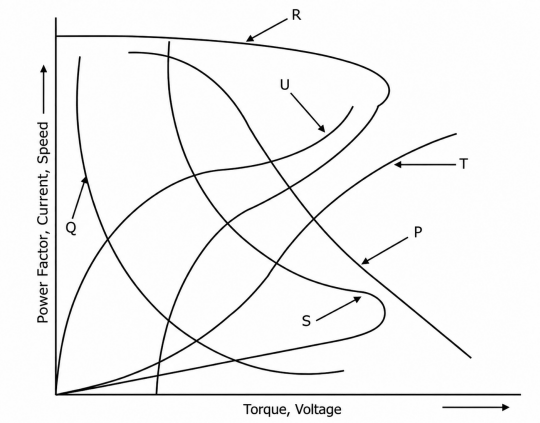


Fig. Q4.95.3

(a) Speed–torque characteristics of induction machine under motoring operation.
(b) Current–torque characteristics of a DC series motor.
(c) Power factor variation with voltage of an induction motor under no-load operation.
(d) Speed–torque characteristics of induction machine under DC injection dynamic braking operation.
(e) Speed–torque characteristics of DC series motor.

A. $a-R,\ b-Q,\ c-S,\ d-T,\ e-P$

B. $a-R,\ b-U,\ c-P,\ d-S,\ e-Q$

C. $a-P,\ b-Q,\ c-S,\ d-R,\ e-T$

D. $a-T,\ b-P,\ c-Q,\ d-R,\ e-S$

Q4.95.4 NAT 1995 5M

A 50 Hz, 440 V, three-phase, 4-pole induction motor develops half the rated torque at 1490 rpm. With the applied voltage magnitude remaining at the rated value. What should be its frequency if the motor has to develop the same torque at 1600 rpm? Neglect stator and rotor winding resistances, leakage reactances and iron losses.

Q4.94.1 MCQ 1994 1M

Skew is used in induction motors in order to reduce torque due to

A. Time harmonics **B.** Space harmonics
C. Slot harmonics **D.** Reverse rotating fields

Q4.94.2 NAT 1994

A 3-phase induction motor coupled to a pump is operating at normal speed. If one line gets disconnected, the motor stops. (True/False)

Q4.94.3 Descriptive 1994 4M

A six-pole, 50 Hz induction motor rotating at 1400 rpm is in _____ mode.

Q4.94.4 Descriptive 1994 4M

In a variable-frequency induction motor drive, the voltage must be varied _____ to the frequency.

Q4.93.1 MCQ 1993 1M

A three phase slip ring induction motor is fed from the rotor side with stator winding short circuited. The frequency of the currents flowing in the short circuited stator is

A. slip frequency

B. supply frequency

C. frequency corresponding to rotor speed

D. zero

Q4.91.1 MSQ 1991 2M

In a poly-phase squirrel-cage induction motor, increased starting torque can be obtained by

A. Increasing the frequency of operation

B. Using deep-bar rotors

C. Increasing the number of poles

D. Using a double-cage rotor

Q4.91.2 MCQ 1991 5M

A prime mover drives a 6 pole, 3-phase induction frequency converter. The converter is connected to 60 Hz, 3-phase supply on the primary. If the prime mover speed is 3000 rpm, the frequencies of the possible outputs from the converter are

A. 120 Hz, 60 Hz
B. 90 Hz, 210 Hz
C. 186 Hz, 86 Hz
D. 180 Hz, 210 Hz

Q4.91.3 NAT 1991 5M

A 5 kW, 400 V, 50 Hz, 4 pole, delta connected three-phase induction motor is supplied by a cable of negligible inductance. On starting the motor using a star-delta starter, it is found that the starting torque is the same on star as well as delta connection, due to the voltage drop in the feeder resistance. The equivalent circuit parameters of the motor are as follows:

$r_1 = 1\,\Omega, \quad x_1 = 4.5\,\Omega, \quad r_2' = 1.4\,\Omega, \quad x_2' = 4.5\,\Omega$

Determine the feeder resistance.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q4.26.1 0.09	Q4.26.2 A	Q4.25.1 C
Q4.25.2 A	Q4.24.1 0.26 to 0.29	Q4.24.2 C
Q4.23.1 760	Q4.23.2 A	Q4.22.1 585
Q4.22.2 1733	Q4.22.3 0.63	Q4.21.1 B
Q4.21.2 0.52	Q4.21.3 90	Q4.20.1 4.21
Q4.19.1 14 to 16	Q4.19.2 C	Q4.17.1 C
Q4.17.2 70.0533	Q4.17.3 C	Q4.16.1 C
Q4.16.2 31.80	Q4.15.1 1.192	Q4.15.2 401.66
Q4.15.3 0.17	Q4.14.1 3.33	Q4.14.2 C
Q4.14.3 C	Q4.14.4 B	Q4.14.5 A
Q4.14.6 B	Q4.13.1 D	Q4.13.2 B
Q4.12.1 D	Q4.12.2 B	Q4.11.1 *
Q4.10.1 A	Q4.10.2 C	Q4.09.1 A
Q4.08.1 C	Q4.08.2 B	Q4.08.3 D
Q4.08.4 C	Q4.07.1 A	Q4.07.2 D
Q4.07.3 C	Q4.07.4 B	Q4.07.5 C
Q4.06.1 D	Q4.06.2 C	Q4.06.3 A
Q4.06.4 B	Q4.05.1 A	Q4.05.2 B
Q4.05.3 D	Q4.05.4 B	Q4.05.5 A
Q4.04.1 C	Q4.04.2 C	Q4.04.3 A
Q4.03.1 D	Q4.03.2 A	Q4.03.3 B
Q4.03.4 C	Q4.02.1 A	Q4.02.2 C
Q4.02.3 1040,268.66	Q4.00.1 C	Q4.00.2 B
Q4.00.3 D	Q4.00.4 1100.7	Q4.99.1 C
Q4.99.2 6.89,0.348	Q4.98.1 D	Q4.98.2 A:Q,B:P,C:S
Q4.98.3 0.27	Q4.97.1 240,800	Q4.97.2 *
Q4.97.3 B	Q4.96.1 C	Q4.96.2 C
Q4.96.3 A	Q4.96.4 a:T,b:P,C:Q	Q4.96.5 C,D
Q4.95.1 C	Q4.95.2 4	Q4.95.3 B
Q4.95.4 53.72	Q4.94.1 C	Q4.94.2 False
Q4.94.3 GENERATING	Q4.94.4 DIRECTLY PROPOR-TIONAL	Q4.93.1 A
Q4.91.1 B, D	Q4.91.2 B	Q4.91.3 5.377

UNIT

V

Synchronous Machines

Topics

A. Working Principle, Equivalent Circuits,Torques of Cylindrical and Salient Pole Machines
B. Electrical Power in Alternators
C. Behaviour at Over,Under and Nominal Excitation
D. Voltage Regulation of Synchronous Generator
E. Parallel Operation of Generators/ Droop Characteristics
F. Starting of Synchronous Motors
G. Stability of Synchronous Machines
H. Hunting Phenomenon
I. Open Circuit Characteristics and Short Circuit Characteristics(OCC/SCC)
J. Armature Winding and Slot Harmonics

Unit Snapshot*

Stream: EE	**Questions:** 80	**MCQ:** 55	**MSQ:** 2	**NAT:** 23
1M: 34	**2M:** 37	**Descriptive:** 9	**Avg Weightage** ~3M	**Range:** 1M(2018)–5M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 1M

A certain application requires power at a frequency of 16.67 Hz, while the available grid frequency is 50 Hz. A 3-phase synchronous motor connected to the 50 Hz, 3-phase grid, driving a synchronous generator, is used for this application.

Which one of the following combinations is a suitable choice for the number of poles in the motor and generator, respectively?

A. 2, 6 **B.** 6, 2 **C.** 4, 6 **D.** 6, 4

Q5.26.2 MCQ 2026 1M

The figure shows the single-line diagram of a synchronous generator delivering $P = 50$ MW of power at unity power factor to an infinite bus.

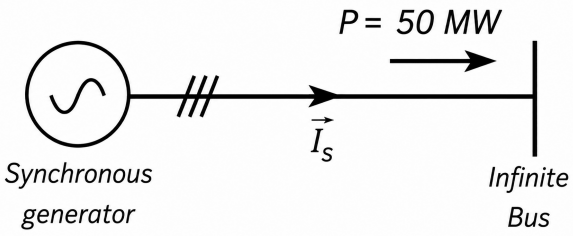


Fig. Q5.26.2

I_s denotes the stator current phasor. If the field excitation is increased, which one of the following options correctly describes its effect on the stator current, power factor, and load angle of the machine?

A. Stator current increases, power factor becomes lagging, load angle remains the same

B. Stator current decreases, power factor becomes leading, load angle remains the same

C. Stator current increases, power factor becomes lagging, load angle decreases

D. Stator current increases, power factor becomes leading, load angle increases

Q5.25.1 NAT 2025 1M

The induced emf in a 3.3 kV, 4 pole, 3-phase star connected synchronous motor is considered to be equal and in phase with the terminal voltage under no load condition. On application of a mechanical load, the induced emf phasor is deflected by an angle of 2° mechanical with respect to the terminal voltage phasor. If the synchronous reactance is 2 Ω, and stator resistance is negligible, then the motor armature current magnitude, in ampere, during loaded condition is closest to _____ *(round off to two decimal places).*

Q5.25.2 MSQ 2025 1M

Instrument(s) required to synchronize an alternator to the grid is/are

- ☐ **A.** Voltmeter
- ☐ **B.** Wattmeter
- ☐ **C.** Synchroscope
- ☐ **D.** Stroboscope

Q5.24.1 MCQ 2024 2M

A 3-phase, 11 kV, 10 MVA synchronous generator is connected to an inductive load of power factor $\frac{\sqrt{3}}{2}$ via a lossless line with a per-phase inductive reactance of 5 Ω. The per-phase synchronous reactance of the generator is 30 Ω. If the generator is producing the rated current at the rated voltage, then the power factor at the terminal of the generator is:

A. 0.63 lagging **B.** 0.87 lagging
C. 0.63 leading **D.** 0.87 leading

Q5.23.1 NAT 2023 2M

A three-phase synchronous motor with synchronous impedance of $0.1 + j0.3$ per unit per phase has a static stability limit of 2.5 per unit. The corresponding excitation voltage in per unit is _____ (Round off to 2 decimal places).

Q5.22.1 NAT 2022 2M

A star-connected 3-phase, 400 V, 50 kVA, 50 Hz synchronous motor has a synchronous reactance of 1 Ω per phase with negligible armature resistance. The shaft load on the motor is 10 kW while the power factor is 0.8 leading. The loss in the motor is 2 kW. The magnitude of the per phase excitation emf of the motor, in volts, is _____. *(round off to nearest integer).*

Q5.20.1 MCQ 2020 1M

A three-phase cylindrical rotor synchronous generator has a synchronous reactance X_s and a negligible armature resistance. The magnitude of per-phase terminal voltage is V_A and the magnitude of per-phase induced emf is E_A. Considering the following two statements, P and Q,

P: For any three-phase balanced leading load connected across the terminals of this synchronous generator, V_A is always more than E_A

Q: For any three-phase balanced lagging load connected across the terminals of this synchronous generator, V_A is always less than E_A

Which of the following options is correct?

A. P is false and Q is true.

B. P is true and Q is false.

C. P is false and Q is false.

D. P is true and Q is true.

Q5.20.2 MCQ 2020 2M

The figure below shows the per-phase Open Circuit Characteristics (measured in V) and Short Circuit Characteristics (measured in A) of a 14 kVA, 400 V, 50 Hz, 4-pole, 3-phase, delta-connected alternator, driven at 1500 rpm. The field current, I_f is measured in A. Readings taken are marked as respective (x, y) coordinates in the figure. Ratio of the unsaturated and saturated synchronous impedances $(Z_{s(\text{unsat})}/Z_{s(\text{sat})})$ of the alternator is closest to:

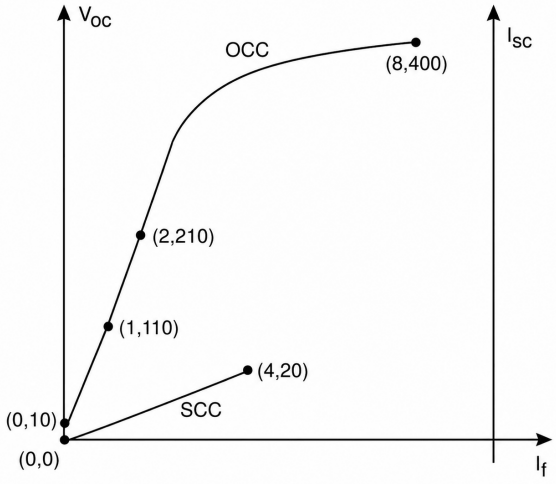


Fig. Q5.20.2

A. 2.100 **B.** 2.025 **C.** 2.000 **D.** 1.000

Q5.20.3 NAT 2020 2M

A cylindrical rotor synchronous generator with constant real power output and constant terminal voltage is supplying 100 A current to a 0.9 lagging power factor load. An ideal reactor is now connected in parallel with the load, as a result of which the total lagging reactive power requirement of the load is twice the previous value while the real power remains unchanged. The armature current is now _____ A (rounded off to 2 decimal places).

Q5.19.1 MCQ 2019 1M

A three-phase synchronous motor draws 200 A from the line at unity power factor at rated load. Considering the same line voltage and load, the line current at a power factor of 0.5 leading is

A. 100 A **B.** 200 A **C.** 300 A **D.** 400 A

Q5.19.2 NAT 2019 2M

A 220 V (line), three-phase, Y-connected, synchronous motor has a synchronous impedance of $(0.25+j2.5)\ \Omega$/phase. The motor draws the rated current of 10 A at 0.8 pf leading. The rms value of line-to-line internal voltage in volts (round off to two decimal places) is _____.

Q5.18.1 MCQ 2018 1M

In a salient pole synchronous motor, the developed reluctance torque attains the maximum value when the load angle in electrical degrees is

A. 0 **B.** 45 **C.** 60 **D.** 90

Q5.17.1 NAT 2017 1M

A three-phase, 50 Hz, star-connected cylindrical-rotor synchronous machine is running as a motor. The machine is operated from a 6.6 kV grid and draws current at unity power factor (UPF). The synchronous reactance of the motor is 30 Ω per phase. The load angle is 30°. The power delivered to the motor in kW is _____. (Give the answer up to one decimal place).

Q5.17.2 NAT 2017 2M

Two parallel connected, three-phase, 50 Hz, 11 kV, star-connected synchronous machines A and B, are operating as synchronous condensers. They together supply 50 MVAR to a 11 kV grid. Current supplied by both the machines are equal. Synchronous reactances of machine A and machine

B are 1 Ω and 3 Ω, respectively. Assuming the magnetic circuit to be linear, the ratio of excitation current of machine A to that of machine B is _____. (Give the answer up to two decimal places.)

Q5.17.3 MCQ 2017 1M

If a synchronous motor is running at a leading power factor, its excitation induced voltage (E_f) is

A. equal to terminal voltage V_t

B. higher than the terminal voltage V_t

C. less than terminal voltage V_t

D. dependent upon supply voltage V_t

Q5.17.4 NAT 2017 1M

A 25 kVA, 400 V, Δ-connected, 3-phase, cylindrical rotor synchronous generator requires a field current of 5 A to maintain the rated armature current under short-circuit condition. For the same field current, the open-circuit voltage is 360 V. Neglecting the armature resistance and magnetic saturation, its voltage regulation (in % with respect to terminal voltage), when the generator delivers the rated load at 0.8 pf leading, at rated terminal voltage is _____.

Q5.16.1 NAT 2016 2M

A three-phase, 50 Hz salient-pole synchronous motor has a per-phase direct-axis reactance (X_d) of 0.8 pu and a per-phase quadrature-axis reactance (X_q) of 0.6 pu. Resistance of the machine is negligible. It is drawing full-load current at 0.8 pf (leading). When the terminal voltage is 1 pu, per-phase induced voltage, in pu, is _____.

Q5.16.2 MCQ 2016 1M

Consider a system consisting of a synchronous generator working at a lagging power factor, a synchronous motor working at an overexcited condition and a directly grid-connected induction generator. Consider capacitive VAr to be a source and inductive VAr to be a sink of reactive power. Which one of the following statements is TRUE?

A. Synchronous motor and synchronous generator are sources and induction generator is a sink of reactive power.

B. Synchronous motor and induction generator are sources and synchronous generator is a sink of reactive power.

C. Synchronous motor is a source and induction generator and synchronous generator are sinks of reactive power.

D. All are sources of reactive power.

Q5.15.1 NAT 2015 2M

A three-phase, 11 kV, 50 Hz, 2 pole, star connected, cylindrical rotor synchronous motor is connected to an 11 kV, 50 Hz source. Its synchronous reactance is 50 Ω per phase, and its stator resistance is negligible. The motor has a constant field excitation. At a particular load torque, its stator current is 100 A at unity power factor. If the load torque is increased so that the stator current is 120 A, then the load angle (in degrees) at this load is _____.

Q5.14.1 NAT 2014 1M

A star connected 400 V, 50 Hz, 4 pole synchronous machine gave the following open circuit and short circuit test results:

Open circuit test: V_{oc} = 400 V (rms, line-to-line) at field current, I_f = 2.3 A

Short circuit test: I_{sc} = 10 A (rms, phase) at field current, I_f = 1.5 A

The value of per phase synchronous impedance in Ω at rated voltage is _____.

Q5.14.2 MCQ 2014 2M

A three phase synchronous generator is to be connected to the infinite bus. The lamps are connected as shown in the figure for the synchronization. The phase sequence of bus voltage is R-Y-B and that of incoming generator voltage is R′-Y′-B′.

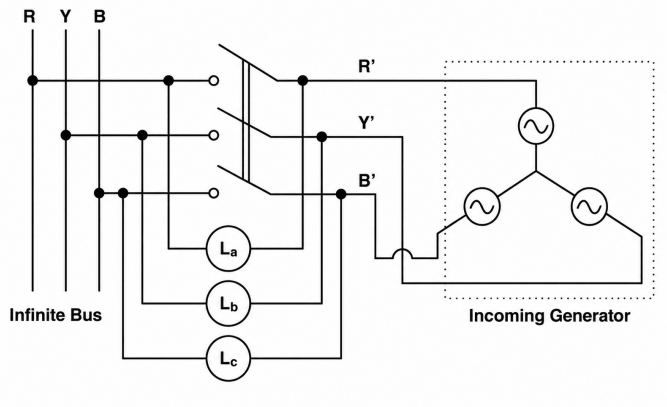


Fig. Q5.14.2

It was found that the lamps are becoming dark in the sequence L_a-L_b-L_c. It means that the phase sequence of incoming generator is

A. opposite to infinite bus and its frequency is more than infinite bus

B. opposite to infinite bus but its frequency is less than infinite bus

C. same as infinite bus and its frequency is more than infinite bus

D. same as infinite bus and its frequency is less than infinite bus

Q5.14.3 MCQ 2014 2M

A 20-pole alternator is having 180 identical stator slots with 6 conductors in each slot. All the coils of a phase are in series. If the coils are connected to realize single-phase winding, the generated voltage is V_1. If the coils are reconnected to realize three-phase star-connected winding, the generated phase voltage is V_2. Assuming full pitch, single-layer winding, the ratio V_1/V_2 is

A. $\frac{1}{\sqrt{3}}$ **B.** $\frac{1}{2}$ **C.** $\sqrt{3}$ **D.** 2

Q5.14.4 NAT 2014 2M

A synchronous generator is connected to an infinite bus with excitation voltage $E_f = 1.3$ pu. The generator has a synchronous reactance of 1.1 pu and is delivering real power (P) of 0.6 pu to the bus. Assume the infinite bus voltage to be 1.0 pu. Neglect stator resistance. The reactive power (Q) in pu supplied by the generator to the bus under this condition is _____.

Q5.14.5 NAT 2014 2M

There are two generators in a power system. No-load frequencies of the generators are 51.5 Hz and 51 Hz, respectively, and both are having droop constant of 1 Hz/MW. Total load in the system is 2.5 MW. Assuming that the generators are operating under their respective droop characteristics, the frequency of the power system in Hz in the steady state is _____.

Q5.14.6 MCQ 2014 1M

In a synchronous machine, hunting is predominantly damped by

A. mechanical losses in the rotor

B. iron losses in the rotor

C. copper losses in the stator

D. copper losses in the rotor

Q5.14.7 NAT 2014 2M

A non-salient pole synchronous generator having synchronous reactance of 0.8 pu is supplying 1 pu power to a unity power factor load at a terminal voltage of 1.1 pu. Neglecting the armature resistance, the angle of the voltage behind the synchronous reactance with respect to the angle of the terminal voltage in degrees is _____.

Q5.11.1 MCQ 2011 1M

A three-phase, salient pole synchronous motor is connected to an infinite bus. It is operated at no load at normal excitation. The field excitation of the motor is first reduced to zero and then increased in the reverse direction gradually. Then the armature current

A. increases continuously

B. first increases and then decreases steeply

C. first decreases and then increases steeply

D. remains constant

Q5.11.2 MCQ 2011 2M

The direct axis and quadrature axis reactances of a salient pole alternator are 1.2 p.u and 1.0 p.u respectively. The armature resistance is negligible. If this alternator is delivering rated kVA at upf and at rated voltage then its power angle is

A. 30° **B.** 45° **C.** 60° **D.** 90°

Q5.09.1 MCQ 2009 1M

A field excitation of 20 A in a certain alternator results in an armature current of 400 A in short circuit and a terminal voltage of 2000 V on open circuit. The magnitude of the internal voltage drop within the machine at a load current of 200 A is

A. 1 V **B.** 10 V **C.** 100 V **D.** 1000 V

Common Data Questions

Statement for Linked Questions (Q5.08.1 and Q5.08.2): A synchronous motor is connected to an infinite bus at 1.0 pu voltage and draws 0.6 pu current at unity power factor. Its synchronous reactance is 1.0 pu and resistance is negligible.

Q5.08.1 MCQ 2008 2M

The excitation voltage (E) and load angle (δ) will respectively be

A. 0.8 pu and 36.86° lead

B. 0.8 pu and 36.86° lag

C. 1.17 pu and 30.96° lead

D. 1.17 pu and 30.96° lag

Q5.08.2 MCQ 2008 2M

Keeping the excitation voltage same, the load on the motor is increased such that the motor current increases by 20%. The operating power factor will become

A. 0.995 lagging **B.** 0.995 leading
C. 0.791 lagging **D.** 0.848 leading

Q5.08.3 MCQ 2008 1M

Distributed winding and short chording employed in AC machines will result in

A. increase in emf and reduction in harmonics.

B. reduction in emf and increase in harmonics.

C. increase in both emf and harmonics.

D. reduction in both emf and harmonics.

Q5.07.1 MCQ 2007 2M

A three-phase synchronous motor connected to ac mains is running at full load and unity power factor. If its shaft load is reduced by half, with field current held constant, its new power factor will be

A. unity **B.** leading
C. lagging **D.** dependent on machine parameters

Q5.07.2 MCQ 2007 2M

A 100 kVA, 415 V (line), star-connected synchronous machine generates rated open circuit voltage of 415 V at a field current of 15 A. The short circuit armature current at a field current of 10 A is equal to the rated armature current. The per unit saturated synchronous reactance is

A. 1.731 **B.** 1.5 **C.** 0.666 **D.** 0.577

Q5.07.3 MCQ 2007 2M

An isolated 50 Hz synchronous generator is rated at 15 MW, which is also the maximum continuous power limit of its prime mover. It is equipped with a speed governor with 5% droop. Initially, the generator is feeding three loads of 4 MW, each at 50 Hz. One of these loads is programmed to trip permanently if the frequency falls below 48 Hz. If an additional load of 3.5 MW is connected, then the frequency will settle down to

A. 49.417 Hz **B.** 49.917 Hz **C.** 50.083 Hz **D.** 50.583 Hz

Q5.06.1 MCQ 2006 2M

A synchronous generator is feeding a zero power factor (lagging) load at rated current. The armature reaction is

A. magnetizing **B.** demagnetizing
C. cross-magnetizing **D.** ineffective

Q5.06.2 MCQ 2006 2M

A 3 phase, 400 V, 5 kW, star connected synchronous motor having an internal reactance of 10 Ω is operating at 50% load, unity pf. Now, the excitation is increased by 1%. What will be the new load in percent if the power factor is to be kept the same? Neglect all losses and consider a linear magnetic circuit.

A. 67.9% **B.** 56.9% **C.** 51% **D.** 50%

Common Data Questions

Statement for Linked Questions (Q5.06.3, Q5.06.4 and Q5.06.5): A 4 pole, 50 Hz, synchronous generator has 48 slots in which a double layer winding is housed. Each coil has 10 turns and is short pitched by an angle to 36° electrical. The fundamental flux per pole is 0.025 Wb.

Q5.06.3 MCQ 2006 2M

The line-to-line induced emf (in volts), for a three phase star connection is approximately

A. 808 **B.** 888 **C.** 1400 **D.** 1538

Q5.06.4 MCQ 2006 2M

The line-to-line induced emf (in volts), for a three phase connection is approximately

A. 1143 **B.** 1332 **C.** 1617 **D.** 1791

Q5.06.5 MCQ 2006 2M

The fifth harmonic component of phase emf (in volts), for a three phase star connection is

A. 0 **B.** 269 **C.** 281 **D.** 808

Q5.05.1 MCQ 2005 2M

In relation to the synchronous machines, which one of the following statements is false?

A. In salient pole machines, the direct-axis synchronous reactance is greater than the quadrature-axis synchronous reactance

B. The damper bars help the synchronous motor self-start

C. Short circuit ratio is the ratio of the field current required to produce the rated voltage on open circuit to the rated armature current

D. The V-curve of a synchronous motor represents the variation in the armature current with field excitation, at a given output power

Common Data Questions

Statement for Linked Questions (Q5.05.2 and Q5.05.3): Statement for Linked Answer Questions 83a & 83b: A 1000 kVA, 6.6 kV, 3-phase star connected cylindrical pole synchronous generator has a synchronous reactance of 20 Ω. Neglect the armature resistance and consider operation at full load and unity power factor.

Q5.05.2 MCQ 2005 1M

The induced emf (line-to-line) is close to

A. 5.5 kV **B.** 7.2 kV **C.** 9.6 kV **D.** 12.5 kV

Q5.05.3 MCQ 2005 1M

The power (or torque) angle is close to

A. 13.9° **B.** 18.3° **C.** 24.6° **D.** 33.0°

Q5.04.1 MCQ 2004 2M

Two 3-phase, Y-connected alternators are to be paralleled to a set of common bus bars. The armature has a per phase synchronous reactance of 1.7 Ω and negligible armature resistance. The line voltage of the first machines is adjusted to 3300 V and that of the second machine is adjusted to 3200 V. the machine voltages are in phase at the instant they are paralleled. Under this condition, the synchronizing current per phase will be

A. 16.98 A **B.** 29.41 A **C.** 33.96 A **D.** 58.82 A

Q5.04.2 MCQ 2004 2M

A 400 V, 50 kVA, 0.8 pf leading Δ-connected, 50 Hz synchronous machine has a synchronous reactance of 2 Ω and negligible armature resistance. The friction and windage losses are 2 kW and the core loss is 0.8 kW. The shaft is supplying 9 kW load at a power factor of 0.8 leading. The line current drawn is

A. 12.29 A **B.** 16.24 A **C.** 21.29 A **D.** 36.88 A

Q5.04.3 MCQ 2004 2M

A 500 MW 3-phase Y-connected synchronous generator has a rated voltage of 21.5 kV at 0.85 pf. The line current when operating at full load rated conditions will be

A. 13.43 kA **B.** 15.79 kA **C.** 23.25 kA **D.** 27.36 kA

Q5.04.4 MCQ 2004 2M

A hydraulic turbine having rated speed of 250 rpm is connected to a synchronous generator. In order to produce power at 50 Hz, the number of poles required in the generator are

A. 6 **B.** 12 **C.** 16 **D.** 24

Q5.03.1 MCQ 2003 2M

A 4-pole, 3-phase, double layer winding is housed in a 36-slot stator for an ac machine with 60° phase spread. Coil span is 7 slot pitches. Number of slots in which top and bottom layers belong to different phases is

A. 24 **B.** 18 **C.** 12 **D.** 0

Q5.03.2 MCQ 2003 1M

A stand-alone engine driven Synchronous generator is feeding a partly inductive load. A capacitor is now connected across the load to completely nullify the inductive current. For this operating condition.

A. the field current and fuel input have to be reduced

B. the field current and fuel input have to be increased

C. the field current has to be increased and fuel input left unaltered

D. the field current has to be reduced and fuel input left unaltered

Q5.03.3 MCQ 2003 1M

Curves X and Y in Fig.Q9 denote open circuit and full-load zero power factor (zpf) characteristics of asynchronous generator. Q is a point on the zpf characteristics at 1.0 p.u. voltage. The vertical distance PQ in Fig.Q9 gives the voltage drop across

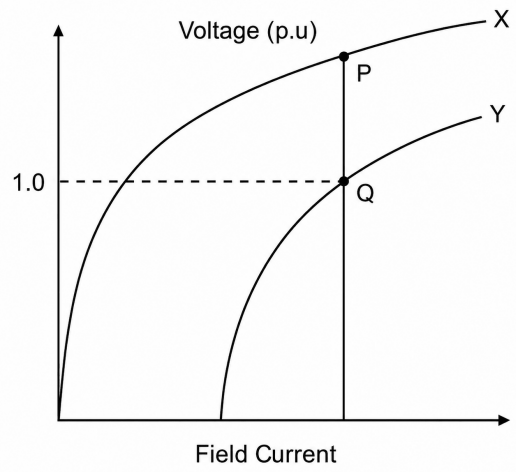


Fig. Q5.03.3

A. Synchronous reactance

B. Magnetizing reactance

C. Potier reactance

D. Leakage reactance

Q5.02.1 MCQ 2002 2M

The flux per pole in a synchronous motor with the field circuit ON and the stator disconnected from the supply is found to be 25 mWb. When the stator is connected to the rated supply with the field excitation unchanged, the flux per pole in the machine is found to be 20 mWb while the motor is running on no load. Assuming no load losses to be zero, the no load current down by the motor from the supply

A. lags the supply voltage

B. leads the supply voltage

C. is in phase with the supply voltage

D. is zero

Q5.02.2 NAT 2002 2M

A 415 V, 2 pole, 3 phase, 50 Hz, star connected, non-salient pole synchronous motor has synchronous reactance of 2 Ω per phase and negligible stator resistance. At a particular field excitation, it draws 20 A at unity power factor from a 415 V, 3 phase, 50 Hz supply. The mechanical load on the motor is now increased till the stator current is equal to 50 A. The field excitation remains unchanged. Determine:

(a) the per phase open circuit voltage E_o

(b) the developed power for the new operating condition and corresponding power factor.

Q5.01.1 MCQ 2001 1M

It is desirable to eliminate 5^{th} harmonic voltage from the phase voltage of an alternator. The coils should be short-pitched by an electrical angle of

A. 30° **B.** 36° **C.** 72° **D.** 18°

Q5.01.2 MCQ 2001 1M

Fig.P1.8 shows the magnetization curves of an alternator at rated armature current, unity power factor and also at no load. The magnetization curve for rated armature current, 0.8 power factor leading is given by

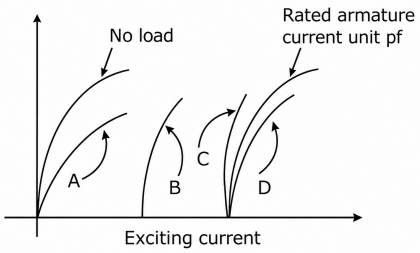


Fig. Q5.01.2

A. curve A **B.** curve B **C.** curve C **D.** curve D

Q5.01.3 MCQ 2001 2M

A star-connected 440 V, 50 Hz alternators has per phase synchronous reactance of 10Ω. It supplies a balanced capacitive load current of 20 A, as shown in the per phase equivalent circuit of Fig.2.9. It is desirable to have zero voltage regulation. The load power factor should be

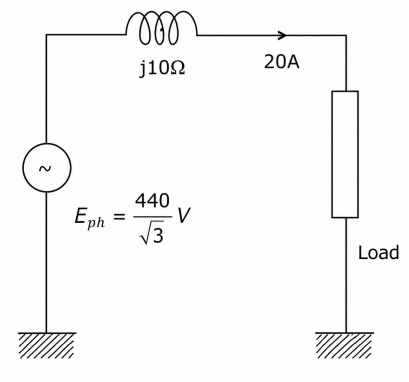


Fig. Q5.01.3

A. 0.82 **B.** 0.47 **C.** 0.39 **D.** 0.92

Q5.01.4 NAT 2001 5M

A 50 kW synchronous motor is tested by driving it by another motor. When the excitation is not switched on, the driving motor takes 800 W. When the armature is short-circuited and the rated armature current of 10 A is passed through it, the driving motor requires 2500 W. On open-circuiting the armature with rated excitation, the driving motor takes 1800 W. Calculate the efficiency of the synchronous motor at 50% load. Neglect the losses in the driving motor.

Q5.01.5 NAT 2001 5M

Two identical synchronous generators, each of 100 MVA, are working in parallel supplying 100 MVA at 0.8 lagging p.f. at rated voltage. Initially the machines are sharing load equally. If the field current of first generator is reduced by 5% and of the second generator increased by 5%, find the sharing of load (MW and MVAR) between the generators. Assume $X_d = X_q = 0.8$ p.u., no field saturation and rated voltage across load. Reasonable approximations may be made.

Q5.00.1 MCQ 2000 1M

The phase sequence of a three-phase alternator will reverse if

A. the field current is reversed keeping the direction of rotation same

B. the field current remains the same but the direction of rotation is reversed

C. the field current is reversed and the number of poles is doubled

D. the number of poles is doubled without reversing the field current

Q5.00.2 MCQ 2000 1M

A single-phase, 2000 V alternator has armature resistance and reactance of 0.8 ohms and 4.94 ohms respectively. The voltage regulation of the alternator at 100 A load at 0.8 leading power-factor is:

A. 7% **B.** −8.9% **C.** 14% **D.** 0%

Q5.00.3 NAT 2000 5M

A 2300 V, 3-phase synchronous motor driving a pump is provided with a line ammeter and a field rheostat. When the rheostat is adjusted such that the ac line current is minimum. The ammeter reads 8.8 A. What is the power being delivered to the pump, neglecting losses? How should the rheostat be adjusted so that the motor operates at 0.8 leading power factor? How many kVARs is the motor supplying to the system at this new power factor?

Q5.99.1 MCQ 1999 1M

Higher synchronous reactance is preferred in the present day alternators, because one can have

A. reduced sub-transient currents

B. reduced harmonic currents

C. reduced transient currents

D. higher voltage regulation with load

Q5.99.2 MCQ 1999 2M

A three phase alternator is wound with a 60 degree phase-spread armature windings and develops 300 kVA. If the armature is reconnected utilizing all the coils for single phase operation with a phase spread of 180 degrees, the new rating of the machine is

A. 100 kVA **B.** 200 kVA **C.** 250 kVA **D.** 300 kVA

Q5.99.3 NAT 1999 5M

A 5 MVA, 11 kV, 3-phase star connected alternator is synchronized to the bus bars and is operating with an induced EMF of 125% of the rated voltage. If the load current is 500 A, what is the power factor of operation? The machine has a synchronous reactance of 5Ω and negligible resistance per phase.

Q5.98.1 MCQ 1998 1M

A synchronous generator connected to an infinite bus is overexcited. Considering only the reactive power, from the point of view of the system, the machine acts as

A. a capacitor **B.** an inductor
C. a resistor **D.** None of the above

Q5.98.2 MCQ 1998 2M

The armature of a single phase alternator is completely wound with T single turn coils distributed uniformly. The induced voltage in each turn is 2 V (rms). The emf of the whole winding is

A. $2T$ volt **B.** $1.11T$ volt
C. $1.414T$ volt **D.** $1.273T$ volt

Q5.97.1 NAT 1997 2M

A single-phase alternator has a synchronous reactance of 2 Ω and negligible resistance. If it supplies 10 A to a purely capacitive load at 200 V, determine the generated emf and the voltage regulation (%).

Q5.96.1 MCQ 1996 1M

A cylindrical-rotor synchronous motor is connected to the supply with its field winding short-circuited. It will

A. not start

B. start but not run at synchronous speed

C. start as an induction motor and then run as synchronous motor

D. start and run as a synchronous motor

Q5.96.2 MCQ 1996 1M

In Figure, the characteristic that corresponds to the variation of synchronous reactance of a synchronous motor with field current is

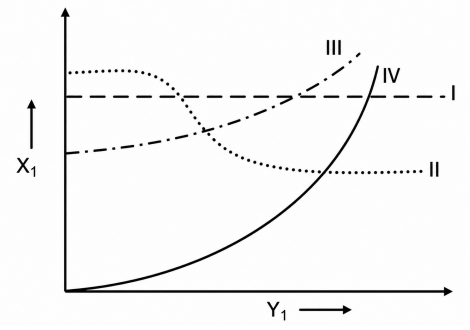


Fig. Q5.96.2

A. curve I **B.** curve II
C. curve III **D.** curve IV

Q5.96.3 MCQ 1996 1M

During hunting of synchronous motor

A. negative phase sequence currents are generated

B. harmonics are developed in the armature circuit

C. damper bar develops torque

D. field excitation increases

Q5.95.1 MCQ 1995 1M

A synchronous motor on load draws a current at a leading power factor angle ϕ. If the internal power factor angle – which is the phase angle between the excitation e.m.f. and the current in the time phasor diagram is ψ, then the air gap excitation m.m.f lags the armature m.m.f by

A. ψ **B.** $\frac{\pi}{2} + \psi$ **C.** $\frac{\pi}{2} - \psi$ **D.** $\psi + \phi$

Q5.95.2 NAT 1995 1M

The distribution factor for a 36 slot stator with three-phase, 8-pole winding, having 120° phase spread, is _____

Q5.95.3 NAT 1995 5M

A three-phase, 50 Hz, 415 V, synchronous machine operates at rated voltage and at a leading power factor of 0.9. Shaft power is 15 kW and the excitation emf is 400 V. If per phase resistance is 0.5 ohm, find the synchronous reactance. Neglect mechanical losses of the system.

Q5.94.1 MCQ 1994 1M

Match the items on the right hand side with those on left hand side

(a)	Commutation	(P)	Resistive load
(b)	V-curves	(Q)	Inductive load
(c)	Freewheeling diode	(R)	Capacitive load
(d)	Overlap	(S)	Inter pole
		(T)	Source Inductance
		(U)	Synchronous motor

A. $a-S,\ b-U,\ c-Q,\ d-T$

B. $a-T,\ b-U,\ c-R,\ d-S$

C. $a-S,\ b-T,\ c-Q,\ d-U$

D. $a-U,\ b-S,\ c-R,\ d-T$

Q5.93.1 MCQ 1993 2M

A three phase alternator has negligible stator resistance. A short circuit test is conducted on this alternator. At a particular speed a field current of I_{f1} is required to drive the rated armature current. If the speed of the alternator is reduced to half, the field current required to maintain rated armature current

A. would be equal to I_{f1}

B. would be equal to $2I_{f1}$

C. would be equal to $I_{f1}/2$

D. cannot be predicted due to insufficient data

Q5.93.2 MSQ 1993 2M

A synchronous motor operates at 0.8 p.f. lagging. If the field current of the motor is continuously increased

☐ **A.** the power factor decreases upto a certain value of field current and thereafter it increases

☐ **B.** the armature current increases upto a certain value of field current and thereafter it decreases

☐ **C.** the power factor increases upto a certain value of field current and thereafter it decreases

☐ **D.** the armature current decreases upto a certain value of field current and thereafter it increases

Q5.92.1 MCQ 1992 1M

The torque angle of a synchronous machine operating from a constant voltage bus, is usually defined as the space angle between

A. Rotor mmf wave and stator mmf wave

B. Rotor mmf wave and resultant flux density wave

C. Stator mmf wave and resultant flux density wave

D. Stator mmf wave and resultant mmf wave

Q5.92.2 MCQ 1992 5M

Figure depicts the load characteristics of an isolated three-phase alternator, running at constant speed. Match the following sets of operating conditions with the given characteristics. Disregard the effects of saliency, saturation and stator resistance.

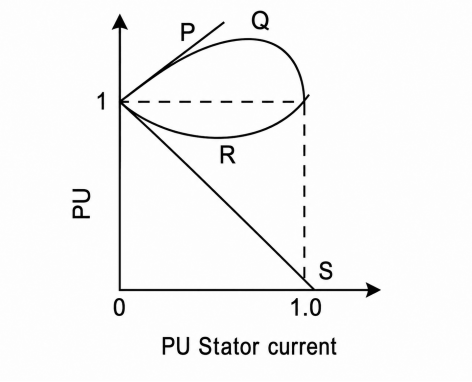


Fig. Q5.92.2

(a) Constant excitation and non-zero leading power-factor
(b) Constant excitation and zero power-factor, leading
(c) Constant terminal voltage and zero power-factor, leading
(d) Constant terminal voltage and non-zero leading power-factor

A. $P-b,\ Q-a,\ R-d,\ S-c$

B. $P-d,\ Q-a,\ R-b,\ S-c$

C. $P-c,\ Q-b,\ R-d,\ S-a$

D. $P-b,\ Q-c,\ R-a,\ S-d$

Q5.91.1 MCQ 1991 1M

Damper winding is provided in a poly-phase synchronous motor in order to

A. Damp out the noise of the machine

B. Prevent hunting

C. Provide starting torque

D. Provide a cylindrical structure to reduce wind friction

Q5.91.2 NAT 1991 5M

A 10 kVA, 380 V, 4-pole, 50 Hz, star-connected cylindrical rotor alternator has a stator resistance and synchronous reactance of 1 Ω and 15 Ω respectively. It supplies a load of 8 kW at rated voltage and 0.8 power factor lagging.

(a) Draw a phasor diagram of operation

(b) Express the resistance and synchronous reactance in per unit values with the machine rating as the base

(c) Calculate the percentage regulation

(d) What is the terminal voltage if the load is suddenly removed (with the speed and excitation unaltered)?

Answer Key

Q5.26.1 B	Q5.26.2 C	Q5.25.1 66.25 to 66.75
Q5.25.2 A,C	Q5.24.1 A	Q5.23.1 1.6022
Q5.22.1 245	Q5.20.1 A	Q5.20.2 A
Q5.20.3 125.29	Q5.19.1 D	Q5.19.2 245 to 246
Q5.18.1 B	Q5.17.1 838.3	Q5.17.2 0.74 to 0.75
Q5.17.3 B	Q5.17.4 -14.55	Q5.16.1 1.608

Q5.16.2 A	Q5.15.1 47.26	Q5.14.1 15.06
Q5.14.2 A	Q5.14.3 D	Q5.14.4 0.10 to 0.12
Q5.14.5 50	Q5.14.6 D	Q5.14.7 33.47
Q5.11.1 B	Q5.11.2 B	Q5.09.1 D
Q5.08.1 D	Q5.08.2 A	Q5.08.3 D
Q5.07.1 B	Q5.07.2 C	Q5.07.3 C

Q5.06.1 B	Q5.06.2 A	Q5.06.3 C
Q5.06.4 C	Q5.06.5 A	Q5.05.1 C
Q5.05.2 B	Q5.05.3 C	Q5.04.1 A
Q5.04.2 C	Q5.04.3 B	Q5.04.4 D
Q5.03.1 A	Q5.03.2 D	Q5.03.3 A
Q5.02.1 B	Q5.02.2 242.92, 35.382, 0.985	Q5.01.1 B

Q5.01.2 C	Q5.01.3 D	Q5.01.4 91.80 to 91.85
Q5.01.5 A:35.32, 26.49 B:44.6, 33.45	Q5.00.1 B	Q5.00.2 B
Q5.00.3 35.056, 26.29	Q5.99.1 C	Q5.99.2 B
Q5.99.3 0.8556	Q5.98.1 A	Q5.98.2 D
Q5.97.1 180, −10	Q5.96.1 B	Q5.96.2 B
Q5.96.3 C	Q5.95.1 C	Q5.95.2 0.844
Q5.95.3 2.73	Q5.94.1 A	Q5.93.1 A
Q5.93.2 C,D	Q5.92.1 A	Q5.92.2 A
Q5.91.1 C	Q5.91.2 0.069, 1.038, 85.54, 705.07	

UNIT

VI

Miscellaneous Topics

Topics

A. Single Phase/Universal Motors

B. Special machines

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 25	**MCQ:** 23	**MSQ:** 0	**NAT:** 2
1M: 14	**2M:** 11	**Descriptive:** 0	**Avg Weightage** ~1M	**Range:** 0M(2014)–2M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q6.23.1 MCQ 2023 1M

A 10-pole, 50 Hz, 240 V, single phase induction motor runs at 540 RPM while driving rated load. The frequency of induced rotor currents due to backward field is

A. 100 Hz **B.** 95 Hz **C.** 10 Hz **D.** 5 Hz

Q6.22.1 MCQ 2022 1M

The type of single-phase induction motor, expected to have the maximum power factor during steady state running condition, is

A. split phase (resistance start).

B. shaded pole.

C. capacitor start.

D. capacitor start, capacitor run.

Q6.18.1 NAT 2018 2M

The equivalent circuit of a single phase induction motor is shown in the figure, where the parameters are $R_1 = R_2' = X_1 = X_{l2}' = 12\,\Omega$, $X_m = 240\,\Omega$ and s is the slip. At no-load, the motor speed can be approximated to be the synchronous speed. The no-load lagging power factor of the motor is _____ (up to 3 decimal places).

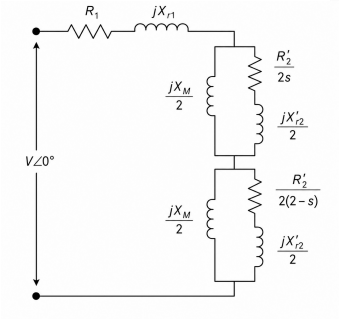


Fig. Q6.18.1

Q6.17.1 NAT 2017 2M

A 375 W, 230 V, 50 Hz, capacitor start single-phase induction motor has the following constants for the main and auxiliary windings (at starting): $Z_m = (12.50 + j15.75)\,\Omega$ (main winding), $Z_a = (24.50 + j12.75)\,\Omega$ (auxiliary winding). Neglecting the magnetizing branch, the value of the capacitance (in μF) to be added in series with the auxiliary winding to obtain maximum torque at starting is _____.

Q6.16.1 MCQ 2016 1M

The direction of rotation of a single-phase capacitor run induction motor is reversed by

A. interchanging the terminals of the AC supply.

B. interchanging the terminals of the capacitor.

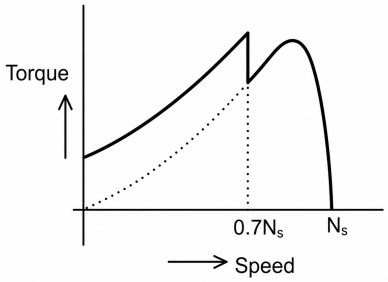
C. interchanging the terminals of the auxiliary winding.

D. interchanging the terminals of both the windings.

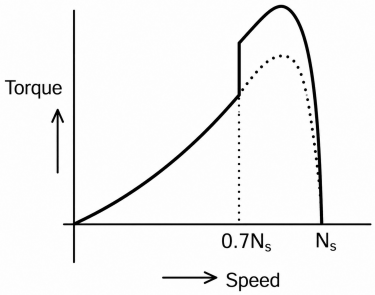
Q6.14.1 MCQ 2014 1M

A single phase induction motor is provided with capacitor and centrifugal switch in series with auxiliary winding. The switch is expected to operate at a speed of $0.7\,N_s$, but due to malfunctioning the switch fails to operate. The torque-speed characteristic of the motor is represented by

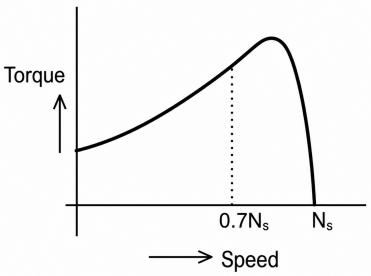
A.

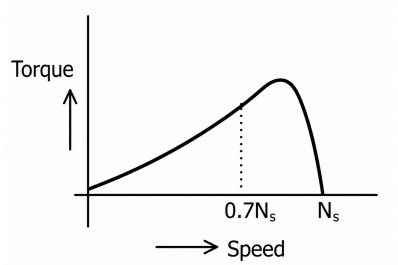


B.



C.





D.

Q6.12.1 MCQ 2012 1M

A two-phase load draws the following phase currents:

$$i_1(t) = I_m \sin(\omega t - \phi_1), \qquad i_2(t) = I_m \cos(\omega t - \phi_2).$$

These currents are balanced if ϕ_1 is equal to

A. $-\phi_2$ **B.** ϕ_2 **C.** $(\pi/2 - \phi_2)$ **D.** $(\pi/2 + \phi_2)$

Q6.09.1 MCQ 2009 2M

A 220 V, 50 Hz, single-phase induction motor has the following connection diagram and winding orientations shown. MM' is the axis of the main stator winding (M_1M_2) and AA' is that of the auxiliary winding (A_1A_2). Directions of the winding axes indicate direction of flux when currents in the windings are in the directions shown. Parameters of each winding are indicated. When switch S is closed, the motor

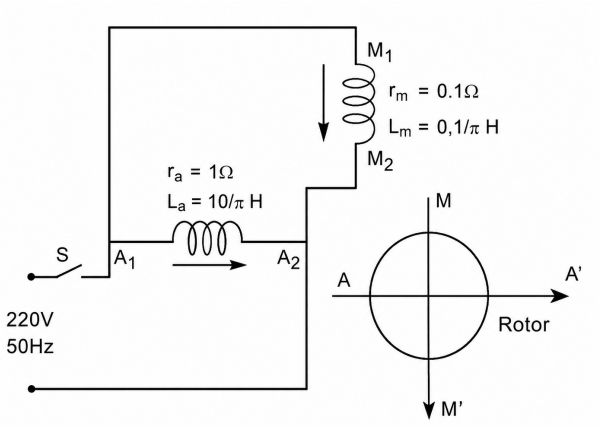


Fig. Q6.09.1

A. rotates clockwise

B. rotates anticlockwise

C. does not rotate

D. rotates momentarily and comes to a halt

Q6.08.1 MCQ 2008 2M

A 230 V, 50 Hz, 4-pole, single-phase induction motor is rotating in the clockwise (forward) direction at a speed of 1425 rpm. If the rotor resistance at standstill is 7.8 Ω, then the effective rotor resistance in the backward branch of the equivalent circuit will be

A. 2 Ω **B.** 4 Ω **C.** 78 Ω **D.** 156 Ω

Q6.08.2 MCQ 2008 1M

In a stepper motor, the detent torque means

A. Minimum of the static torque with the phase winding excited.

B. Maximum of the static torque with the phase winding excited.

C. Minimum of the static torque with the phase winding unexcited.

D. Maximum of the static torque with the phase winding unexcited.

Q6.07.1 MCQ 2007 2M

A three-phase, three-stack, variable reluctance step motor has 20 poles on each rotor and stator stack. The step angle of this step motor is

A. 3° **B.** 6° **C.** 9° **D.** 18°

Q6.06.1 MCQ 2006 1M

For a single phase capacitor start induction motor which of the following statements is valid?

A. The capacitor is used for power factor improvement

B. The direction of rotation can be changed by reversing the main winding terminals

C. The direction of rotation cannot be changed

D. The direction of rotation can be changed by interchanging the supply terminals

Q6.05.1 MCQ 2005 2M

In a single phase induction motor driving a fan load, the reason for having a high resistance rotor is to achieve

A. Low starting torque **B.** quick acceleration
C. high efficiency **D.** reduced size

Q6.04.1 MCQ 2004 1M

The type of single-phase induction motor having the highest power factor at full load is

A. shaded pole type **B.** split-phase type
C. capacitor-start type **D.** capacitor-run type

Q6.04.2 MCQ 2004 2M

A single-phase, 230 V, 50 Hz, 4 pole, capacitor-start induction motor has the following stand still impedances.

Main winding $Z_m = 6.0 + j4.0\,\Omega$

Auxiliary winding $Z_a = 8.0 + j6.0\,\Omega$

The value of the starting capacitor required to produce 90° phase difference between the currents in the main and auxiliary windings will be

A. 176.84 μF **B.** 187.24 μF
C. 265.26 μF **D.** 280.86 μF

Q6.04.3 MCQ 2004 2M

For a 1.8°, 2-phase bipolar stepper motor, the stepping rate is 100 steps/second. The rotational speed of the motor in rpm is

A. 15 **B.** 30 **C.** 60 **D.** 90

Q6.04.4 MCQ 2004 1M

For a given stepper motor, the following torque has the highest numerical value.

A. Detent torque **B.** Pull-in torque
C. Pull-out torque **D.** Holding torque

Q6.04.5 MCQ 2004 2M

A rotating electrical machine having its self-inductances of both the stator and the rotor windings, independent of the rotor position will definitely not develop

A. Starting torque **B.** Synchronizing torque
C. Hysteresis torque **D.** Reluctance torque

Q6.03.1 MCQ 2003 2M

A single phase induction motor with only the main winding excited would exhibit the following response at synchronous speed

A. Rotor current is zero

B. Rotor current is non-zero and is at slip frequency

C. Forward and backward rotating fields are equal

D. Forward rotating field is more than the backward rotating field

Q6.03.2 MCQ 2003 1M

Group I lists different applications and Group II lists the motors for these applications. Match the application with the most suitable motor and choose the right combination among the choices given thereafter

Group I		Group II	
P	Food mixer	1	Permanent magnet dc motor
Q	Cassette tape recorder	2	Single phase induction motor
R	Domestic water pump	3	Universal motor
S	Escalator	4	Three phase induction motor
		5	DC series motor
		6	Stepper motor

A. $P-3,\ Q-6,\ R-4,\ S-5$

B. $P-1,\ Q-3,\ R-2,\ S-4$

C. $P-3,\ Q-1,\ R-2,\ S-4$

D. $P-3,\ Q-2,\ R-1,\ S-4$

Q6.00.1 MCQ 2000 1M

A 1.8° step, 4-pole stepper motor has a total of 40 teeth on 8 poles of stator. The number of rotor teeth for this motor will be

A. 40 **B.** 50 **C.** 100 **D.** 80

Q6.99.1 MCQ 1999 1M

Starting torque can be obtained in the case of a single phase induction motor with identical main and auxiliary windings by connecting

A. a capacitor across the mains

B. a capacitor in series with the machine

C. a capacitor in series with the auxiliary winding

D. the main and the auxiliary windings in series

Q6.96.1 MCQ 1996 1M

In case of a split phase motor, the phase shift between currents in the two windings is around

A. 30 degrees
B. 70 degrees
C. 90 degrees
D. 120 degrees

Q6.96.2 MCQ 1996 1M

The torque speed characteristic of a repulsion motor resembles which of the following dc motor characteristic?

A. Separately excited
B. Shunt
C. Series
D. Compound

Q6.96.3 MCQ 1996 2M

Which type of motor is most suitable for computer printer drive?

A. Reluctance motor
B. Hysteresis motor
C. Shaded pole motor
D. Stepper motor

Answer Key

Q6.23.1 B	Q6.22.1 D	Q6.18.1 0.1062
Q6.17.1 149.5	Q6.16.1 C	Q6.14.1 C
Q6.12.1 B	Q6.09.1 C	Q6.08.1 A
Q6.08.2 D	Q6.07.1 B	Q6.06.1 B
Q6.05.1 B	Q6.04.1 D	Q6.04.2 A
Q6.04.3 B	Q6.04.4 C	Q6.04.5 D
Q6.03.1 B	Q6.03.2 C	Q6.00.1 B
Q6.99.1 C	Q6.96.1 A	Q6.96.2 C
Q6.96.3 D		

Also in this Series

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NexusX	Network Theory, Analog Electronics, Digital Electronics, Control Systems and Signals & Systems — EC, EE, IN
SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.

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